Entrepreneurship and Intellectual Property Rights

V Semester SEE Question Papers

Course codes: HS351TA, 21HS51A, 18HSI51, 16HSI51, HS361TA

- Jul/Aug 2026 (HS361TA)
- Feb/Mar 2026 (HS351TA)
- Aug 2025 (HS361TA)
- Feb/Mar 2025 (HS351TA)
- Apr/May 2024 (21HS51A)
- Apr 2024 (18HSI51)
- Jun 2023 Makeup (18HSI51)
- Apr 2023 (16HSI51)
- Mar/Apr 2023 (18HSI51)
- Jan/Feb 2021 (16HSI51)
- Jan/Feb 2021 (18HSI51)
- Dec 2020/Jan 2021 (18HSI51)
- Nov/Dec 2019 (16HSI51)
- Fast Track Jul 2019 (16HSI51)

HS361TA

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B.E. Regular / Supplementary Examination July / August 2026

ENTREPRENEURSHIP AND INTELLECTUAL PROPERTY RIGHTS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	State any two scopes of entrepreneurship.	02	2	1
	1.2	How does entrepreneurship contribute to national development?	02	1	1
	1.3	Differentiate between an entrepreneur and an intrapreneur.	02	1	1
	1.4	Outline the concept of break-even analysis.	02	2	3
	1.5	What factors should an entrepreneur consider before choosing a source of finance?	02	2	2
	1.6	Mention any two objectives of Trademark protection.	02	2	4
	1.7	Give an example of a non-registrable mark.	02	2	4
	1.8	What is the validity period of a registered Design in India?	01	1	2
	1.9	What happens if a Trade Secret becomes public?	01	2	2
	1.10	Mention any two forms of intellectual property rights available to businesses.	02	1	3
	1.11	In the context of ideation, what is the primary role of integrating engineering principles?	02	2	2

PART-B

2	a	A textile business owner continued using traditional methods despite competitors adopting automated production systems. After observing the success of automation over several years, he gradually introduced automated machines. (i) Identify the type of entrepreneur. (ii) State two characteristics of this entrepreneur. (iii) Analyze the advantages and disadvantages of such entrepreneurial behavior.	08	3	2
	b	Ravi works as a project manager in a manufacturing company. He developed an AI-based production monitoring system that improved efficiency by 25%. The company provided resources and support for implementation. (i) Is Ravi an entrepreneur or an intrapreneur? Justify your answer. (ii) Differentiate between entrepreneur and intrapreneur with respect to this case. (iii) Evaluate the benefits gained by the company through Ravi's initiative.	08	4	2
3	a	A startup wants to launch an electric bicycle rental service in Bengaluru. Describe the steps involved in conducting market feasibility and demand analysis.	08	4	2

	b	Rationalize the process of developing a Business Model and a Minimum Viable Product (MVP). How do these help in validating a startup idea?	08	3	3
		OR			
4	a	Apply Porter's Generic Competitive Strategies to a supply-chain business and justify which strategy would help it achieve a competitive advantage. Support your answer with suitable examples.	08	3	4
	b	Analyze how cross-disciplinary collaboration enhances technological innovation in startups.	08	2	2
5	a	Describe the major financial statements and explain their importance in business decision-making.	08	2	3
	b	Discuss the significance of contracts and corporate governance in entrepreneurial ventures.	08	2	3
		OR			
6	a	Develop a marketing mix (4Ps) for a newly launched technology-based startup product.	08	3	1
	b	A startup observed that some employees consistently exceeded performance targets while others struggled to meet expectations. The company had no formal appraisal system. Discuss how performance evaluation can improve employee productivity and organizational effectiveness.	08	4	3
7	a	Discuss the functions of Trade Marks and Eco-Labels.	06	3	4
	b	Define "Patent Transfer". Discuss the various methods by which patent rights can be transferred.	10	3	4
		OR			
8	a	An international company develops a method to extract active compounds from a plant traditionally used in Ayurveda, claiming higher efficiency and stability. Can this method be patented, and how does it relate to the protection of traditional knowledge?	08	3	4
	b	A software company patents an algorithm for data encryption. Another company independently develops a similar algorithm and uses it in its products. Can the first company claim infringement?	08	4	4
9	a	Explain the features of Industrial Design and discuss its importance in product development and marketing.	08	2	2
	b	A furniture manufacturer creates an innovative chair design but does not register it under the Designs Act. A competing company later registers a similar design and claims exclusive ownership. (i) Identify the Intellectual Property involved. (ii) Discuss the legal issues faced by the original manufacturer. (iii) Explain why design registration is important. (iv) What lessons can businesses learn from this case?	08	4	3
		OR			
10	a	A college computer laboratory installs one licensed software package on fifty computers without purchasing additional licenses. Discuss the legal consequences.	08	4	4
	b	Explain the legal framework governing Trade Secret Protection in India. Discuss the challenges in protecting Trade Secrets.	08	2	2

Question Paper ID: 315151

Course Code: HS351TA

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

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Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

Common to AS / CH / CV / EC / EE / EI / ET / IM / ME

Entrepreneurship and Intellectual Property Rights

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Identify one analytical method used for opportunity identification.	02	1	1
1.2	Differentiate between an entrepreneur and an intrapreneur.	02	1	1
1.3	State the main advantage of cross-disciplinary collaboration for technological innovation.	02	2	2
1.4	What is meant by Entrepreneurial Finance and Resource Management?	02	3	3
1.5	Define the 4Ps in entrepreneurial marketing	02	1	1
1.6	Identify the purpose of Targeting in the STP process.	02	2	2
1.7	Mention any two objectives of the Patent System.	02	4	2
1.8	Mention any two methods used for IP Valuation.	02	4	2
1.9	What is Patent Drafting?	02	4	2
1.10	Mr. X has discussed an idea of video games with Mr. Y and Mr. Z. Later, Mr. Y has written a Video games based on discussion. Who will be author and owner of video games?	02	1	1

Part B

Question No	Question	M	CO	BT
2a	Discuss the role of entrepreneurship in promoting innovation and contributing to economic development. Support your answer with real-world examples.	08	1	3

2b	Differentiate between innovative, imitative, and Fabian entrepreneurs with suitable examples. Analyze how each type contributes differently to industrial and economic growth	08	1	4
3a	Define Market Opportunity and explain the key factors to identify potential market opportunities.	08	2	2
3b	Evaluate the advantages and limitations of growth strategies such as mergers, acquisitions, and strategic alliances.	08	2	2
	OR			
4a	Evaluate the importance of technical feasibility through prototype development and proof of concept.	08	2	2
4b	Design a business model canvas for a mobile app providing real-time traffic and parking solutions.	08	3	4
5a	Develop a marketing mix (4Ps) for a newly launched technology-based startup product.	08	1	3
5b	Construct an STP (Segmentation, Targeting, Positioning) model for a startup entering a competitive consumer market.	08	2	4
	OR			
6a	Formulate a Unique Value Proposition (UVP) and branding strategy for a product aimed at millennial customers.	08	2	3
6b	Analyze how CRM tools can enhance customer retention and long-term sales growth in entrepreneurial ventures.	08	1	4
7a	Discuss in detail the concept, scope, and salient features of patents. Explain with examples the kinds of inventions that are patentable under the Indian Patent Law. How does understanding these distinctions help in promoting innovation?	08	4	3
7b	Define “Patent Transfer”. Discuss the various methods by which patent rights can be transferred.	08	4	3
	OR			
8a	Define Eco-Label. How does it differ from a Trade Mark?	08	4	3
8b	Explain the different types of Trade Mark Transfers Under the Act.	08	4	3
9a	Explain the acts which do not constitute Infringement of Copyright.	10	3	4
9b	List the advantages of protecting a design under Industrial design right.	06	2	2
	OR			
10a	Discuss any five works that can be claimed under copyrights.	10	1	1
10b	Identify and explain the key tools and measures available in India to protect trade secrets effectively?	06	2	2

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RV COLLEGE OF ENGINEERING®

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VI Semester B.E. Regular Examinations August – 2025

ENTREPRENEURSHIP AND INTELLECTUAL PROPERTY RIGHTS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	What are the two advantages of being an Intrapreneur with a company?	02	2	1
	1.2	Define Imitative (Adoptive) Entrepreneur.	02	2	1
	1.3	Define Financial Feasibility Analysis.	02	2	2
	1.4	Define Prototype Development.	02	2	2
	1.5	What is the role of CRM in building strong customer relationships and enhancing satisfaction levels?	02	2	2
	1.6	How do startups benefit from venture capital investment?	02	2	3
	1.7	Define Compulsory Licensing.	02	2	4
	1.8	Define Eco-label.	02	2	4
	1.9	How do performer's rights differ from copyright?	02	3	4
	1.10	Differentiate between industrial design and utility patent with example.	02	2	5

PART-B

2	a	Anjali, a 45-year-old teacher, wants to start her own online learning platform. Her friends discourage her, saying she's too old, doesn't have a tech background, and won't be able to compete with big companies. They also tell her she needs a huge investment to start. Questions: (i) Identify and describe any three myths reflected in the friends' opinions. (ii) What advice would you give Anjali to overcome these misconceptions? (iii) Mention two real-life entrepreneurs who succeeded despite similar challenges.	08	3	1
	b	How can engineering colleges promote entrepreneurship among students? Propose initiatives or program that link technical skills with business development.	08	2	1
3	a	Discuss the significance of break-even analysis for startups and emerging businesses.	08	4	2
	b	Develop a detailed business plan framework for a new e-commerce startup, incorporating key elements such as market research, financial projections, and operational strategies.	08	4	2
		OR			

4	a	You are part of a student team developing a tech-based startup idea. Discuss how you would approach the process of transforming this idea into a viable business opportunity by addressing the following aspects: (i) Identifying market opportunities and analyzing current trends. (ii) Applying engineering principles during the ideation and design phase. (iii) Promoting cross-disciplinary collaboration for innovation. (iv) Conducting market feasibility and demand analysis. (v) Evaluating technical feasibility through prototype development and proof of concept. (vi) Analyzing financial feasibility, including cost estimation and revenue projections.	10	2	2
	b	Define SWOC analysis and discuss its significance in strategic planning. How can organizations use SWOC analysis to formulate effective business strategies?	06	2	2
5	a	A company wants to launch a new energy drink. Explain how it can use market segmentation, targeting, and positioning to ensure successful market entry.	08	4	3
	b	You are developing a startup offering an innovative eco-friendly packaging solution. Explain how you would create a compelling Unique Value Proposition (UVP) that clearly communicates your product's value to customers. In your answer, discuss: (i) What UVP is and why it is important (ii) The steps you would take to develop a UVP (iii) How to differentiate your product from competitors (iv) How your UVP would influence branding, marketing and customer perception. Support your response with logical reasoning and, where possible with relevant examples.	08	4	3
		OR			
6	a	"Entrepreneurs face several legal and ethical challenges while establishing and running their ventures. Discuss the importance of understanding and managing the following aspects in entrepreneurship: (i) Intellectual Property Rights (IPR) – including patents, trademarks, and copyrights (ii) Contracts – their role, essential elements, and how they protect business interests (iii) Corporate Governance – principles, ethical considerations, and its impact on business sustainability	08	2	3
	b	Examine the concept of Corporate Social Responsibility (CSR) and discuss its significance for entrepreneurs in ensuring ethical business practices. Include real-world examples to illustrate the impact of CSR on business success and reputation. Importance of CSR for Entrepreneurs.	08	3	3
7	a	Explain the criteria for patentability of an invention and describe the step-by-step procedure to file a patent application.	08	3	4
	b	Discuss the relationship between traditional knowledge and patent law. Explain the challenges involved in protecting traditional knowledge through patents and describe alternative ways to safeguard such knowledge.	08	3	4
		OR			
8	a	Explain the concept of passing off and trademark infringement. Discuss the legal remedies available to business affected by these issues. Support your answer with relevant case studies to illustrate how courts have dealt with passing off and trademark infringement disputes.	08	3	4

	b	Describe the absolute grounds for which a trademark registration can be denied.	08	3	4
9	a	Discuss the legal tools available in India for protecting trade secrets. How do these tools help business safeguard their confidential information?	08	3	3
	b	Explain the concept of passing off and trademark infringement. Discuss the legal remedies available to businesses affected by these issues. Support your answer with relevant case studies to illustrate how courts have dealt with passing off and Trade Mark.	08	3	5
OR					
10	a	Examine how copyright law strikes a balance between protecting creators' rights and serving the public interest. What challenges arise in enforcing copyright in today's digital environment?	08	3	5
	b	Define industrial design and highlight its main characteristics. Outline the process to secure design protection and explain the reasons why design rights may be revoked.	08	3	5

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Regular Examinations Feb/March-2025

COMMON TO AS / BT / CH / CV / IEM / ME

ENTREPRENEURSHIP AND INTELLECTUAL PROPERTY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

PART-A

			M	BT	CO
1	1.1	List four types of entrepreneurs.	02	1	1
	1.2	Define: Entrepreneur.	02	2	1
	1.3	Identify the Activities involved in Market Demand Assessment.	02	4	2
	1.4	How can a business plan help in risk management?	02	4	2
	1.5	Explain the two types of promotional strategies.	02	2	3
	1.6	_____ is a fundamental framework in marketing that helps businesses identity and reach their most valuable customers.	02	1	3
	1.7	Give an example which brings about the fact that a mark cannot be registered as trademark if it consists exclusively of goods.	02	4	5
	1.8	List any two different forms of Trademark (Any Two).	02	1	5
	1.9	What are the criteria for a design to qualify for registration as an industrial design?	02	2	4
	1.10	Define performer's Rights?	02	3	5

PART-B

2	a	Justify the importance of entrepreneurship.	08	5	1
	b	What factors contributed to Flipkart's success in the Indian market.	05	4	1
	c	What challenges did Flipkart face during its growth phase?	03	3	1
3	a	A company's gross profit margin is increasing, but its net profit margin is decline. What could be the potential reasons?	08	4	2
	b	List the key Contents of a Business Plan (Any 8)	08	1	3
		OR			
4	a	Explain the key components of financial feasibility analysis?	10	2	2
	b	An entrepreneur wants to launch a niche product. What steps should they take to access market feasibility?	06	4	2
5	a	What is meant by Market Segmentation? Give its importance.	04	2	2
	b	A clothing brand wants to launch a new line targeting eco-conscious millennials. How should they approach market segmentation?	04	5	2
	c	Define *SWOC* Analysis. Explain its significance in strategic Management.	08	4	2

		OR			
6	a	What are the Types of Insurance as Risk Management tools?	08	2	2
	b	What are the objectives of financial management? Investigate financial management tools & Techniques**.**	08	5	2
7	a	Explain the tools to protect a trade secret?	08	1	4
	b	Explain the Inventions which are not patentable under the Indian Patent Act.	08	5	4
		OR			
8	a	Discuss the Trade mark infringement among the corporate classes in India with Two Case examples	08	2	5
	b	A company owns the trademark “Perfect Clean” for cleaning products. Over time, customers start using “Perfect Clean” as a generic term for all cleaning products. How can the company prevent the loss of its trademark rights?	08	5	5
9	a	List & explain steps involved for obtaining Design Protection under the *ID* Act.	08	1	4
	b	Copy Right protects expression but not the ideas. Discuss.	08	5	4
		OR			
10	a	A digital artist uses an AI tool to generate artwork and claims copyright over the output. Can they enforce copyright on this work?	06	5	5
	b	Explain the acts which do not constitute Infringement of copyright under the act.	10	1	4

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Regular Examinations April/May-2024

COMMON TO AIML/CSE/ISE/ECE/EEE/EI/ET

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

		PART-A	M	BT	CO
1	1.1	When the patentee agrees to give another person certain defined right in the patent with immediate effect by a document and not by an agreement the assignment is termed as _____.	01	2	1
	1.2	An assignment of an existing patent through an _____ which has been duly registered is called legal assignment.	01	1	2
	1.3	Whether the assignee of a true inventor can ask for Patent?	01	4	1
	1.4	_____ of the Act makes it clear that the proprietor of a certification trademark should not himself carry on a trade in the goods of the kind certified or a trade of the provision of services of the kind certified.	01	1	1
	1.5	The term Entrepreneur is derived from the term “entreprendre” which means _____.	01	1	4
	1.6	Symbol of Maharaja of Air India is _____.	01	4	1
	1.7	What is the term for the act of registering someone else’s domain name with the intent to profit from its sale to the rightful owner?	01	1	2
	1.8	What does DRM stand for in the context of protecting intellectual property rights?	01	1	1
	1.9	How does watermarking contribute to protecting intellectual property rights in digital media?	01	4	2
	1.10	In India, the literary work is protected until _____.	01	1	1
	1.11	What symbol is commonly used to indicate copyright protection?	01	2	1
	1.12	What is the term used for gaining unauthorized access to a computer system or network?	01	2	2
	1.13	A student photographer takes a picture using a school-owned camera as part of a class assignment. The student never signed any agreement to give away his copyrights and the student handbook doesn’t say anything about who gets it. Who owns the copyright?	01	4	2
	1.14	_____ is an application of information and communication technology (ICT) for delivering Government Service.	01	1	1
	1.15	Susie’s department is implementing many projects. She finds herself starting and stopping work on one task to go and work on another task, and then return to the work on the original task. Susie is experiencing:	01	2	3
	1.16	_____ Monitors all internet and other network activity, looking for suspicious data and preventing unauthorized access.	01	2	2

	1.17	Which international agreement provides a framework for the protection of intellectual property rights, including in the digital environment?	01	4	2
	1.18	What is a subscription business model? Give an example.	01	1	4
	1.19	Give the classification of projects.	02	1	4

PART-B

2	a	What details should a complete specification contain?	08	1	1
	b	What is the Rice patent? Explain the Indian Government's response to the Rice patent.	08	2	2
3	a	Explain the Tools to protect a trade secret.	08	2	2
	b	Illustrate the Trade mark infringement/'passing off' case in the context of cookie manufacturer Malaysian.	08	3	2
		OR			
4	a	Illustrate and explain the different categories of NON-REGISTRABLE TRADEMARKS.	08	3	2
	b	Illustrate any two cases of Trade Mark Violation in India.	08	3	2
5	a	Describe the procedure for obtaining Design Protection.	10	2	2
	b	Illustrate any two case examples of your choice for Design Protection.	06	3	1
		OR			
6	a	Compare between Electronic Signature and Digital Signature (Any 8 differences).	08	3	2
	b	Illustrate the Infringement of Copy Right with suitable Case examples.	08	3	1
7	a	Analyze the different Entrepreneurial Development Models.	08	3	4
	b	Illustrate with appropriate examples myths of entrepreneurship.	08	3	3
		OR			
8	a	Explain the challenges encountered by the women in their entrepreneurial journey.	08	2	4
	b	Determine the measures required for Capacity Building for Entrepreneurship in India.	08	3	3
9	a	Analyze the reasons as to why some Business Plans fail. (Ten reasons)	10	3	3
	b	Justify and elaborate using relevant real-world examples of successful business model innovation.	06	4	4
		OR			
10	a	Discuss the steps on procedure for Setting up an enterprise in Indian context.	08	4	3
	b	Illustrate the different types of business models with examples.	08	3	4

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Apr 2024

COMMON TO EE / CS / ET / IS / BT / EC

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

			M	CO	BT
1	1.1	The broadcasting organizations like TV, Radio have a term of copyright for _____.	01	1	2
	1.2	If a person invents a new process for recording music, he or she will likely to apply for a _____.	01	1	2
	1.3	An invention has to be _____ and _____ in order to patentable.	02	2	2
	1.4	_____ is the period of limitation for filing suit for infringement of trademark in India.	01	1	2
	1.5	If the term of the years of the assignment of patents is not mentioned, then it is assigned for _____ years.	01	2	1
	1.6	Symbol of Maharaja of Air India is a _____.	01	1	2
	1.7	The trademark is registered and can be renewed every time for a period of _____.	01	2	2
	1.8	WIPO stands for _____.	01	1	2
	1.9	Does inventions relating to atomic energy are patentable or not?	01	2	1
	1.10	Software is protected as intellectual property in India primarily under the law of _____.	01	1	2
	1.11	Validity of patent is _____ years from the date of registration.	01	2	1
	1.12	The Information Technology came into effect in the year _____.	01	3	2
	1.13	_____ firm is associate the most with Design Thinking.	01	2	2
	1.14	Design Thinking typically helps in _____.	01	3	2
	1.15	In order to register Industrial Design, design should be new and _____.	01	3	2
	1.16	Whether the shape and configuration of spring can be registered as Industrial Design in India? Analyze.	02	4	2
	1.17	_____ is an important source of idea generation due to their familiarity with the needs of market.	02	3	2

PART-B

2	a	List and explain different types of “Intellectual property rights”.	08	1	2
	b	Explain the essential features of granting patent under the Indian Patent Act.	08	2	3

3	a	What cannot be patented under the Indian Patent Act?	08	1	2
	b	Differentiate between patent and trade secrets.	08	2	2
		OR			
4	a	Identify the grounds for refusal of registration of a Trademark in India. What kind of Trademarks can be registered under the (Indian) Trade Marks Act, 1999?	08	2	2
	b	Discuss the infringement of Trademark with two decided case studies.	08	2	2
5	a	"Copyright protects the expression, not the ideas". Explain this statement with suitable illustrations.	08	2	3
	b	Explain the essential features of infringement of Copyright with two decided case studies.	08	2	2
		OR			
6	a	Define the term Industrial Design and explain the essential features of Industrial Design.	08	2	2
	b	Define Cyber Crimes. Explain the commonly categorized types of Cyber Crimes.	08	1	2
7	a	What do you understand by concept of entrepreneurship? Explain the characteristics of a successful entrepreneur. What is design thinking? Describe the design thinking process.	08	1	2
	b	Identify your own entrepreneurship style based on your personality traits, strengths and weaknesses.	08	2	3
8	a	Explain the method of Customer Centric selling skill in detail.	08	2	2
	b	Identify and explain the central ideas and principles of design thinking.	08	2	3

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations June 2023 (Makeup)

COMMON TO EE / CS / ET / IS / BT / EC

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Does Trademarks assure quality of the product?	01
	1.2	Traditional Knowledge is limited to any specific field of knowledge. True/False	01
	1.3	The broadcasting organizations like *TV*, Radio have a term of copyright for _____.	01
	1.4	A method of growing plants on roof top is patentable or not.	01
	1.5	If a company develops a new technology that improves its main product, what type of intellectual property can they use to stop others from copying their invention?	01
	1.6	If you file provisional specification, the complete specification is required to be filed within _____ months.	01
	1.7	_____ is a symbol generally used to identify a particular product, which indicates its source.	01
	1.8	Section 66*A* of *IT* Act, 2000 has been struck down by Supreme Court on 24th March, 2015 in _____ case.	01
	1.9	Whether the shape and configuration of spring can be registered as Industrial Design in India? Analyze	02
	1.10	If an employee learns a better way of auditing the expenses of a company from his previous employer, which he uses for auditing in the present company, then is it infringement of trade secret?	02
	1.11	A Biology Teacher in a College collects the articles written by researchers on cloning and circulates the same amongst his students. Is he guilty of infringement?	02
	1.12	Mr.Binu gets a message to share the credit card details, passwords by masquerading as trustworthy source is called as _____.	01
	1.13	Customer delight experiences frequently make their way onto social media through _____ means.	01
	1.14	List any two myths related to entrepreneurs.	02
	1.15	List two sales skill to become as effective entrepreneur.	02

PART-B

2	a	Can any invention be patented after publication or display in the public exhibition? Does the Patent Office keep information of the invention secret?	08
	b	Does the Patent Office help to select a patent attorney or agent to make patent search or to prepare and prosecute patent application? Is it mandatory to obtain prior permission from the Patent Office to file application for patent outside India or abroad?	08
3	a	List the difference between Registered and Unregistered Trademarks. What are the benefits of Trademark Registration?	08
	b	Explain the absolute grounds for refusal of registration of Trade mark.	08
		OR	
4	a	Define Trademark. Explain the method of registering trademark under the Act.	08
	b	Explain the essential features of infringement of Trade Mark with two cases.	08
5	a	Describe the Provisions of Information Technology Act, 2000 dealing with Cyber crime.	08
	b	Does plagiarism in research manuscript amount to copyright infringement? Evaluate the statement in the context of academic work. Also deliberate on the exception to Copyright infringement and Remedies for Copyright infringement in Indian Context.	08
		OR	
6	a	Copy right protects the expression, not the ideas. Justify this statement with suitable illustrations.	08
	b	Is registration compulsory for obtaining design protection? What designs are not eligible for registration under the Designs Act, 2000?	08
7	a	Evaluate the theory of David McClelland's with respect to the characteristics trait and challenges of an Entrepreneurship.	08
	b	"Entrepreneurs require high levels of creativity in all the processes they conduct". In the light of the above statement, bring out the relevance of design thinking approach in encouraging critical thinking and fostering innovation for successful and sustainable entrepreneurial activity.	08
8	a	What are the limitations of personal selling? Discuss the method of customer centric selling skill with suitable examples.	08
	b	"The reward is in the risk." "If there is no risk, there is no reward." "Life is either a daring adventure or nothing at all." All these statements are telling about the risk. Enumerate on "Risk taking as a positive trait in entrepreneurial context."	08

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations April-2023

COMMON SUBJECT

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	A scientist gets an idea on how to make biodegradable plastics. How can this idea be patented by the scientist?	02
	1.2	The patent Act 1970 does not convey any monopoly right to the Inventors Assignee if the assignee is inventors close relative. Comment.	02
	1.3	Can a person register his name as a Trademark? Why?	02
	1.4	The value of the trademark depends upon the _____ associated with it.	01
	1.5	_____ is the period of limitations for filing suit for infringement of trademark in India.	01
	1.6	List any two characteristics of effective entrepreneurs.	02
	1.7	Can opposition party leader Mehul claim copyright over his speech delivered in parliament which is written by his personal secretary ?	02
	1.8	Define industrial design.	02
	1.9	Identify any two entrepreneurial myths.	02
	1.10	Jack, a college final year student, is from a well off business family. He has spotted on opportunity in his hometown related to the production of electricity using wind. Can he become an entrepreneur and turn this into a venture?	01
	1.11	What is the meaning of the term “Contextual enquiry” with regard to empathize phase in design thinking.	02
	1.12	Failure rate in the first year of engineering seems to be more and reduces as the students progress to higher semesters. How would you propose to empathize with the first year students in order to obtain insights into numerous problems they might have? (Any two)	01

PART-B

2	a	In order to be patentable, an invention must be worthy of a patent grant. The patent worthiness of an invention is assessed by the government through patentability requirements. Explain the essential requirements of patentability.	08

	b	Enumerate the procedure for obtaining a patent in India.	08
3	a	List the factors to be taken into consideration while decoding the question of deceptive similarity with two decided cases.	08
	b	List and explain the differences between the Trade mark and passing off.	08
		OR	
4	a	Explain the absolute grounds for refusal of registration of Trade mark under the act.	08
	b	Explain the criteria and purpose for ecolabels ? Does eco labeling affect a consumer choice?	08
5	a	Define copyright fair use. What are copyright exceptions such as fair use under the Copyright Act?	08
	b	Define cyber crime. Explain the different kinds of cybercrimes under the Information Technology Act?	08
		OR	
6	a	Explain the different ways of transferring the copyrights.	08
	b	Explain the piracy of industrial design with two cases.	08
7	a	List and explain any seven specific step in entrepreneurial journey.	08
	b	Explain the ways to make effective communication.	08
8	a	What is design thinking? Describe the design thinking process.	08
	b	Explain the steps towards customer centric selling skill in detail.	08

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Mar/Apr 2023

COMMON TO EE / CS / ET / IS / BT / EC

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	On what basis, the total royalty on the book sold is computed under the copyright agreement?	01
	1.2	If a person uses someone's song as a background in his movie, then such an act could be an example for _____.	01
	1.3	International copyright is governed by the _____ Convention.	01
	1.4	A business arrangement where one party allows another party to use a business name and sell its products or services is known as _____.	01
	1.5	List the characteristics of an Entrepreneur?	02
	1.6	Why are small businesses important to country's economy?	02
	1.7	From a college computer, person '*A*' learned how to copy programs. Person '*B*' asked '*A*' to copy a program for his home use. What would be the most ethical response?	02
	1.8	What are the knowledge requirements of an entrepreneur?	02
	1.9	What should be the main worry of a company's founder who asks for capital in exchange for equity shares in their venture?	02
	1.10	List the motivations to be an entrepreneur.(any two)	02
	1.11	What are: a) Manufacturing Enterprises b) Service Enterprises	02
	1.12	List the characteristics of a good trade mark.	02

PART-B

2	a	Intellectual property can be categorized into artistic or commercial as per the nature of work. Justify.	08
	b	Turmeric is a well-known remedy for various injuries and beauty treatments. State the nature of *IPR* protection given by law.	08
3	a	Briefly explain the procedure of registering a trademark.	08
	b	Briefly explain the rights conferred to the person after the trademark registration.	08
		OR	

4	a	Briefly explain the chances of refusal of a trademark application.	08
	b	Briefly explain the exceptions to the trademark rights granted to a person.	08
5	a	Briefly explain the need for protecting an industrial design.	08
	b	Briefly explain the procedure of protecting computer software.	08
		OR	
6	a	Consider you are into a media business and planning to start a new broadcasting unit. Discuss about the rights you may exercise.	08
	b	Briefly explain the constraints of the design that cannot be protected by Industrial Design rights.	08
7	a	Entrepreneurial talents are the backbone of Indian economy. Justify.	08
	b	Briefly explain how an entrepreneur perceives opportunity, plan, organize resources and succeed the goal set.	08
8	a	Briefly explain any one success and a failure story of an entrepreneur in India.	08
	b	Briefly explain how design thinking helps the entrepreneur's success.	08

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Jan/Feb-2021

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	If a patent application is filed but a patent has not yet been issued, the applicant usually places the words patent _____ on the article.	01
	1.2	The royalty should not exceed _____ % of the net-ex-factory sale price in bulk of the patented article in respect of medicine or food.	01
	1.3	If you file provisional specification, the complete specification is required to be filed within _____ months.	01
	1.4	The term of an Indian patent is _____ years from the date of filing in respect of food and drugs.	01
	1.5	_____ is a product formula, pattern, design, and compilation of data, customer list or other business secret.	01
	1.6	Hall mark is an example for _____ mark.	01
	1.7	For licensing of Trademarks, the application for recordal of the registered user in India should be made within _____ months from the date of the agreement.	01
	1.8	An assignment where the owner transfers the rights and value of the trademark as associated with the product it sells, is known as _____.	01
	1.9	Through what mode does the central government can acquire a patent from the patentee for public purpose?	01
	1.10	If you write an original story, what type of intellectual property gives you the right to decide who can make and sell copies of your work?	01
	1.11	As per Design Thinking, empathy means _____.	01
	1.12	Finding fresh and creative solutions to problems is known as _____.	01
	1.13	In the concept of design thinking, two common categories of prototypes used by designers are _____ and _____.	02
	1.14	List any two myths about entrepreneurship.	02
	1.15	Compare Patent and Copyright.	02
	1.16	Differentiate between Copyrights and Industrial designs.	02

PART-B

2	a	Explain different methods of transferring the Patent Rights.	08
	b	Classify and explain the Intellectual Property Rights with suitable examples.	08
3	a	Differentiate between Infringement and Passing off.	05
	b	List the factors to be considered for deceptive similarity in infringement of Trademark.	03
	c	Explain the different modes of transferring the Trademark.	08
		OR	
4	a	Identify and explain the grounds for refusal of registration of a trademark in India.	08
	b	A British manufacturer of biscuits has been selling cookies in Malaysia for more than 20 *years* and has had a registered trademark 'ChipsMore' for these goods during that time. Two years ago, a Malaysian company started manufacturing and selling cookies under the brand "ChipsPlus'. Analyze the case study and comment on Infringement and Passing of.	08
5	a	List and explain the essential characteristics of a design.	04
	b	Describe the process for registration of the Industrial design.	06
	c	Explain the acts which do not constitute Infringement of Copyright.	06
		OR	
6	a	List and explain different types of Cyber-crimes.	08
	b	List and explain the rights conferred by Copyrights.	04
	c	List and explain the rights conferred by Broadcasting Reproduction Rights.	04
7	a	Identify and explain common characteristics of successful Entrepreneurs.	08
	b	List and explain any four strengths and weaknesses of an Entrepreneur.	08
8	a	Explain the first three steps of Design Thinking process with a suitable example.	10
	b	Why people are interested in becoming Entrepreneurs? Give reasons.	06

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Jan/Feb 2021

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	What is a Patent?	02
	1.2	List the criteria of patentability?	02
	1.3	Expand the following acronyms. a) $WIPO$ b) $TRIPS$	02
	1.4	When referring to copyrighted materials, what is meant by the term "Fair use"?	02
	1.5	List two examples each of companies which have letter mark and symbol mark as their trademark.	02
	1.6	Identify any two tools to protect Trade secret.	02
	1.7	$AICTE$ has mandated that you earn 100 activity points through community service as part of partial fulfillment of your $B.E$ degree Award for batches that started from $2018-19$. You want to work on an assignment that impacts people in a genuine way. So, you ask your mentor and some of your close friends about what can be done? One of them suggest that, there are lots of "old age homes" (Vriddhashram) where elderly people find shelter. Noticeable characteristics are that they seem to be sad most of the time with drooping shoulders. Identify any two niche ideas that come to you off the top of your mind that can bring smiles on their faces.	02
	1.8	For one of your assignment on design thinking, you are working with kids having hearing disability. Identify any two activities you can plan to empathize or immerse yourself in order to get a better understanding of what they go through on a normal day before you can design a useful solution for them.	02
	1.9	Caitlin is creating a new product for university students. She takes a design –thinking approach. Her first step is addressing who she is creating the product for and conducts research on understanding this target market. What is this step called in the design thinking process?	01
	1.10	What is the meaning of the term "Contextual enquiry" with regard to emphatize phase in design thinking?	01

	1.11	Design thinking is sometimes visualized as a linear process, but it is actually :_____.	01
	1.12	Jack, a college final year student, is from a well –off business family. He has spotted an opportunity in his hometown related to the production of electricity using wind. Can he become an entrepreneur and turn this into a venture?	01

PART-B

2	a	Describe the advantages of trade secrets compared to patenting?	10
	b	What is patent information? Briefly explain the significance of using the patent information. How does a patent document help in $R\,\&\,D$.	06
3	a	What is Trade Mark? What are the essential features of trade mark?	06
	b	Explain Infringement of Trademark with two case studies.	10
		OR	
4	a	Explain the absolute grounds for refusal of registration of Trademark.	08
	b	PayPal has accused PayTM of copying its logo to grow its own user base. The US based e-wallet, a pioneer in digital transactions, claims that PayTM had no other reason to use the same two-tone colour scheme in trademark infringement complaint with the Indian Trademark office. As it comes at a time when PayTM is enjoying massive gains from the demonetization of large notes, the complaint does seem rather opportunistic; however this is not what court will be analyzing. Though the colours of a logo may not mean anything to the common observer, smaller matters have been at the centre of trademark infringement cases in the past. By understanding the above case, explain deceptive similarity and the conditions on which it will be decided using similar case examples.	08
5	a	Distinguish between Industrial Design and Copyright.	05
	b	Define Cyber-crime. Explain the different kinds of cyber –crimes.	06
	c	Explain the rights conferred by copyrights.	05
		OR	
6	a	Explain the procedure to obtain Design rights.	08
	b	You happen to interact with a junior from your program in your college who has the required skills to be copywriter and industrial designer. He happens to be part of $DEBSOC, CARV$ and takes keen interest in contributing to production design in kannada sangha events. i) Explain him regarding the career choice he can pursue with these skills. ii) Comment on the skills you noticed which are required to excel in the above mentioned field.	08
7	a	Explain the ways to make effective communication.	08
	b	Describe the common characteristics of entrepreneurs.	08

8	a	Select a theme of your choice from the following and use design thinking principles/steps to solve. Emphasis should be on process of building empathy in potential users, creating a needs statement, brainstorming and ideation and generating concepts. (prototype building and testing the ideas is not included in the scope). (your thought process is being examined) i) Portable drinking water in public places and reduce *PET* bottles. ii) Reduction of food waste in lavish Indian weddings and other programs. iii) Wet and dry waste segregation for effective recycling and disposal.	06
	b	Customer is an unpaid employee as in case of an *ATM* vestibule. Position yourself in year 2008 and look forward to improve the facilities that would enable you to provide better services with respect to safety, smooth transaction by understanding the needs of the customer.	05
	c	Take a topic or a problem over which you want to brainstorm with your group of 4 people. Quote the problem definition. What would be your inputs to the problem genesis and solution space? Summarize.	05

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RV COLLEGE OF ENGINEERING®

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V Semester B.E. Dec-20/Jan-21 Examinations

DEPARTMENT OF INDUSTRIAL ENGINEERING AND MANAGEMENT

18HSI51 / 61 – INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

		PART-A	M	BT	CO
1	1.1	The broadcasting organizations like TV, Radio have a term of copyright for _____.	01	–	–
	1.2	Plan of a building can be protected by _____.	01	–	–
	1.3	A scientist involved in space research discusses a scientific principle with his colleague. The colleague uses the principle and invents a working device based on the principle. A third colleague who was instrumental in suggesting the manner of putting in place the arrangement of the mechanical devices claims himself to be the true and first inventor and applies for the patent. Is he entitled to apply for the patent? In case not, who out of the three, can apply for a patent.	02	–	–
	1.4	Collective Mark is registered by _____.	01	–	–
	1.5	A scientist in DRDO develops anti-radar chemical which, when coated on aircrafts, would make them invisible from the screen of the radar. Before his employment he was made to sign a contract which provided that all the products of his research would belong to the employer that is the DRDO. Who in the given case is authorized to apply for the patent?	02	–	–
	1.6	Symbol of Maharaja of Air India is _____.	01	–	–
	1.7	Imagine a cricketer sets up his own company to sell his own range of clothes. What type of intellectual property can he use to show that the clothes are made by his company?	01	–	–
	1.8	Does Trademarks assure quality of the product.	01	–	–
	1.9	Information Technology Amendment Act came into force on _____.	01	–	–
	1.10	Are GM (genetically modified) crops are patentable?	01	–	–
	1.11	Computer program is considered as one of the _____ Intellectual Property.	01	–	–
	1.12	The process of moving from concrete observations and stories to more abstract themes that span across groups of people is _____.	01	–	–
	1.13	_____ firm is associated the most with Design Thinking.	01	–	–
	1.14	Design Thinking typically helps in _____.	01	–	–
	1.15	Caitlin is creating a new product for university students. She takes a design-thinking approach. Her first step is addressing who she is creating the product for and conducts research on understanding this target market. What is this step in the design thinking process?	01	–	–
	1.16	In order to register Industrial Design, Design should be new and _____.	01	–	–

	1.17	Whether the shape and configuration of spring can be registered as Industrial Design in India? Analyze.	01	–	–
	1.18	_____ is an important source of idea generation due to their familiarity with the needs of market?	01	–	–

PART-B

2	a	Describe the inception and growth of World Intellectual property Organization by mentioning its role & importance for protecting the new invention.	08	–	–
	b	In order to be patentable, an invention must be worthy of a patent grant. The patent worthiness of an invention is assessed by the government through patentability requirements. Explain the essential requirements of patentability.	08	–	–
3	a	Compulsory Licensing allows governments to license third parties (that is, parties other than the patent holders) to produce and market a patented product or process without the consent of patent owners. Discuss in brief.	08	–	–
	b	Explain the different modes of Protection of Trade Secrets in relations to the role of Employees in Organizations, highlighting their main obligations as an employee.	08	–	–
		OR			
4	a	Identify the grounds for refusal of registration of a Trademark in India. What kind of Trademarks can be registered under the (Indian) Trade Marks Act, 1999?	08	–	–
	b	How does "Likelihood of Confusion" factor into Infringement of Trade Mark. Explain with examples of any two case situations.	08	–	–
5	a	"Passing off: a fall back for potentially registerable marks that were not registered or failed to renew a registration". Explain this statement with two case situations.	08	–	–
	b	Copy Right protects the expression, not the ideas. Explain this statement with suitable illustrations.	08	–	–
		OR			
6	a	Define the term Industrial Design and explain the essential features of Industrial Design.	08	–	–
	b	Define Cyber Crimes. Explain the commonly categorized types of Cyber Crimes.	08	–	–
7	a	Discuss the barriers which cause communication breakdown in an organizational setting.	08	–	–
	b	Identify your own entrepreneurship style based on your personality traits, strengths, and weaknesses.	08	–	–
8	a	Explain the method of Customer Centric selling skill in detail.	08	–	–
	b	Identify and explain the central ideas & principles of design thinking.	08	–	–

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Nov/Dec-19

Common CS / EE / EC / IS / TE / AS

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	If a person invents a new process for recording music, he or she will likely apply for a _____.	01
	1.2	Intellectual Property Rights (*IPR*) protect the use of information and ideas that are of _____.	01
	1.3	If somebody wants to develop a new method of horticulture, can he get the patent for the same?	02
	1.4	Trade-secret means _____.	01
	1.5	An invention has to be _____ and _____ in order to be patentable.	02
	1.6	Hall Mark is _____ mark.	01
	1.7	The value of the trade mark depends upon the _____ associated with it.	01
	1.8	_____ is the period of limitation for filing suit for infringement of trademark in India.	01
	1.9	_____ is an international treaty administered by *WIPO* to facilitate international trademark registration.	01
	1.10	Plan of a building can be protected by _____	01
	1.11	If the term of the years of the creative work assignment is not mentioned, then it is assigned for _____ years	01
	1.12	If an employee in a company develops a program, would this employee own the copyright?	02
	1.13	Compulsory license will be granted only after the expiry of _____ years.	01
	1.14	In a successful copyright infringement case, the plaintiff can recover _____	01
	1.15	The word entrepreneur has been taken from the _____ language.	01
	1.16	List two sales skill necessary to become an effective entrepreneur.	01
	1.17	Why should an entrepreneur do a feasibility study for starting a new venture?	01

PART-B

2	a	What is Intellectual Property? Explain different types of Intellectual property.	08
	b	Describe the procedure for obtaining a patent.	08
3	a	Explain the factors to be considered for deceptive similarity in infringement of Trade Mark.	08
	b	Define Trade Mark? Explain the features of registrable and non-registrable Trademark.	08
		OR	
4	a	Discuss the functions and features of Trade Mark with illustrations.	08
	b	Explain the infringement of Trademark with two decided case studies.	08
5	a	What is the object of registration designs? Mention the types of designs not registrable under the Act?	08
	b	"The philosophical justification for including computer programmes within the definition of 'literary work' has been that computer programmes are also products of intellectual skill like any other Literary work". Discuss	08
		OR	
6	a	Explain the rights of performer and broad casting organization under law of copyright.	08
	b	Can copyright be assigned? If so, what is the mode of assigning copyright?	08
7	a	What do you understand by concept of entrepreneurship? Explain the characteristics of a successful entrepreneur.	08
	b	List the six myths about Entrepreneurship. Write a note on any one of the successful entrepreneurs.	08
8	a	List and explain the characteristics of different entrepreneurial styles.	08
	b	What is design Thinking? Describe the design thinking process.	08

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RV COLLEGE OF ENGINEERING®

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V / VI Semester B. E. Fasttrack Examinations Jul-19

Common to all branches

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	X made minor improvements over his patent expired drug which is used to cure lucemia or cancer disease. Can he/she is eligible to hold fresh patent on the drug with minor improvement to make the drug easy to swallow.	02
	1.2	Can a manufacturer register the trademark 'Comfort' for his footwear?	01
	1.3	Give the reason why generic word cannot be allowed to register as a trademark.	01
	1.4	In India _____ is issued if the price of the invention (patented article) is not affordable to common man.	01
	1.5	How do you protect the computer software which has functional relationship with the computer in India?	01
	1.6	Can employer allow to make application for patent over intellectual effort of his employee?	01
	1.7	Can a design for chocolate easter egg consisting of an inner layer of white chocolate surrounded by an outer layer of dark chocolate be registrable?	01
	1.8	Mention the essential requirement of a design to be registered under Industrial Design Act.	02
	1.9	X invented UT (ultra – technology) scan machine to diagnose heardrum disease. Can he register "UT – Scanning" as a trademark for this machine?	01
	1.10	_____ is the tool for cybercrime.	01
	1.11	The term _____ is used for corporate entrepreneurship.	01
	1.12	_____ is the means by which the entrepreneur either creates new wealth producing resources or endows existing resources with enhanced potential for creating wealth.	01
	1.13	Name any two myths about entrepreneurship.	02
	1.14	_____ are the reliable source for products and services, value added enterprise.	01
	1.15	_____ will focus on environmentally friendly products from entrepreneurs.	01
	1.16	What is customer profile?	01
	1.17	Personal selling is the one method of _____ activity available to entrepreneurs.	01

PART-B

2	a	Discuss the grounds on which conferment or granting of patent can be opposed.	10
	b	Why trade secrets are kept in secrecy? How trade secrets are protected in India?	06
3	a	Can Jasmine fragrance be registered as a trademark for perfume in India?	05
	b	Discuss the characteristics of good trademark.	05
	c	Explain the procedure for protection of unregistered trademark against infringement.	06
		OR	
4	a	Define the term trademark. Explain the functions of trademark.	05
	b	Discuss the distinctiveness of trademark with illustrations.	05
	c	Discuss the characteristics of deceptive similarity. What are the factors that the courts consider while determining deceptive similarity.	06
5	a	Explain the rights which are conferred under copyright Act in India.	05
	b	Discuss the different subject matter of copyright in which copyright existed.	06
	c	Why a design need to be registered under Design Act? Give reasons.	05
		OR	
6	a	Define Cybercrime. Discuss how traditional crime is different from cybercrime.	06
	b	Discuss who is the owner of copyright in different subject matter of copyright.	10
7	a	Distinguish between Entrepreneur and Intrapreneur.	02
	b	Discuss entrepreneurial DNA needed to be possessed by the entrepreneur.	08
	c	Explain how entrepreneurship has influenced economic development of a country.	06
8	a	What is Design Thinking?	04
	b	Discuss why customers profile need to be developed as a basic element for marketing research.	06
	c	Discuss the importance of risk taking in entrepreneurship.	06

Operations Management

V Semester SEE Question Papers

Course codes: IM352IA, 21IM53, 18IM54, 16IM55

- Feb/Mar 2026 (IM352IA)
- Feb/Mar 2025 (IM352IA)
- Feb/Mar 2024 (21IM53)
- Mar/Apr 2023 (18IM54)
- Jan/Feb 2021 (16IM55)
- Jan/Feb 2021 (18IM54)
- Fast Track Oct 2020 (16IM55)
- Nov/Dec 2019 (16IM55)
- Fast Track Jul 2019 (16IM55)
- Nov/Dec 2018 (16IM55)

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Question Paper ID: 320247

Course Code: IM352IA

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

INDUSTRIAL ENGINEERING AND MANAGEMENT

Operations Management

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Differentiate between goods and services.	02	1	2
1.2	What is meant by productivity in operations?	02	1	2
1.3	Differentiate between job shop and flow shop.	02	3	2
1.4	Differentiate between long-term and short-term capacity decisions.	02	3	2
1.5	What is meant by exponential smoothing?	02	3	2
1.6	What is the role of forecasting in capacity planning?	02	3	2
1.7	State the concept of Theory of Constraints (TOC).	02	3	2
1.8	What is meant by load balancing in operations?	02	3	2
1.9	What is sequencing?	02	3	2
1.10	What is bottleneck scheduling?	02	3	2

Part B

Question No	Question	M	CO	BT
2a	The owner of a small manufacturing business has patented a new device for washing dishes and cleaning dirty kitchen sinks. Before trying to commercialize the device and add it to his or her existing product line, the owner wants reasonable assurance of success. Variable costs are estimated at $7 per unit produced and sold. Fixed costs are about $56,000 per year. i. If the selling price is set at $25, how many units must be produced and sold to break even? ii. Forecasted sales for the first year are 10,000 units if the price is reduced to $15. With this pricing strategy, what would be the product's total contribution to profits in the first year?	08	1	3
2b	With suitable example, Distinguish between manufacturing processes and service processes.	08	1	2
3a	Compare process structures in services and manufacturing.	08	3	4
3b	Explain long-term capacity planning. Describe capacity timing strategies and capacity sizing strategies with suitable explanations.	08	3	4
	OR			
4a	Explain timing and sizing strategies in capacity planning with diagrams.	08	3	4
4b	How do the process strategies of eBay and McDonald's differ, and how do their choices relate to customer introduced variability?	08	2	2

<table>
<tr><td>5a</td><td>Explain different forecasting methods used in operations management.</td><td>08</td><td>3</td><td>3</td></tr>
<tr><td>5b</td><td>The monthly demand for units manufactured by the Acme Rocket Company has been as follows:

<table><tr><td>Month</td><td>Units</td><td>Month</td><td>Units</td></tr><tr><td>May</td><td>100</td><td>September</td><td>105</td></tr><tr><td>June</td><td>80</td><td>October</td><td>110</td></tr><tr><td>July</td><td>110</td><td>November</td><td>125</td></tr><tr><td>August</td><td>115</td><td>December</td><td>120</td></tr></table>
i. Use the exponential smoothing method to forecast the number of units for June to January. The initial forecast for May was 105 units; α = 0.2.

ii. Calculate the absolute percentage error for each month from June through December and the MAD and MAPE of forecast error as of the end of December.</td><td>08</td><td>3</td><td>3</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>6a</td><td>A firm uses three forecasting models: Moving Average, Exponential Smoothing, and linear regression. Explain how forecast error is calculated and used to identify the most accurate forecasting model.</td><td>08</td><td>3</td><td>2</td></tr>
<tr><td>6b</td><td>A lathe manufacturer had noted the sales invoices from January to August and noticed the sales during each month as 10, 11, 10, 11, 14, 15, 14 & 15 units respectively. Calculate the forecast for all the periods.

(i) If he has followed 4 month weighted moving average (with weights 0.4, 0.3, 0.2 & 0.1).

(ii) If he has followed SES with α = 0.2 and his estimated sales for January is 11.</td><td>08</td><td>3</td><td>3</td></tr>
<tr><td>7a</td><td>Evaluate resource planning approaches for service providers.</td><td>08</td><td>4</td><td>5</td></tr>
<tr><td>7b</td><td>A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowance built into time standards) is best. It currently has three copy machines. Based on the following table of information, determine how many machines are needed at the copy center.

<table><tr><td>Item</td><td>Client X</td><td>Client Y</td></tr><tr><td>Annual demand forecast (copies)</td><td>2,000</td><td>6,000</td></tr><tr><td>Standard processing time (hour/copy)</td><td>0.5</td><td>0.7</td></tr><tr><td>Average lot size (copies per report)</td><td>20</td><td>30</td></tr><tr><td>Standard setup time (hours)</td><td>0.25</td><td>0.40</td></tr></table></td><td>08</td><td>3</td><td>4</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>8a</td><td>Explain the MRP explosion process with a neat example.</td><td>08</td><td>3</td><td>3</td></tr>
<tr><td>8b</td><td>Explain how the Theory of Constraints is applied to product mix decisions.</td><td>08</td><td>3</td><td>3</td></tr>
<tr><td>9a</td><td>With suitable example, differentiate between forward scheduling and backward scheduling</td><td>08</td><td>4</td><td>2</td></tr>
<tr><td>9b</td><td>Given job processing data, identify the bottleneck machine and schedule jobs to minimize idle time.

<table><tr><td>Machine</td><td>Available Time</td><td>Required Time</td></tr><tr><td>M1</td><td>40</td><td>35</td></tr><tr><td>M2</td><td>40</td><td>45</td></tr><tr><td>M3</td><td>40</td><td>30</td></tr></table></td><td>08</td><td>3</td><td>4</td></tr>
</table>

OR

10a

Consider the following single machine scheduling problem.

Job, j	Processing time, t_j	Due date, d_j
1	12	17
2	10	14
3	10	12
4	9	20
5	14	18
6	17	27

Determine the sequence which will minimize the maximum lateness (L_{max}). Also, determine the L_{max} with respect to the optimal sequence.

08 | 4 | 3

10b

Consider the following problem in single machine scheduling with independent jobs.

Job (j)	1	2	3	4	5	6	7	8
Processing time (tj)	5	12	8	10	3	15	8	6
Wj	2	1	1	2	3	4	2	3

(i) Find the optimal sequence, which will minimize the mean flow time and also obtain the corresponding minimum mean flow time.

(ii) Determine the sequence, which will minimize the weighted mean flow time. Also find the corresponding weighted mean flow time.

08 | 4 | 3

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Regular Examinations Feb/March-2025

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Distinguish between the terms 'competitive priorities' and 'competitive capabilities'.	02	2	1
	1.2	What is meant by 'nested process' in operations management?	02	2	1
	1.3	The product-process matrix brings together three elements. List them.	02	1	2
	1.4	What is meant by the concept 'economies of scale' in operations capacity planning?	02	2	2
	1.5	List any four basic patterns of demand time series.	02	1	3
	1.6	What is 'forecast error' in the context of demand forecasting?	02	2	3
	1.7	What is meant by 'bottleneck' in the context of constraint management?	02	2	3
	1.8	List the inputs required for MRP explosion.	02	1	3
	1.9	What is shortest processing time sequencing in scheduling?	02	2	4
	1.10	Expand the acronym 'CDS heuristic'.	02	1	4

PART-B

2	a	Identify nine competitive priorities used in operations strategy and explain how a consistent pattern of decisions can develop organizational capabilities.	10	2	1
	b	Describe the process view of operations in typical organizations with the help of a block diagram.	06	2	1
3	a	Explain the four inventory strategies that should be coordinated with process choice in a manufacturing environment. Provide example for each of the strategy.	10	3	2
	b	Provide a note on the three elements in a customer-contract matrix in service processes.	06	2	2
		OR			
4	a	Operations managers must examine three dimensions of capacity strategy before making capacity decisions. What are they? Explain each with suitable examples.	10	3	2
	b	Enumerate the four steps of systematic approach to long term capacity decisions.	06	2	2

5	a	Compute a 3 week moving average forecast for the arrival of medical clinical patients in week 4. The number of arrivals for the past 3weeks was as follows. $Week$ | $Patients$ 1 | 400 2 | 380 3 | 411 If the actual number of patient's arrival in week 4 is 415, what is the forecast error for week 4? Compute the forecast for week 5.	10	3	3
	b	Provide a note on the following types of error along with the formula used to compute the error. i) Mean absolute deviation ii) Mean absolute percentage error.	06	2	3
		OR			
6	a	The manager of a pizza shop must forecast the weekly demand for the special pizzas so that he can order pizza shells weekly. Recent demand has been as follows: $Week$ | $Pizzas$ | $Week$ | $Pizzas$ $June$ 2 | 50 | $June$ 23 | 56 $June$ 9 | 65 | $June$ 30 | 55 $June$ 16 | 52 | $June$ 7 | 60 Forecast the demand for pizza for June 23 to July 14 by using the simple moving average method with $n = 3$. Then repeat the forecast by using the weighted moving average method with $n = 3$ and weights of $0.50; 0.30$ and 0.20 with 0.50 applying to the most recent demand.	10	4	3
	b	"Big data characterized by 3V's holds a lot of promise for predicting the demand for products and services". Elaborate on 3V's and big data role in forecasting.	06	3	3
7	a	A plant manager received marketing's latest forecast of broadcaster roles for the next year. She wants its production line to be designed to make 2400 spreads per week atleast the next 3 months. The plant will operate 40 hours per week. i) What should be the line's cycle time? ii) What is the smallest number of workstations that she could hope for in designing the line for this cycle time? iii) Suppose she finds a solution that requires only 5 stations, what would be the line's efficiency?	10	4	3
	b	List the 7 key principles of 'theory of constraints'.	06	1	3
		OR			
8	a	Create a bill of materials for a ceiling fan. Provide part number, quantity, unit measure and description for each item.	10	5	3
	b	What is meant by lot sizing rule in the context of MRP explosion? Differentiate between the fixed order quantity rule and periodic order quantity rule.	06	2	3
9	a	Consider the following single machine schedule problem. $Job\ (i)$ | 1 | 2 | 3 | 4 | 5 $Processing\ time\ (t_j)$ | 15 | 4 | 5 | 14 | 8 Find the optimal sequence which will minimize the mean flow time and also obtain the mean flow time.	10	3	4

	b	State the context in which minimizing the make span is important in single machine model of scheduling. Also mention the formula for computing the mean makespan.	06	3	4
		OR			
10	a	Consider the following 2 machines and 6 jobs flow shop scheduling problem. Using Johnson's algorithm, obtain the optimal sequence which will minimize the makespan. Also represent the makespan with a $GANTT$ chart.	16	4	4

$Job\ (i)$	$Machine\ 1$	$Machine\ 2$
1	5	4
2	2	3
3	13	14
4	10	1
5	8	9
6	12	11

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations April/May-2024

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Differentiate between process and operation.	02	2	1
	1.2	List any two supply chain business processes.	02	1	1
	1.3	List any two strategies used to manage production and inventory.	02	1	1
	1.4	Highlight any two dimensions of capacity strategy before making capacity decisions.	02	1	1
	1.5	Mention any two qualitative methods of forecasting.	02	1	2
	1.6	Highlight the importance of forecast errors.	02	2	2
	1.7	List any two challenges faced during ERP implementation.	02	1	2
	1.8	Mention any two inputs used in MRP explosion diagram.	02	1	2
	1.9	List any two priority sequencing rules commonly used in practice.	02	1	1
	1.10	What is flow time in scheduling problem?	02	2	2

PART-B

2	a	Differentiate between competitive priorities and competitive capabilities with suitable examples.	08	2	1
	b	IKEA shows how important operations management is for its own success and the success of any type of organization. In this view identify some of the activities that IKEA's operations managers are involved in through highlighting some eight important aspects.	08	3	2
3	a	List and explain operations process characteristics. Also briefly discuss about implication of four characteristics of operations processes.	08	3	2
	b	How does the degree of customer contact, customization and process characteristics help in structuring processes in a service organization?	08	3	2
		OR			
4	a	Explain the concept of economies of scale when planning capacity for long term implementation.	08	2	2
	b	The best choice for a manufacturing process depends on the volume and degree of customization required of the process. Justify the above statement with the help of a suitable matrix.	08	3	3

5	a	PM Computer Services assembles customized personal computers from generic parts. Formed and operated by part-time UMass Lowell students Paulette Tyler and Maureen Becker, the company has had steady growth since it started. The company assembles computers mostly at night, using part-time students. Paulette and Maureen purchase generic computer parts in volume at a discount from a variety of sources whenever they see a good deal. Thus, they need a good forecast of demand for their computers so that they will know how many parts to purchase and stock. They have compiled demand data for the last 12 months as reported below.			

Period	Month	Demand	Period	Month	Demand
1	Jan	37	7	Jul	43
2	Feb	40	8	Aug	47
3	Mar	41	9	Sep	56
4	Apr	37	10	Oct	52
5	May	45	11	Nov	55
6	Jun	51	12	Dec	54

i. Use exponential smoothing with smoothing parameter $\alpha = 0.3$ to compute the demand forecast for January (Period 13).

ii. Use exponential smoothing with smoothing parameter $\alpha = 0.5$ to compute the demand forecast for January (Period 13). 10 3 3

b With the help of suitable examples explain any two qualitative methods of forecasting. 06 3 3

OR

6 a The Instant Paper Clip Office Supply Company sells and delivers office supplies to companies, schools, and agencies within a 50-mile radius of its warehouse. The office supply business is competitive, and the ability to deliver orders promptly is a big factor in getting new customers and maintaining old ones. (Offices typically order not when they run low on supplies, but when they completely run out. As a result, they need their orders immediately.) The manager of the company wants to be certain that enough drivers and vehicles are available to deliver orders promptly and that they have adequate inventory in stock. Therefore, the manager wants to be able to forecast the demand for deliveries during the next month. From the records of previous orders, management has accumulated the following data for the past 10 months:

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
Orders	120	90	100	75	110	50	75	130	110	90

a) Compute the monthly demand forecast for February through November using the naive method.

b) Compute the monthly demand forecast for April through November using a 3-month moving average.

c) Compute the monthly demand forecast for June through November using a 5-month moving average.

d) Compute the monthly demand forecast for April through November using a 3-month weighted moving average. Use weights of 0.5, 0.33, and 0.17, with the heavier weights on the more recent months. 10 3 3

b Examine the various factors affecting demand forecasting. 06 2 3

7	a	With the help of suitable examples explain seven key principles of the theory of constraints.	08	3	3
	b	What are the key challenges associated with ERP implementation? How can organizations mitigate these challenges?	08	4	4
		OR			
8	a	With the help of suitable MRP explosion diagram highlight the important inputs to MRP.	08	2	3
	b	How do you manage bottlenecks in manufacturing and service processes? Explain with the help of case example.	08	4	2
9	a	There are seven jobs, each of which has to be processed on two machines. A and B in the order AB. Processing times are given in the following table:			

Job	Machine A	Machine B
1	3	8
2	12	10
3	15	10
4	6	6
5	10	12
6	11	1
7	9	3

		Determine a sequence of these jobs that will minimize the total elapsed time. Also find total elapsed time and idle time for machines A and B.	10	4	5
	b	Give Johnson's procedure for determining an optimal sequence for processing 'n' items on two machines. Give justification of the rule used in the procedure.	06	2	4
		OR			
10	a	Consider the following 2 machines and 6 jobs flow shop scheduling problem using Johnson's algorithm. Obtain the optimal sequence which will minimize the make span. Processing times are in minutes.			

Jobs	Machine 1	Machine 2
1	5	4
2	2	3
3	13	14
4	10	1
5	8	1
6	12	11

	b	What is the no passing rule in a sequencing algorithm? Explain the principal assumptions made while dealing with sequencing problems.	06	2	3

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations March /April-2023

Industrial Engineering and Management

OPEARATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Explain the term value-added.	02
	1.2	List the nine competitive priorities.	02
	1.3	What is the objective of a process strategy?	02
	1.4	List the four process strategies.	02
	1.5	How is capacity planning measured.	02
	1.6	What are the five steps of theory of constraints?	02
	1.7	What are the causal forecasting methods?	02
	1.8	What is the purpose of a tracking signal and how it is calculated?	02
	1.9	What is the key difference between ERP and MRP?	02
	1.10	List the inputs for MPS.	02

PART-B

2	a	Explain the term operations management and Identify the three major functional areas of business organizations and briefly describe how they are interrelated?	08
	b	Describe the new trends and challenges in operations management.	08
3	a	Describe the four basic process strategies.	08
	b	List the different tools for process analysis and design and explain them.	08
		OR	
4	a	Define strategic fit. Describe the elements of strategic fit.	08
	b	What is six sigma model of quality improvement? Describe the five steps of six sigma improvement model.	08
5	a	Following data is given for a rocket propulsion: • Expansion for industry of 1 ton capacity curing furnace. • Furnace time= 30mins • System efficiency= 80% • Required output=16 tons per 8 hours • Land efficiency is estimated at 50% of system capacity. i) Determine the number of furnaces required? ii) Estimate the percentage of time the furnace will be idle.	08

	b	The Peach tree Airport in Atlanta serves light aircraft. It has a single runway and one air traffic controller to land planes. It takes on airplane 12 minutes to land and clear the runway. Planers arrive at the airport at the rate of 4 per hour. i) Determine the average number of planes that will stack up waiting to land. ii) Find the average time a plane must wait in line before it can land. iii) Calculate average time it takes a plane to clear the runway.	08
		OR	
6	a	How are bottlenecks removed in manufacturing? Explain.	08
	b	Green Grass's plant manager just received marketing's latest forecasts of Big Broadcaster sales for the next year. She wants its production line to be designed to make 2,400 spreaders per week. The plant will operate 40 hours per week. i) What should be the line's cycle time? ii) What is the smallest number of workstations that she could hope for in designing the line for this cycle time? iii) Suppose that she finds a solution that requires only five stations, what would be the line's efficiency?	08
7	a	Given the data below, what is the simple linear regression model that can be used to predict sales in future weeks?	

$Week$	$sales$
1	150
2	157
3	162
4	166
5	177

			08
	b	What is Delphi method? Describe its main advantage and limitations.	08
8	a	Solve the following problem of 5 jobs through 2 machines A and B in the order $A-B$. the processing time of each jobs on a specific machine in hours is given in the table below.	

Job	$Machines$	
	A	B
1	6	8
2	7	3
3	12	9
4	5	15
5	10	11

		Determine the sequence which minimizes the total elapsed time and also find idle time of each machine.	08
	b	List and explain the inputs and outputs of MRP.	08

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Differentiate between internal and external customer with respect to process view.	02
	1.2	What are the four core processes nested in Operations Management?	02
	1.3	Identify four basic process decisions in Operations Management?	02
	1.4	What are two different but complementary philosophies for process design?	02
	1.5	List the measures of capacity utilization.	02
	1.6	List the steps of a Systematic Approach to Long-Term Capacity Decisions in manufacturing plants.	02
	1.7	Differentiate between constraint and bottleneck.	02
	1.8	What is Tracking Signal in forecasting? Write the formula for Tracking Signal.	02
	1.9	What are Inventory Reduction Tactics or levers?	02
	1.10	What is aggregation? List the three dimensions that companies perform aggregation.	02

PART-B

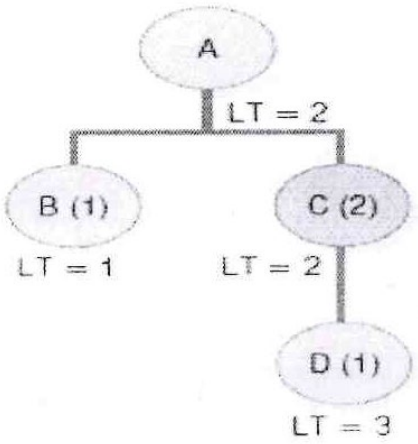
2	a	Developing a customer-driven operations strategy begins with corporate strategy, which coordinates the firm's overall goals with its core processes. It determines the markets the firm will serve and the responses the firm will make to changes in the environment. Identify and discuss the factors Connection between Corporate strategy and key operations management decisions with block diagram.	08
	b	Calculate the productivity for the following operations: a) Three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week. b) A team of workers makes 400 units of a product, which is sold in the market for $10 each. The accounting department reports that for this job the actual costs are $400 for labor, $1,000 for materials, and $300 for overhead.	08

3	a	Identify and discuss with block diagram the important four decisions that are essential for an effective process design.	08
	b	Identify and discuss the strategies used for structuring the process in Manufacturing.	08
		OR	
4	a	The manager should understand how the four major process decisions tie together, so as to spot ways of improving poorly designed processes. The choices should fit the situation and each other. When the fit is more strategic, the process will be more effective. As industrial engineer you are required to identify, examine and discuss the services and manufacturing processes, looking for ways to test for strategic fit.	08
	b	Identify and discuss the key elements of process reengineering.	08
5	a	Operation managers must examine three dimensions of capacity strategy before making capacity Decisions. Identify and discuss those three dimensions of capacity strategy before making capacity decisions.	08
	b	A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowance built into time standards) is best. It currently has three copy machines. Based on the following table of information, determine how many machines are needed at the copy center.	08

Item	Client X	Client Y
Annual demand forecast (copies)	2000	6000
Standard processing time (hour/copy)	0.5	0.7
Average lot size (copies per report)	30	30
Standard setup time (hours)	0.25	0.40

		OR	
6	a	What is Theory of Constraints (TOC)? Identify seven key principles of the Theory of Constraints.	08
	b	Green Grass's plant manager just received marketing's latest forecasts of Big Broadcaster sales for the next year. She wants its production line to be designed to make 2,400 spreaders per week for at least the next 3 months. The plant will operate 40 hours per week. a) What should be the production line's cycle time? b) What is the smallest number of workstations that she could hope for in designing the line for this cycle time? c) Suppose that she finds a solution that requires only five stations, what would be the line's efficiency?	08

7	a	The forecast process itself, typically done on a monthly basis, consists of structured steps. These steps often are facilitated by someone who might be called a demand manager, forecast analyst, or demand / supply planner. However, many other people are typically involved before the plan for the month is authorized. Identify and discuss those structured steps for forecasting process.	08
	b	A plant makes monthly shipments of electric drills to a wholesaler in average lot sizes of 280 drills. The wholesaler's average demand is 70 drills a week, and the lead time from the plant is 3 weeks. The wholesaler must pay for the inventory from the moment the plant makes a shipment. If the wholesaler is willing to increase its purchase quantity to 350 units, the plant will give priority to the wholesaler and guarantee a lead time of only 2 weeks. What is the effect on the wholesaler's cycle and pipeline inventories?	08
8	a	Briefly discuss the different types of Operations Planning and Scheduling across the Organization.	08
	b	The MPS start quantities for product A calls for the assembly department to begin final assembly according to the following schedule: 100 units in week 2; 200 unit in week 4; 120 units in week 6; 180 units in week 7; and 60 units in week 8. Develop a material requirements plan for the net 8 week for items B, C, and D. The BOM for A is shown in figure and data from the inventory records are shown in table.	08



Inventory Recorded Data			
	Items		
Data Category	B	C	D
Lot Sizing Rule	POQ(P=3)	L4L	FOQ=500 units
Lead Time	1 week	2 weeks	3 weeks
Scheduled Receipts	None	200 (week 1)	None
Beginning Inventory (On Hand)	20	0	425

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Jan/Feb–21

Industrial Engineering and Management

OPEARATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Calculate the productivity for the following operations: Three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week.	02
	1.2	A hospital is considering a new procedure to be offered at Rs. 200 per patient. The fixed cost per year would be Rs.1,00,000 with total variable cost of Rs.100 per patient. What is the break-even quantity for this service?	02
	1.3	State three approaches to inventory that should be coordinated with process choice.	02
	1.4	List the five techniques that are effective for documenting and evaluating processes.	02
	1.5	Write the formula to calculate utilization of any equipment.	02
	1.6	Differentiate between backlog and back order.	02
	1.7	Give one example each for Dependent and Independent demand.	02
	1.8	Define Line balancing.	02
	1.9	Define Value Stream Mapping (VSM). Write any 4 symbols used to draw VSM.	02
	1.10	Differentiate between push and pull workflow.	02

PART-B

2	a	Consider amazon.com, whose Website enjoys millions of “hits” each day and puts customers in touch with millions of services and products. What are amazon.com’s competitive priorities and what should its operations strategy focus on?	06

	b	A retailer must decide whether to build a small or large facility at a new location. Demand at the location can be either low or high, with probabilities estimated to be 0.4 and 0.6 respectively. If a small facility is built and demand proves to be high, the manager may choose not to expand (payoff = $223,000) or to expand (payoff = $270,000). If the small facility is built and demand is low, there is no reason to expand and the payoff is $200,000.If a large facility is built and demand proves to be low, the choice is to do nothing ($40,000) or to stimulate maximum demand through local advertising. The response to advertising may be either modest or sizable, with their probabilities estimated to be 0.3 and 0.7 respectively. If it is modest, the payoff is estimated to be only $20,000; the payoff grows to $220,000 if the response is sizable. Finally, if a large facility is built and demand turns out to be high, the payoff is $800,000. Draw a decision tree. Then analyze it to determine the expected payoff for each decision and event node. Which alternative- building a small facility or building a large facility-has the higher expected payoff?	10
3	a	How do the process strategies of eBay and McDonald's differ, and how do their choices relate to customer-introduced variability?	06
	b	A work sampling study was conducted to determine the proportion of the time a worker is idle. The following information was gathered on a random basis.	

Day	Number of times worker idle	Total number of observations
Monday	17	44
Tuesday	18	56
Wednesday	14	48
Thursday	16	60

i) Based on these preliminary results, what percent of the time is the worker working?

ii) If the idle time is judged to be excessive, what additional categories might you add to a follow-up work sampling study to identify the root causes? 10

OR

4 a Performance criteria important in creating a layout can go well beyond communication and materials handling. Identify the types of layout performance criteria that might be most important in the following settings.

i) Airport
ii) Bank
iii) Classroom
iv) Product designers' office 08

b Vera Johnson and Merris Williams manufacture vanishing cream. Their packaging process has four steps: (1) mix, (2) fill, (3) cap, and (4) label. They have had the reported process failures analyzed, which shows the following:

Process Failure		Frequency
Lumps of unmixed product		7
Over or under filled jars		18
Jar lids did not seal		6
Labels rumpled or missing		29
	Total	60

08

5	a	Green Grass's plant manager just received marketing's latest forecasts of Big Broadcaster sales for the next year. She wants its production line to be designed to make 2,400 spreaders per week for atleast the next three months. The plant will operate 40 hours per week. i) What should be the line's cycle time? ii) What is the smallest number of workstations that she could hope for in designing the line for this cycle time?	08
	b	A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowance built into time standards) is best. It currently has three copy machines. Base on the following table of information; determine how many machines are needed at the copy center.	

Items	Client X	Client Y
Annual demand forecast (copies)	2000	6000
Standard processing time (hr/copy)	0.5	0.7
Average lot size (copies/report)	20	30
Standard setup time (hours)	0.25	0.4

08

OR

6	a	Take a process that you encounter on a daily basis, such as the lunch cafeteria or the journey from your home to school/ work, and identify the bottlenecks that limit the throughput of this process.	06
	b	The Dahlia Medical Center has 30 labour rooms, 15 combination labour and delivery rooms, 3 delivery rooms, and 1 special delivery room reserved for complicated births. All these facilities operate around the clock. Time spent in labour room varies from hours to days, with an average of about a day. The average uncomplicated delivery requires about 1 hour in a delivery room. During an exceptionally busy 3-day period, 109 healthy babies were born at Dahlia Medical center. Sixty babies were born in separate labour and delivery rooms, 45 were born in combined labour and delivery rooms, and only 4 babies required a labour room and the complicated-delivery room. Which of the facilities (labour rooms, combination labour and delivery rooms, or delivery rooms) had the greatest utilization rate?	10
7	a	The monthly demand for units manufactured by the Acme Rocket Company has been as follows:	

Month	May	June	July	Aug	Sep	Oct	Nov	Dec
Units	100	80	110	115	105	110	125	120

		Use the exponential smoothing method to forecast the number of units for June to January. The initial forecast for May was 105 units; a = 0.2.	10
	b	Consider the master flight schedule of a major airline, such as Air New Zealand. Discuss the ways in which it is analogous to a master production schedule for a manufacturer.	06

8	a	Explain the eight types of wastes.	06
	b	Consider the following 3 machines and 5 jobs flow shop problem.	

Job, i	Processing Time		
	Machine 1	Machine 2	Machine3
1	10	10	12
2	12	8	20
3	16	6	14
4	12	4	20
5	20	8	8

i) Check whether the Johnson's algorithm can be extended to this problem.

ii) If yes, find the optimal sequence and the corresponding makespan. 10

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Fast track Examinations Oct-2020

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Define an operations strategy	02
	1.2	Diagrammatically represent the integration between the three main functional areas of business, namely operations, finance and marketing.	02
	1.3	What are the four major process related decisions pertaining to operations management?(List them)	02
	1.4	List any four aspects of continuous flow processes.	02
	1.5	Define the terms capacity and utilization.	02
	1.6	List any four of the seven key principles of the theory of constraints.	02
	1.7	Give the precise definition of ordering cost and setup cost.	02
	1.8	Identify and list the five basic patterns of most demand time series.	02
	1.9	Clearly define the sales and operations plan(S & OP)	02
	1.10	What is the meaning of MRP explosion?	02

PART-B

2	a	Define operations management. Explain with the help of a block diagram the process view of operations management and write how processes work.	06
	b	A manager is trying to decide whether to build a small, medium or large facility. Demand can be low, average or high with estimated probabilities being $0.25, 0.40$ and 0.35 respectively. A small facility is expected to earn an after tax net present value of just \$18,000 if demand is low. If demand is average, the small facility is expected to earn \$ 75,000; It can be increased to medium size to earn a net present value of \$60,000.If demand is high, the small facility is expected to earn \$75,000 and can be expanded to medium size to earn \$60,000 or to large size to earn \$ 125,000 and medium sized facility is expected to lose an estimated \$25,000 if demand is low and earn \$140,000 if demand is average. If the demand is high the medium sized facility is expected to earn a net present value of \$150,000; It can be expanded to a large size for a net pay off of \$145,000.	

		If a large facility is built and demand is high earnings are expected to be \$220,000.If demand is average for the large facility the present value is expected to be \$125,000; if demand is low, the facility is expected to lose \$60,000. i) Draw a decision tree for this problem? ii) What should the management do to achieve the highest expected pay off?	10
3	a	Explain why processes exist everywhere in all organizations. Take the following supply chain activities and give a brief description for each. i) Warehousing ii) Customer service iii) Logistics.	08
	b	Briefly discuss on the process strategy decisions. Use a block diagrammatic representation to bring out the connections among major process strategy decisions for effective processes.	08
		OR	
4	a	Define the following terms clearly: i) Customer increment ii) Resource flexibility iii) Capital intensity iv) Economics of scope.	08
	b	Position a process on the customer-contact matrix or product process matrix. Bring out the characteristics of the following types of process choices: i) Job processes ii) Batch processes iii) Line processes iv) Continuous flow processes.	08
5	a	Outline the systematic approach to long term capacity decisions using the four steps to arrive at the capacity related decisions	08
	b	A copy center in an office building prepares band reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to make, collate and bind each copy depends on among other factors the number of pages. The center operates 250 days per year, with one 8 hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowances built in to the time standards) is best. It correctly has 3(three) copy machines. Based on the following table of information, determine how many machines are needed at the copy center.	

Item	*Client X*	*Client Y*
Annual demand forecast(Copies)	2,000	6,000
Standard processing time(Hours/copy)	0.5	0.7
Average lot size (copies per report)	20	30
Standard setup time(hours)	0.25	0.40

08

OR

6 a Johnson logs wants to setup a line to serve 60 customers per hour. The work elements and their precedence relationship are shown in the following table 6(a)

i) What is the minimum number of stations?

ii) How many stations are required using the longest work element decision rule?

iii) Suppose that a solution requiring five stations is obtained what is the efficiency?

Table 6(a)-Johnson logs assembly line data

Work element	*Time(Sec)*	*Immediate predecessor*
A	40	*None*
B	30	A
C	50	A
D	40	B
E	6	B
F	25	C
G	15	C
H	20	D, E
I	18	F, G
J	30	H, I
Total	274	

08

b Clearly outline the central ideas enshrined in the theory of constraints. List the key principles of the theory of constraints. List any two benefits of applying TOC to practical scenarios. 08

7 a The monthly demand for units manufactured by the $ACME$ company has been as follows:

Month	*Units*	*Month*	*Units*
May	100	*September*	105
June	80	*October*	110
July	110	*November*	125
August	115	*December*	120

i) Use the exponential smoothing method to forecast the number of units for June to January. The initial forecast for May was 105 units $\alpha = 0.2$

ii) Calculate the absolute percentage error for each month from, June through December and also the MAD and $MAPE$ of forecast error as of the end of December. 08

b Nelson's hardware store stocks a 19.2 volt cordless drill that is a popular seller. Annual demand is 5000 units. The ordering cost is \$15 and the inventory holding cost is \$4/unit/year.

i) What is the economic order quantity?

ii) What is the total annual cost for this inventory item? 08

8	a	Clearly bring out the distinction between the two basic planning strategies of chase strategy and level strategy.	04
	b	A company manufactures wooden decors. Management schedules overtime every week end to reduce the back logs on most popular models. The automatic routing machine is used to cut certain types of edges on the desktops. The following orders need to be scheduled for the routing machine.	

$Order$	$Time\ since\ order\ arrived(hours\ ago)$	$Estimated\ machine\ time(hours)$	$Due\ date\ (hours\ from\ now)$
1	1	9	20
2	0	7	21
3	6	8	12
4	5	3	8
5	3	12	18

The due date reflect the needs for the order to be at its next operation

i) Develop separate schedules by using $FCFS$ and EDD rules. Compare the schedules on the basis of average flow time and average past due hours.

ii) Comment on the performance of these two rules relative to these measures. 08

	c	Briefly explain the concepts related to MRP explosion with an example	04

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Differentiate between internal and external customer with respect to process view.	02
	1.2	What are the four core processes nested in Operations Management?	02
	1.3	Identify four basic process decisions in Operations Management?	02
	1.4	What are two different but complementary philosophies for process design?	02
	1.5	List the measures of capacity utilization.	02
	1.6	List the steps of Systematic approach to Long-Term capacity Decisions related to manufacturing plants.	02
	1.7	Differentiate between constraint and bottleneck.	02
	1.8	What is Tracking signal in forecasting? Write the formula for tracking signal.	02
	1.9	What are Inventory reduction tactics or levers?	02
	1.10	What is aggregation? List the three dimensions that companies perform aggregation.	02

PART-B

2	a	Developing a customer-driven operations strategy begins with corporate strategy, which coordinates the firm's overall goals with its core processes. It determines the markets the firm will serve and the responses the firm will make to changes in the environment. Identify and discuss the factors connection between corporate strategy and key operations management decisions with block diagram.	08
	b	Calculate the productivity for the following operations: i) Three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week. ii) A team of workers makes 400 units of a product, which is sold in the market for \$10 each. The accounting department reports that for this job the actual costs are \$400 for labour, \$1000 for materials and \$300 for overhead.	08
3	a	Identify and discuss with block diagram the important four decisions that are best understood as an effective process design.	08
	b	Identify and discuss the strategies used for structuring the process in manufacturing.	08

OR

4 a The manager should understand how the four major process decisions tie together, so as to spot ways of improving poorly designed processes. The choices should fit the situation and each other. When the fit is more strategic, the process will be more effective.
As an industrial engineer you are required to identify, examine and discuss the services and manufacturing processes, looking for ways to test for strategic fit. 08

b Identify and discuss the key elements of process reengineering. 08

5 a Operations managers must examine three dimensions of capacity strategy before making capacity decisions.
Identify and discuss those three dimensions of capacity strategy before making capacity decisions. 08

b A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15%(beyond the allowance built into time standards) is best. It currently has three copy machines. Based on the following table of information, determine how many machines are needed at the copy center.

$Item$	$Client\ X$	$Client\ Y$
$Annual\ demand\ forecast(copies)$	2,000	6,000
$Standard\ processing\ time(hour/copy)$	0.5	0.7
$Average\ lot\ size(copies\ per\ report)$	20	30
$Standard\ setup\ time(hours)$	0.25	0.40

08

OR

6 a What is Theory of Constraints (TOC)? Identify seven key principles of the Theory of Constraints 08

b Green Grass's plant manager just received marketing's latest forecasts of Big Broadcaster sales for the next year. She wants its production line to be designed to make 2,400 spreaders per week for at least the next 3 months. The plant will operate 40 hours per week.

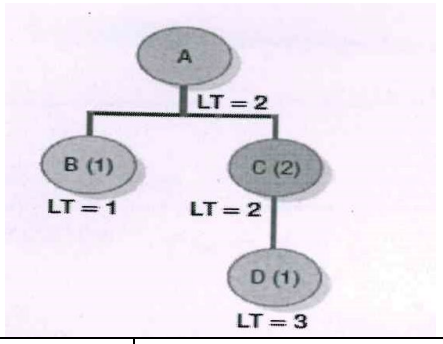
i) What should be the line's cycle time?

ii) What is the smallest number of workstations that she could hope for in designing the line for this cycle time?

iii) Suppose that she finds a solution that requires only five stations. What would be the line's efficiency? 08

7 a The forecast process itself, typically done on a monthly basis, consists of structured steps. These steps often are facilitated by someone who might be called a demand manger, forecast analyst, or demand / supply planner. However, many other people are typically involved before the plan for the month is authorized. Identify and discuss those structured steps for forecasting process. 08

	b	A plant makes monthly shipments of electric drills to a wholesaler in average lot sizes of 280 drills. The wholesaler's average demand is 70 drills a week, and the lead time from the plant is 3 weeks. The wholesaler must pay for the inventory from the moment the plant makes a shipment. If the wholesaler is willing to increase its purchase quantity to 350 units, the plant will give priority to the wholesaler and guarantee a lead time of only 2 weeks. What is the effect on the wholesaler's cycle and pipeline inventories?	08
8	a	Briefly discuss the different types of Operation Planning and scheduling across the Organization.	08
	b	The MPS start quantities for product A calls for the assembly department to begin final assembly according to the following schedule:100 units in week 2; 200 units in week 4; 120 units in week 6; 180 units in week 7; and 60 units in week 8. Develop a material requirements plan for the next 8 weeks for items B, C and D. The BOM for A is shown in Figure and data from the inventory records are shown in Table.	



	Item		
Data category	B	C	D
$Lot\ sizing\ rule$	$POQ(P=3)$	$L4L$	$FoQ = 500 units$
$Lead\ time(LT)$	$1\ week$	$2\ weeks$	$3\ weeks$
$Scheduled\ receipts$	$None$	$200(week\ 1)$	$None$
$Beginning\ (on\ hand)\ inventory$	20	0	425

08

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RV COLLEGE OF ENGINEERING®

(Autonomous Institution affiliated to VTU)

V Semester B. E.Fast Track ExaminationsJuly-19

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Diagrammatically represent the interactions between different functional areas of business-mainly marketing, operations and finance.	02
	1.2	List out any four of the key differences between manufacturing and service processes.	02
	1.3	List out the four major process strategy decisions.	02
	1.4	Bring out any four distinctions between job process and batch process.	02
	1.5	List out any two of the long term capacity and short term constraint management capacity decisions as part of overall capacity management.	02
	1.6	Articulate unambiguously and cleanly any four of the seven key principles of the theory of constraints.	02
	1.7	Write down the formula for safety stock with the assumptions that demand distribution and lead time distribution are measured in the same units. Demand and lead time are independent. Explain the notations used in the formula.	02
	1.8	List out any four of the judgment methods for demand forecasting with a one line description for each of these.	02
	1.9	Define the following terms clearly. i. Sales and operations plan ii. Resource plan	02
	1.10	Clearly define the periodic order quantity and lot for lot lot-sizing role to bring out its working mechanics.	02

PART-B

2	a	How does the process view and the supply chain view of operations management, help us in understanding the need and importance of operations management as a strategic function in the organization? Elaborate on these two perspectives of operations management.	08

	b	The owner of a small manufacturing business has patented a new device for washing dishes and cleaning dirty kitchen sinks. Before trying to commercialize the device and add it to his or her existing product line, the owner wants reasonable assurance of success. Variable lots are estimated at $7 per unit produced and sold. Fixed lots are about$500,000 *per year*. i) Forecasted sales for the first year are 10,000 units. If the price is reduced to$15. With this pricing strategy, what would be the products total contribution to profits in the first year? ii) If the selling price is set at $25, how many units must be produced and sold to break even? Use both algebraic and graphical approaches.	08

3	a	Construct the product-process matrix and show in this matrix process characteristic categories, level of product customization and the elements of volume. Elaborate on the type of manufacturing process structure that exist in various types of organization briefly.	08
	b	"After a process is documented, metrics data collected and disconnects identified, the process analyst or design team puts together a set of changes that will make the process better". Compare this with doctor diagnosing the patient and recommending treatments based on the diagnosis. Explain how questioning and brainstorming will assist the process designer in analyzing the gaps in the process. Take an example to support your answer.	08
		OR	
4	a	Develop the customer-contact matrix and indicate the elements of degree of customer contact, customization and process characteristic. Elaborate in the process divergence and flow based on the customer-contact matrix. How does this matrix provide the starting point for evaluating and improving a process?	08
	b	Capital intensity is the mix of equipment and human skills in the process. Carefully examine the advantages and disadvantages of automating manufacturing process. Distinguish between fixed automation and flexible automation. Give two examples for the fixed and flexible automation respectively. Justify automation in the context of capital intensity.	08

5	a	Outline the procedure for carrying out the systematic approach to long term capacity decisions. As part of the approach, discuss some of the qualitative and quantitative methods for evaluating the alternatives for capacity decisions.	08
	b	You have been asked to put together a capacity plan for a critical operation at the Surefoot Sandal company. Your capacity measure is number of machines. Three products (men's, women's and children's sandal) are manufactured. The time standards (processing and setup), lot sizes and demand forecast are given in the Table 5b below. The firm operates 8-hour shifts, 5 days per week, 50 weeks per year. Experience shows capacity cushion of 5% is sufficient.	

Table 5b: Data pertaining to products

Product	Time standards		Lot size (pairs/lot)	Demand forecast (pairs/year)
	Processing (hours/pair)	Setup (hours/pair)		
Men's sandal	0.05	0.5	240	80,000
Women's sandal	0.10	2.2	180	60,000
Children's sandal	0.02	3.8	360	120,000

i) If the operation currently has two machines, what is the capacity gap?

ii) How many machines are needed? 08

OR

6 a Elaborate on the concept known as economies of scale. Explain how the concept of economies of scale can drive cost down, when output increases. Highlight the four reasons on the cost impact of economies of scale. Discuss briefly diseconomies of scale. 08

b In order to meet holiday demand, pizza house requires a production line that is capable of producing 50 pizzas per week, while operating only 40 hours per week. There are only 4 steps required to produce a single pizza with respective processing times of 5min, 5min, 45min and 15min.

i) What share is the line's cycle time?

ii) What is the smartest number of workstations pizza house could hope for in designing the line considering this cycle time?

iii) Suppose that pizza house finds a solution that requires only four stations. What would be the efficiency of this line? 08

7 a The office supply shop estimates that the average demand for a popular ball point pen is 12,000 pens per week with a standard deviation of 3000 pens. The current inventory policy calls for a replenishment orders of 156000 pens. The average lead time from the distributor is 5weeks, with a standard deviation of 2 weeks. If management wants a 95percent cycle service level, what should the reorder point be? 05

b Nelson's hardware stocks a 19.2volt cordlessdrill that is a popular seller. Annual demand is 5000units, the ordering cost is $15 and the inventory holding cost is $4/unit/year.

i) What is the economic order quantity

ii) What is the total annual cost for this inventory item? 05

c Arrival of patients into a clinic has been noted down. The number of arrivals for the past 3 weeks is as follows.

Week	Patient arrivals
1	400
2	380
3	411

		It is now the end of week 3, so the arrival number of patients is known to be 411 patients. Using $alpha = 0.10$ ($\alpha = 0.10$) i) Calculate the exponential smoothing forecast for week 4 ii) What was the forecast error for week 4 if the actual demand turns out to be 415? iii) What is the forecast for week 5?	06

8	a	Represent in the form of block diagram, the managerial inputs that go into developing sales and operations plan, from the functional areas of operations, distribution and marketing, materials, accounting and finance, engineering and human resource. Discuss the supply options for operations planning and scheduling.	08
	b	The Mowry machine shop still has 5 jobs to be processed as of 8AM today (day 23) at its bottlenecks operations. The day when the order arrived, processing time and promised due dates are given in the following table- Table 8b. The jobs are listed in the order of arrivals.	

Table 8b

Job	Time since order arrived (days ago)	Processing time (days)	Due date (days from now)
A	13	8	36
B	15	4	42
C	17	4	39
D	22	3	53
E	24	10	45

i) Develop separate schedules by using FCFS and EDD rules. Compare the schedules on the basis of average flow time and average days past due.

ii) Which rule gives the best schedule, in your judgment? Why? 08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-18

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Distinguish between order winner and order qualifier.	02
	1.2	Write the formula to calculate break even quantity with proper notations.	02
	1.3	List the three major techniques used for effectively documenting and evaluating processes.	02
	1.4	List the five step procedure for six sigma process improvement.	02
	1.5	Draw the graph to show the economics and diseconomies of scale by taking hospital as an example.	02
	1.6	Explain Theory of Constraints.	02
	1.7	Give one example for Backorders and stock outs.	02
	1.8	Distinguish between dependent and independent item.	02
	1.9	Name the priority sequencing rules that are commonly used in work station. Any 4.	02
	1.10	Define Bill of Material and develop the BOM for chair.	02

PART-B

2	a	Adele Weiss manages the campus flower shop. Flowers must be ordered three days in advance from her supplier. Although valentine's day is fast approaching, sales are almost entirely last-minute, impulse purchases. Advance sales are so small that Weiss has no way to estimate the probability of low (25 dozen), medium (60 dozen) or high (130 dozens) demand for red roses on the big day. She buys roses for $Rs15$/dozen and sells them for $Rs\,40$/dozen. Construct a pay of table. Which decision is indicated by each of the following decision criteria? i) Maximin ii) Maximax iii) Laplace iv) Minimax regret.	10
	b	Describe the role of operations in an organization with the help of a block diagram.	06

3 a Identify the commonly used approaches for effectively redesigning and managing processes. 08

b Compare and contrast the two commonly used strategies for change. 08

OR

4 a Examine the process structure in manufacturing and how to position a manufacturing process on the product process matrix. 08

b Discuss how process decisions should strategically fit together. 08

5 a Identify a systematic four step approach for determining long-term capacity requirements. 08

b Green Grass's plant manager just received marketing's latest forecasts of Big broadcaster sales for the next year. She wants its production line to be designed to make 2,400 spreaders/seek for at least the next three months. The plant will operate 40 hrs/week.

i) What should bet the line's cycle time

ii) What is the smallest number of workstations that she could hope for in designing the line for this cycle time

iii) Suppose that she finds a solution that requires only 5 stations what would be the line's efficiency? 08

OR

6 a A company is setting up an assembly line to produce 192units/8-hr shift. The following table identifies the work elements, times and immediate predecessors

$Work\ Element$	A	B	C	D	E	F	G	H	I	J
$Time(Sec)$	40	80	30	25	20	15	120	145	130	115
$Immediate\ Predecessor$	$-$	A	D,E,F	B	B	B	A	G	H	C,I

i) What is the desired cycle time?

ii) What is the theoretical minimum number of stations?

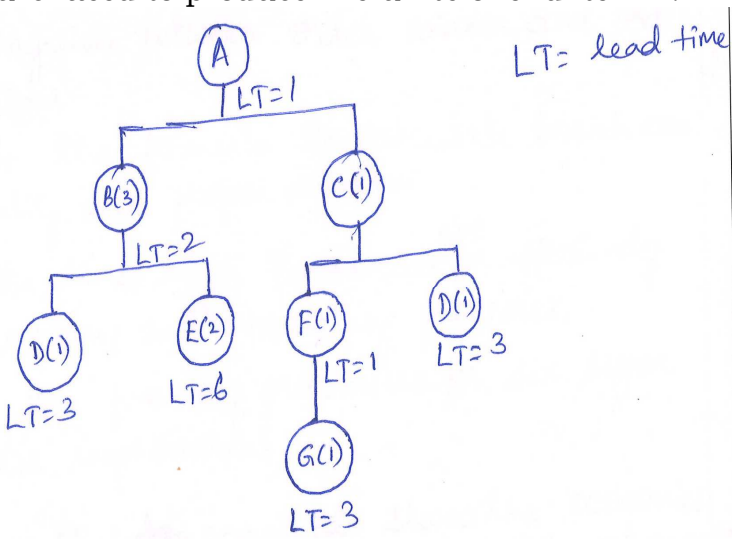
iii) What are the efficiency and balance delay of the solution found? 08

b Briefly explain the four principle reasons which tells why economies of scale can drive costs down when output increases. 08

7 a The following table shows the actual sales of upholstered chairs for a furniture manufacturer and the forecasts made for each of the last 8 months. Calculate CFE, MSE, σ, $MAD, MAPE$ for this product

$Month(t)$	1	2	3	4	5	6	7	8
$Demand(D_t)$	200	240	300	270	230	260	210	275
$Forecast(F_t)$	225	220	285	290	250	240	250	240

08

	b	A museum of natural history opened a gift shop 2 years ago. Managing inventories has become a problem. Low inventory turnover is squeezing profit margins and causing cash-flow problems. One of the top selling $SKU's$ in the container group at the museum's gift shop is a bird feeder. Sales are 18 units/wk and the supplier charges $Rs\,60$/unit. The cost of placing an order with the supplier is $Rs\,45$. Annual holding cost is 25% of a feeder's value, and the museum operates 52 weeks/yr. management chose a 390 unit cot size so that new orders could be placed less frequently. What is the annual cycle inventory cost of the current policy of using a 390 unit lot size? Would a lot size of 468 be better?	08
8	a	Compare the chase planning strategy to the level planning strategy for developing sales and operations.	08
	b	BOM for product A is given below in fig 8b. If there is no existing inventory and no scheduled receipts, how many units of items G, E and D must be purchased to produce five units of end item A?  Fig 8b	08

Finance Accounting and Costing

V Semester SEE Question Papers

Course codes: IM254TA, 12IM5A4

- Feb/Mar 2026 (IM254TA)
- Feb/Mar 2025 (IM254TA)
- Nov/Dec 2017 (12IM5A4)
- Fast Track Jul 2017 (12IM5A4)
- Nov/Dec 2016 (12IM5A4)
- Fast Track Jul 2016 (12IM5A4)

Question Paper ID: 217618

Course Code: IM254TA

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

INDUSTRIAL ENGINEERING AND MANAGEMENT

Finance Accounting and Costing

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	List two qualitative characteristics of accounting information.	02	1	1
1.2	Comment on the Matching Concept with the help of an example.	02	1	1
1.3	Why does the single-entry system make it difficult to detect fraud?	02	1	2
1.4	Record the transaction: *Cash received from Rohan ₹5,000* in a Journal entry.	02	2	3
1.5	How does depreciation affect profit and asset value?	02	1	2
1.6	Name any two expenses recorded in the Profit & Loss Account.	02	1	2
1.7	Define cost.	01	1	1
1.8	Classify costs based on traceability.	02	1	2
1.9	What is a cost sheet?	01	4	1
1.10	Mention any two causes of material price variance.	02	4	1
1.11	List the components of standard cost.	02	4	1

Part B

Question No	Question	M	CO	BT
2a	Critically evaluate the role of accounting in promoting corporate transparency and ethical governance with an example.	10	4	5
2b	Analyze how the Going Concern Concept influences depreciation and asset valuation.	06	2	3

<table>
<tr><td>3a</td><td>From the following ledger balances, prepare a Trial Balance.

Ledger Balances: Cash in Hand ₹6,000, Cash at Bank ₹22,000, Purchases ₹45,000, Sales ₹78,000, Furniture ₹15,000, Rent ₹5,000, Capital ₹60,000</td><td>06</td><td>2</td><td>3</td></tr>
<tr><td>3b</td><td>Explain the steps to rectify the imbalance in trial balance.</td><td>10</td><td>4</td><td>3</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>4a</td><td>For the following balances, prepare a trial balance.

<table>
<tr><th>Particulars (Dr)</th><th>Amount (₹)</th><th>Particulars (Cr)</th><th>Amount (₹)</th></tr>
<tr><td>Sales</td><td>1,07,800</td><td>Capital</td><td>1,00,000</td></tr>
<tr><td>Sales Returns</td><td>2,800</td><td>Discount received</td><td>1,800</td></tr>
<tr><td>Purchases</td><td>62,400</td><td>Plant & Mach.</td><td>35,000</td></tr>
<tr><td>Purchase Return</td><td>2,400</td><td>Bank overdraft</td><td>20,000</td></tr>
<tr><td>Opening Stock</td><td>15,000</td><td>Trade expenses</td><td>1,500</td></tr>
<tr><td>Wages</td><td>10,600</td><td>Furniture</td><td>5,000</td></tr>
<tr><td>Sundry Creditors</td><td>32,500</td><td>Loose Tools</td><td>3,200</td></tr>
<tr><td>Wages Outstanding</td><td>1,400</td><td>Insurance</td><td>500</td></tr>
<tr><td>Carriage outward</td><td>1,400</td><td>Drawings</td><td>3,000</td></tr>
<tr><td>Coal and Gas</td><td>2,500</td><td>Bank charges</td><td>500</td></tr>
<tr><td>Commission</td><td>1,000</td><td>Salaries</td><td>4,500</td></tr>
<tr><td>Factory Building</td><td>1,00,000</td><td>Plant depreciation</td><td>2,000</td></tr>
<tr><td>Sundry Debtors</td><td>15,000</td><td></td><td></td></tr>
</table></td><td>08</td><td>2</td><td>3</td></tr>
<tr><td>4b</td><td>Why is ledger known as the primary book or the principal -book of accounts? Can profit of the business and its financial position be known without maintaining ledger?</td><td>08</td><td>1</td><td>3</td></tr>
<tr><td>5a</td><td>Explain the objectives and limitations of financial statements.</td><td>08</td><td>1</td><td>1</td></tr>
<tr><td>5b</td><td>Explain why Trial Balance is prepared and its limitations.</td><td>08</td><td>1</td><td>1</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>6a</td><td>Explain how depreciation affects:
i) Profit
ii) Asset Value
iii) Capital</td><td>06</td><td>4</td><td>4</td></tr>
<tr><td>6b</td><td>Prepare a Trading Account from the following particulars for the year ended 31st March 2018

Opening Stock ₹50,000

Purchases ₹1,30,000

Sales ₹3,15,000

Factory rent ₹10,000

Purchases return ₹10,000</td><td>10</td><td>3</td><td>3</td></tr>
</table>

Sales return ₹15,000

Wages ₹8,000

Carriage on purchases ₹3,500

Manufacturing Exp. ₹8,900

Custom Duty ₹2,100

Motive power ₹2,000

Dock charges ₹3,000

Coal, Gas, and Water ₹3,000

Factory Manager Salary ₹12,000

Commission on Purchase ₹8,000

Value of Closing stock ₹60,000

7	Product A is a product produced after three distinct processes. The following information is obtained from the accounts of the company for a particular period. Production overheads are incurred Rs. 800 and is recovered at 200% of direct wages. Production during the period was 100 kg. There was no opening or closing stock. Prepare Process Accounts assuming that there is no process loss. Particulars | Total Amount (Rs.) | Process I (Rs.) | Process II (Rs.) | Process III (Rs.) Direct Material | 2,200 | 1,800 | 300 | 100 Direct Labor | 400 | 100 | 200 | 100 Direct Expenses | 500 | 300 | — | 200	16	4	3
	OR						
8a	Analyze how improper classification of cost elements (Material, Labour, Overheads) can affect managerial decision-making.	08	4	4
8b	Prepare a Cost Sheet showing Cost of Sales Particulars | Amount (₹) Direct Material | 40,000 Direct Labour | 25,000 Factory OH | 60% of DL Admin OH | 10% of Factory Cost Selling OH | 5% of Cost of Production	08	4	3

9a	KBC bearing ltd is committed to supply 48,000 bearing per annum to KMR fans on a steady daily basis. It is estimated that it costs Rs 1 as inventory holding cost per bearing per month and that the set up cost per run of bearing manufacture is Rs 3200. i) Determine the optimum run size of bearing manufacture? ii)Calculate what should be the interval between two consecutive optimum runs? iii)Find out the minimum inventory cost.	12	4	4
9b	Explain the concept of Standard Costing and state its objectives.	04	4	2
	OR			
10a	Calculate cost Material Variances and price variances. Standard: 10 kg @ ₹5 Actual: 12 kg @ ₹6	08	4	3
10b	Calculate Labour Variances. Standard: 40 hrs @ ₹20 Actual: 45 hrs @ ₹22	08	4	3

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

V Semester B. E. Regular Examinations Feb/March-2025

Industrial Engineering and Management

FINANCIAL ACCOUNTING AND COSTING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of Statistical table is permitted.

PART-A

			M	BT	CO
1	1.1	Define *GAAP* and mention its importance.	02	2	1
	1.2	Differentiate between assets and liabilities.	02	2	1
	1.3	Distinguish between single entry and double entry methods of book keeping.	02	2	2
	1.4	What is the trial balance, and why is it prepared?	02	2	2
	1.5	What is the purpose of preparing a Profit and Loss Account?	02	2	3
	1.6	Name two simple adjustments made while preparing financial statements.	02	1	3
	1.7	List three elements of costing.	02	1	1
	1.8	Differentiate between financial accounting and cost accounting.	02	2	4
	1.9	What is material cost variance, and why is it important?	02	2	1
	1.10	Standard cost of material for output of 2,600 units is 71,500 rupees and actual output is 2,550 units. If material mix variance is 1,095 rupee adverse, what will be the material usage variance?	02	2	4

PART B

2	a	Explain the objectives of Financial Accounting and its significance in business decision –making.	10	2	1
	b	Briefly describe how budgeting helps in cost control and decision-making. Also, mention two types of budgets commonly used in organizations.	06	3	1
3	a	Explain the rules for journalizing the transaction with appropriate illustration.	04	3	1
	b	Journalise the following transactions in the books of trader . Debit Balance on Jan. 1, 1999: Cash in hand *Rs* 8,000, Cash at Bank *Rs* 25,000, stock of Goods *Rs* 20,000, Furniture *Rs* 2,000, and Building *Rs* 10,000. Sundry Debtors: Vijay *Rs* 2,000, Anil *Rs* 1,000, and Madhu *Rs*, 2,000. Credit Balances on Jan. 1, 1999: Sundry Creditors : Anand Rs 5,000, Loan from Bablu *Rs* 10,000 Following were further transaction in the month of January, 1999: 1. Jan. 1, Purchased goods worth *Rs* 5,000 for cash less 20 % trade discount and 5 % cash discount. 2. Jan. 4, Received *Rs* 1,980 from Viajy and allowed him *Rs* 20 as discount. 3. Jan, 6, Purchased goods from Bharath *Rs* 5,000. 4. Jan. 8, Purchased plant from Mukesh for *Rs* 5,000 and paid *Rs* 100 as cartage for bringing the plant to the factory and another *Rs* 200 as installation charges			

5. Jan. 12, Sold goods to Rahim on credit *Rs* 600.
6. Jan. 15 , Rahim became an insolvent and could pay only 50 paise in a rupee.
7. Jan. 18, Sold goods to Ram for cash *Rs* 1,000.
8. Jan. 20, Paid salary to Ratan *Rs* 2,000.
9. Jan. 21, Paid Anand Rs 4,800 in full settlement.
10. Jan. 26, Interest received from Madhu *Rs* 200.
11. Jan. 28, Paid to Bablu interest on loan *Rs* 500.
12. Jan. 31, Sold Goods for cash *Rs* 500.
13. Jan. 31, Withdrew good from business for personal use *Rs* 200.

12 | 4 | 2

OR

4 a Explain the process of posting from Journal to Ledger. 04 | 2 | 2

b Journalise the following transactions , post them in the Ledger and balance the accounts on 31st January.

- Ram started business with capital of *Rs* 10,000.
- He purchased goods from Mohan on credit *Rs* 2,000.
- He paid cash to Mohan *Rs* 1,000.
- He sold goods to Suresh *Rs* 2,000.
- He received cash from Suresh *Rs* 3,000.
- He further purchased goods from Mohan *Rs* 2,000.
- He paid cash to Mohan *Rs* 1,000.
- He further sold goods to Suresh *Rs* 2,000.
- He received cash from Suresh *Rs* 1,000.

12 | 4 | 2

5 a Prepare the Trading Account of Mr. Ramesh for the year ending 31st December , 1998 from the date as follow:

	Rs		*Rs*
Purchase	10,000	*Wages*	4,000
Purchase Returns	2,000	*Carriage charges*	2,000
Sales	20,000	*Stock on* 1.1.1998	4,000
Sales Returns	5,000	*Stock on* 31.12.1998	6,000

06 | 3 | 3

b Find out the cost of merchandise purchased , cost of merchandise sold, cost of merchandise unsold and Gross Profit from the following transactions:

	Rs.
Purchase (3,000 *articles*)	25,000
Freight	1,000
Local Taxes	1,000
Salaries	2,500
Shop Rent	500
Godown Rent	500
Electrical Charges	600
Municipal Taxes	200
Stationary	250
Furniture (*estimated life* 5 *years*)	12,000
Sales (2,700 *articles*)	32,000

10 | 4 | 3

OR

6 a Distinguish between Profit and Loss Account and Balance sheet 04 | 3 | 3

b From the following balance extracted from the book of M/s Rajendra kumar Gupta & Co., pass the necessary closing entries, prepare a Trading and Profit and Loss Account and a Balance Sheet.

Particulars	*Rs*	*Particulars*	*Rs*
Opening Stock	1,250	*Plant and Machinery*	6,230
Sales	11,800	*Returns Outwards*	1,380
Depreciation	667	*Cash in hand*	895

			Particulars	*Rs*	*Particulars*	*Rs*		
			Commission (Cr.)	211	*Salaries*	750		
			Insurance	380	*Debtors*	1,905		
			Carriage Inwards	300	*Discount(Dr.)*	328		
			Furniture	670	*Bill Receivable*	2,730		
			Printing Charges	481	*Wages*	1,589		
			Carriage Outwards	200	*Returns Inwards*	1,659		
			Capital	9,228	*Bank Overdraft*	4,000		
			Creditors	1,780	*Purchases*	8,679		
			Bill payable	541	*Petty Cash in Hand*	47		
					Bad debts	180	12	4	3
7	a	Explain the elements of costing in detail, and discuss their significance in cost accounting.					06	2	1
	b	From the following information prepare a cost sheet Company Desire a margin of 20 % on the cost of sales.							
			Particulars	*Amount in Rs*					
			Direct material – purchased	80,000					
			Direct material – Opening stock	20,000					
			Direct material – Closing stock	25,000					
			Productive wages	22,000					
			Direct expenses	5,000					
			Consumable stores	4,000					
			Factory manager salary	15,000					
			Unproductive wages	7,000					
			Factory overheads	12,000					
			Work – in – progress						
			Opening Stock	13,000					
			Closing Stock	7,000					
			Office and administrative overheads	28,000					
			Opening stock of finished goods	5,000					
			Closing stock of finished goods	10,000					
			Selling and distribution overheads	33,000			10	4	4
					OR				
8	a	Explain how cost accounting helps in managerial decision-making.					04	2	1
	b	The following particulars have been obtained from the cost records for the year. Assume that all products manufactured during the year have been sold to earn a profit of 20% on selling price.							
				Rs.					
			Material used in manufacturing	1,10,000					
			Material used in Primary packing	20,000					
			Material used in selling the product	3,000					
			Material used in factory	1,500					
			Productive Wages	30,000					
			Factory Supervision expenses	4,000					
			Material used for office	2,500					
			Chargeable Expenses	10,000					
			Indirect expenses – factory	2,000					
			Administrative expenses	3,000					
			Depreciation on Factory Building	3,500					
			Depreciation on Office Building	1,500					
			Freight on materials purchased	5,000					
			Depreciation on Delivery Van	1,000					
			Salary paid to the driver of Delivery Van	3,600					
			Advertisement	2,000					
			Bad Debt	1,500			12	4	4

9	a	Explain the components of standard cost and their role in measuring organizational performance.	08	3	4
	b	JVG Ltd. Produces a product and operates a standard costing system and value material and finished goods inventories at standard cost. The information related with the product is as follows: $Particulars$ | $Cost\ per\ unit(Rs)$ $Direct\ Materials$ $(30kg\ at\ Rs.350\ per\ kg)$ | 10,500 $Direct\ labour$ $(5\ hours\ at\ Rs.80\ per\ hour)$ | 400 The actual information for the month just ended is as follows: i) The budgeted and actual production for the month of September 2019 is 1,000 units ii) Direct materials – 5,000 kg at the beginning of the month. The closing balance of direct materials for the month was 10,000 kg. Purchases during the month were made at $Rs.365$ per kg. The actual utilization of direct materials was 7,200 kg more than the budgeted quantity. iii) Direct labour- 5,300 hours were utilized at a cost of Rs .4,34,600. Calculate : x) Direct material price and usage variances. y) Direct labor rate and efficiency variances.	08	3	4
10	a	Discuss the significance of variance analysis in standard costing. How do material cost variance , labor cost variance , and overhead cost variance contribute to effective cost control?	08	3	4
	b	Gama Ltd. Has furnished the following Standard Cost data per unit of production: • Material $10Kg@\ Rs.10$ per Kg • Labor 6 hours @ $Rs.5.50$ per hour • Variable overhead 6 hours @ $Rs.10$ per hour • Fixed overhead Rs 4,50,000 per month (Based on a normal volume of 30,000 labour hours) The actual cost data for the month: Material used 50,000 kilogram at the cost of Rs 5,25,000. • Labour Paid $Rs.1{,}55{,}000$ for 31,000 hours worked. • Variable Overheads $Rs, 2{,}93{,}000$ • Fixed Overheads $Rs.4{,}70{,}000$ • Actual Production 4,800 units Calculate: i) Material Cost Variance ii) Labour cost variable iii) Fixed overhead cost variance iv) Variable Overhead Cost variance	08	3	4

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-17

Industrial Engineering and Management

FINANCIAL ACCOUNTING AND COSTING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

6. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
7. Answer FIVE full questions from Part B.

PART-A

1	1.1	_____ accounting concept states that transactions and events that can be expressed in money terms are recorded in the books of accounts.	01
	1.2	State the matching concept of accounting.	02
	1.3	$XYZ's$ trial balance as at 31st December $20X2$ contains the following information: 12 % bank loan, $Rs.40{,}000$, interest paid $Rs.3800$. Interest debited to profit and loss account is _____.	02
	1.4	Indicate how assets, liabilities and capital are affected by the following transactions with an accounting equation: Purchase of machinery for cash $Rs.300000$.	02
	1.5	Mr. Vasu's trial balance sheet has the following entries Opening stock $Rs.500$, Purchases $Rs.2500$; Sales $Rs.3600$ and Closing stock $Rs.300$ calculate Mr. Vasu's gross profit.	02
	1.6	Cost of floppy disk used for office computer is part of _____ overhead.	01
	1.7	_____ are cost directly/indirectly indentified with product and include direct materials, direct labour and factory over heads.	01
	1.8	In _____ costing, jobs receive varying degrees of skill and attention from each production department or operation.	01
	1.9	Factory cost = _____ + factory overheads.	01
	1.10	Two methods used for calculation of equivalent production are _____ and _____.	02
	1.11	Equivalent units are computed when some units remain _____ at the end of a particular period.	01
	1.12	Material usage variance is the sum of _____ and _____.	02
	1.13	Difference between the standard cost and actual cost is called as _____.	01
	1.14	_____ budget normally consist of: a) Operating budget b) Financial budget and c) Special decision budget.	01

PART-B

2 a Show the effect of the following business transactions on assets, liabilities and capital through accounting equations:

1	*Commenced business with cash*	2000
2	*Goods purchased on credit*	700
3	*Furniture purchased*	300
4	*Paid to creditors*	200
5	*Amount withdrawn by the proprietor*	400
6	*Paid salaries*	200
7	*Sold goods on cash costing Rs.*500	700
8	*Furniture costing Rs.*1000 *is valued at Rs.*900	

10

b Explain the following concepts:

i) Business entity concept
ii) Going concern concept
iii) Revenue recognition concept. 06

OR

3 a Journalize the following transactions:

2014		*Rs.*
Jan 1	*Started business with cash*	50000
Jan 3	*Paid to bank*	40000
Jan 5	*Sold goods to Mohan*	22000
Jan 9	*Good returned by Mohan*	2000
Jan 11	*Goods purchases from Shyam*	31500
Jan 15	*Goods returned to Shyam*	1500
Jan 18	*Bought furniture for office use by cheque*	9000
Jan 22	*Purchased goods for cash*	1000
Jan 22	*Paid cartage*	50
Jan 30	*Paid interest on loan*	500

10

b "Accounting equation holds good under all circumstances", justify this statement with help of six illustrations. 06

4 a From the following balances extracted from the books of Mythri & Co., prepare a trading and profit & loss account and balance sheet as on 31st December 2012.

Particulars	*Rs*	*Particulars*	*Rs*
Opening stock	11000	*Return outwards*	500
Bills receivable	4500	*Trade expenses*	200
Purchases	39000	*Office expenses*	1000
Wages	2800	*Cash in hand*	500
Insurance	700	*Cash at bank*	4750
Sundary debtors	30000	*Rent and taxes*	1100
Carriage inwards	800	*Carriage outwards*	1450
Commission (Dr.)	800	*Sales*	60000
Interest on capital	700	*Bills payable*	3000
Stationary	450	*Creditors*	19650
Return inwards	1300	*Capital*	17900

The stock on 31st December 2012 was valued at *Rs.*25000. 13

b State the debit and credit rules for personal, real and nominal accounts. 03

OR

5 From the following figures prepare Trading and Profit and Loss account for the year ended 31st March 2014 and a balance sheet as on that date:

Capital	86800	*Bad debts*	700
Drawing	15000	*Bad debts provision*	2100
Investments	14000	*Sundry debtors*	40400
Cash	8000	*Sundry creditors*	25700
Rent and insurance	3000	*Furniture*	8000
Opening stock	36600	*Plant and machinery*	50000
Purchases	186000	*Salaries*	11000
Sales	305000	*Advertisement*	4400
Sales return	5000	*Goodwill*	6000
Wages	22000	*Freight*	6300
Carriage	4200	*Commission(Cr.)*	1000

Adjustments:
Stock on 31st march 2014 was $Rs.31500$.
Salary $Rs.1000$ and wages $Rs.2000$ for march 2014 were unpaid.
Rent outstanding amounted to $Rs.600$ and insurance unexpired amounted to $Rs.400$.
Commission amounting to $Rs.200$ has been received in advance.
Write off $Rs.400$ as bad debts, create provision for doubtful debts at 5 % on sundry debtors and provide 2 % provision for discount on debtors and creditors.
Depreciate furniture and plant and machinery by 10 %. 16

6 a The modern manufacturing company submits the following information on march 31st 2011.

	Rs
Sales for the year	2,75,000
Finished goods inventories(beginning)	7,000
Work in progress inventories(beginning)	4,000
Purchase of raw material	1,10,000
Material at the beginning of the year	3,000
Material at the end of the year	4,000
Direct labour	65,000
Factory overheads	60 % *of labour cost*
Finished goods inventories(ending)	8000
Work in progress inventories(ending)	6,000
Selling expenses	10% *of sales*
Administrative expenses	5% *of sales*

Prepare a cost sheet for the year ending 31st march 2011. 10

b Explain in details the various elements which go to make up the total cost of any product. 06

OR

7 Laxmi chemical Ltd processes a range of products including a detergent, "Mysore", which passes through three processes before completion and transfer to the finished goods warehouse. During September, data relating to this product were as follows:

		Process I	***Process II***	***Process III***	***Total***
	Basic Raw Material	6000	-	-	6000
	Direct Raw Material added in process	8500	9500	5500	23500
	Direct wages	4000	6000	12000	22000
	Direct expenses	1200	930	1340	3470
	Production overhead				16500
	Output (units)	9200	8700	7900	
	Normal loss in process of input (%)	10	5	10	
	Scrap value loss per unit	0.20	0.50	1	

The production overhead is absorbed as a percentage of direct wages. There was no stock at the start or at the end of any process.
You are required to prepare the following accounts:
Process I

a) Process II
b) Process III
c) Abnormal loss
d) Abnormal gain. 16

8 The standard material cost to produce a ton of chemical X is given below:

- 300 kg of material A @ $Rs.10$ $per\ kg$
- 400 kg of material B @ $Rs.5$ $per\ kg$
- 500 kg of material C @ $Rs.6$ $per\ kg$

During a particular period, 100 tons of mixture X was produced from the usage of

- 35 tons of material A @ $Rs.9000$ $per\ ton$
- 42 tons of material B @ $Rs.6000$ $per\ ton$
- 53 tons of material C @ $Rs.9000$ $per\ ton$

Calculate material cost, price and usage and mix variances. 16

OR

9 a Calculate price, usage and mix variance from the following data.

Material	*Quantity (kgs)*	*Price (Rs)*	*Total (Rs)*	*Quantity (kgs)*	*Price (Rs)*	*Total (Rs)*
A	4	1	4	2	3.5	7
B	2	2	4	1	2.0	2
C	2	4	8	3	3.0	9

16

10 The cost of an article at the capacity of 5000 units is given under A below. For variation of 25% in capacity above or below this level, the individual expenses vary as indicated under B below.

		A	$B(Percent)$
	$Material\ cost$	25000	$100(variable)$
	$Labour\ cost$	15000	$100(variable)$
	$Power$	1250	$80(semi-variable)$
	$Repairs$	2000	$75(semi-variable)$
	$Stores$	1000	$100(variable)$
	$Inspection$	500	$20(semi-variable)$
	$Administrative\ overheads$	5000	$25(semi-variable)$
	$Selling\ overheads$	3000	$50(semi-variable)$
	$Depreciation$	10000	$100(fixed)$
	$Total$	62750	
	$Cost\ per\ unit$	12.55	

Prepare the production cost budget at 4000 units and 6000 units. 16

OR

11 The following particulars for two periods are available from the records of a manufacturing company.

	$Period\ 1$	$Period\ 2$
$Output\ (units)$	60000	80000
$Costs$:		
$Direct\ materials$	120000	160000
$Direct\ wages$	300000	400000
$Direct\ expenses$	60000	80000
$Consumable\ materials$	15000	20000
$Shop\ labour$	6000	8000
$Maintenance$	8000	10000
$Inspection$	1600	1800
$Depreciation$	10000	10000
$Insurance$	5000	5000
$Salaries$	6000	6000
$Total\ factory\ cost$	531600	700800

Total production at 100 percent capacity is 100,000 units prepare flexible budget for 70 percent and 90 percent capacity. 16

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

FINANCIAL ACCOUNTING AND COSTING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
4. Answer FIVE full questions from Part B.

PART-A

1	1.1	Journal is book of _____ entry.	01
	1.2	The liabilities of a firm are $Rs.6{,}000$ and the capital of the proprietor is $Rs.4{,}000$. The total assets are _____.	02
	1.3	Goodwill is _____ account.	01
	1.4	$XYZ's$ trial balance as on 31st December $20X2$, contains the following information: 12% bank loan $Rs.40{,}000$, interest paid $Rs.3{,}800$. Interest debited to profit and loss account is _____	02
	1.5	Discount allowed appearing in the trial balance sheet are shown on the _____ side of profit and loss account.	01
	1.6	The process of recording the transaction in the journal is called as _____.	01
	1.7	Depreciation shown on the debit side of profit and loss AC as a _____ item.	01
	1.8	_____ are cost directly/indirectly identified with product and include direct materials, direct labor and factory over heads.	01
	1.9	Two broad methods of costing are _____ and _____.	02
	1.10	A flexible budget recognizes the behavior of _____ and _____.	02
	1.11	Material usage variance is the sum of _____ and _____	02
	1.12	_____ budget are concerned with expected cash receipts/disbursement.	01
	1.13	In marginal costing, fixed costs are charged to _____.	01
	1.14	The budgeted fixed overhead for a budgeted production of 10,000 units is $Rs.20{,}000$. For a certain period the actual production was 11,000 units and actual expenditure came to $Rs.24{,}000$. Then the volume variance is $Rs.$ _____.	02

PART-B

2 a Journalize the following transaction

Date	*Particulars*	*Amount (Rs.)*
20X1 *Jan* 1	*Started business with cash*	1,50,000
1	*Paid into bank*	10,000
3	*Bought machinery*	1,000
4	*Purchased goods from Mythri& Co*	5,000
5	*Sold goods to Ram & Co*	2,700
6	*Paid rent by cheque*	400
8	*Paid for advertisement*	100
12	*Withdrew cash for personal use*	350
25	*Paid salaries to staff*	600
30	*Received cash from Ram & Co*	2,600

12

b Accounting equation holds good under all circumstances. Justify this statement with help of four illustrations. 04

OR

3 a Kiran had the following transaction. Prepare an accounting equation to show their effect on assets, liabilities and capital.

i) Commenced business with $Rs.1{,}00{,}000$.
ii) Purchased furniture for cash $Rs.50{,}000$.
iii) Purchased a building for $Rs.1,50,000$ giving $Rs,50{,}000$ in cash and balance through loan.
iv) Sold goods costing $Rs.20,000$ for $Rs.25{,}000$.
v) Purchased old vehicle (for office use) for cash $Rs.20{,}000$.
vi) Paid cash $Rs.5{,}000$ for loan and $Rs.3{,}000$ for interest.
vii) Withdrew cash for personal use $Rs.5{,}000$.
viii) Furniture costing $Rs.10{,}000$ is valued at $Rs.9000$ 13

b State the debit and credit rules for personal, real and nominal accounts. 03

4 a A book keeper has submitted to you the following trial balance wher in the totals of the debit and credit balance are not equal Redraft the trial balance sheet correctly as at 31st *march* 20X2.

Particulars	*Dr. (Rs.).*	*Cr. (Rs).*
Capital	-	7670
Cash in hand	-	30
Purchases	8990	-
Sales	-	11060
Cash at bank	885	-
Fixtures and fittings	225	-
Freehold premises	1500	-
Water and electricity	65	-
Bills receivable	-	825
Return inward	-	30
Salaries	1075	-
Creditors	-	1890
Debtors	5700	-
Stock at 1 *April* 20X1	3000	-
Printing	225	-
Bills payable	1875	-
Rates, taxes and insurance	190	-
Discount received	445	-
Discounts allowed	-	200
	24,175	21705

10

	b	Differentiate between: i) Journal and ledger ii) Trial balance and balance sheet.	06
		OR	
5		From the following trial balance of M/S. Ram & Sons, prepare trading and profit and loss account for the year ending on 31st *March* 2002 and the balance sheet as on the date: Trial balance as on 31st *March* 2002.	

Particulars	Debit (Rs.)	Credit (Rs.)
Opening Stock (1.04.2001)	5000	
Purchases	16750	
Discount allowed	1300	
Wages	6500	
Sales		30000
Salaries	2000	
Travelling expenses	400	
Commission	425	
Carriage inward	275	
Administration expenses	105	
Trade expenses	600	
Interest	250	
Building	5000	
Furniture	200	
Debtors	4250	
Creditors		2100
Capital		13000
Cash	2045	
Total	45100	45100

		Stock on 31st *March* 2002 was *Rs.* 6000.	16
6	a	The modern manufacturing company submits the following information on *March* 31st 2011.	

Particulars	*Rs*
Sales for the year	275000
Finished goods inventories (beginning)	7000
Work in progress inventories (beginning)	4000
Purchase of raw material	110000
Material at the beginning of the year	3000
Material at the end of the year	4000
Direct labor	65000
Factory overheads	60% *of labor cost*
Finished goods inventories (ending)	8000
Work in progress inventories (ending)	6000
Selling expenses	10% *of sales*
Administrative expenses	5% *of sales*

		Prepare a cost sheet for the year ending 31st *March* 2011.	12
	b	Explain in detail the various elements which go to makeup the total cost of any product.	04
		OR	

7	Product B is obtained after it passes through three distinct processes. The following information is obtained from the accounts for the week ending on 31st $March$ 2006. $Particulars$ | $Total\ amount$ | $Process\ I$ | $Process\ II$ | $Process\ III$ $Direct\ material$ | $Rs.7542$ | $Rs.2600$ | $Rs.1980$ | $Rs.2962$ $Direct\ wages$ | $Rs.9000$ | $Rs.2000$ | $Rs.3000$ | $Rs.4000$ $Production\ overheads$ | $Rs.9000$ | | | 1000 units @ $Rs.3$ each were introduced in Process I. there was no stock of materials or work in progress at the beginning or at the end of the period. The output of each process passes direct to next process and finally to finished store .production overheads are recovered on 100% of direct wages. The following additional data are obtained $Particulars$ | $Output\ during\ the\ week$ | $\%\ normal\ loss\ to\ input$ | $Value\ of\ scarp\ per\ unit$ $Process\ I$ | $950\ units$ | 5% | $Rs.2$ $Process\ II$ | $840\ units$ | 10% | $Rs.4$ $Process\ III$ | $750\ units$ | 15% | $Rs.5$ Prepare process cost accounts and abnormal loss and abnormal gain account.	16
8	The standard material cost to produce a ton of chemical X is given below - $300kg$ of material A @ $Rs.10$ per kg - $400\ kg$ of material B @ $Rs.5$ per kg - $500\ kg$ of material C @ $Rs.6$ per kg During a particular period, 100 tons of mixture X was produced from the usage of - 35 tons of material A @ $Rs.9000$ per ton - 42 tons of material B @ $Rs.6000$ per ton - 53 tons of material C @ $Rs.7000$ per ton Calculate material cost, price and usage and mix variances.	16
	OR	
9 a	Standard hours for manufacturing two products, M and N are 15 hours per unit and 20 hours per unit respectively. Both products require identical kind of labor and the standard wages rate per hour is $Rs.5$. In the year $2006, 10000$ units of M and 15,000 units of N were manufactured. The total labor hours actually worked were 4,50,000 and the actual wages bill came to $Rs.23,00,000$. This include 12,000 hours paid for @ $Rs.7$ per hour and 9,400 hours paid for @ $Rs.7.50$ per hour, the balance having been paid @ $Rs.5$ per hour. Calculate labour variances.	10
b	List the different types of variances computed for: i) Material variances and ii) Labour variances.	06
10	A factory engaged in manufacturing plastic toys is working at 40% capacity and produces 10,000 toys per month. The present cost break up for one toy is as under - Material: $Rs.10$ - Labor: $Rs.3$ - Overheads: $Rs.5$ [60% fixed] The selling price is $Rs.20$ per toy. If it is decided to work the factory at 50% capacity, the selling price falls by 3%. At 90% capacity, the selling price falls by 5% accompanied by a similar fall in the price of material. You are required to prepare a statement showing the profits/losses at $40\%, 50\%$ and 90% capacity utilizations.	16

OR

11 For the production of 10,000 electrical home appliances the following are budgeted expenses:

Direct materials	*Rs.* 70
Direct labour	35
Variable overheads	30
Fixed overheads (*Rs.* 1,50,000)	15
Variable expenses (*Direct*)	5
Selling expenses (10% *fixed*)	15
Administration expenses (*Rs.* 50,000 *rigid for all levels of production*)	5
Distribution expenses (20% *fixed*)	5
Total cost of sale per unit	180

Prepare a budget for production of 6,000 7,000 and 8,000 irons showing distinctly marginal cost and total costs. 16

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

FINANCIAL ACCOUNTING AND COSTING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

6. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
7. Answer FIVE full questions from Part B.

PART-A

1	1.1	One of the Journal entries is, Purchased goods from Mr. X on credit for $Rs.30{,}000$. Analyze the transaction and give accounting equation.	02
	1.2	The liabilities of a firm are $Rs.6{,}000$ and the capital of the proprietor is $Rs.4{,}000$. The total assets are _____.	02
	1.3	Goodwill is treated as _____ account.	01
	1.4	Freight inward appearing in trial balance is shown on the _____ side of the Trading account.	01
	1.5	$X's$ trial balance contains the following information: discount received $Rs.1{,}000$, provision for discount on creditors $Rs.1{,}600$. It is desired to maintain a provision for discount on creditors at $Rs.1{,}100$. The amount to be credited to the profit and loss account is _____.	02
	1.6	Prepaid expenses are _____ from the respective indirect expense on the debit side of profit and loss account.	01
	1.7	Factory cost = _____ + factory overheads.	01
	1.8	Based on relationship to production the two categories of cost are _____ and _____.	02
	1.9	Abnormal loss is written on the _____ side of process account.	01
	1.10	Variance represents the difference between _____ cost and _____ cost.	02
	1.11	Based on the three elements of cost, the cost variances are classified into _____, _____ and _____ variance.	02
	1.12	In the context of process costing, _____ units are computed when some units remain incomplete at the end of a particular period.	01
	1.13	The _____ budget is device to help a firm to plan for and control the use of cash.	01
	1.14	_____ loss may be defined as loss of units which is an inherent part of the production process caused by natural or unavoidable causes.	01

PART-B

2 a List the basic concept of accounting. Explain any two in detail. 08

b Kiran had the following transaction. Prepare an accounting equation to show their effect on assets, liabilities and capital.

i) Commenced business with $Rs.1{,}00{,}000$.
ii) Purchased furniture for cash $Rs.50{,}000$.
iii) Purchased a building for $Rs.1{,}50{,}000$ giving $Rs.50{,}000$ in cash and balance through loan.
iv) Sold goods costing $Rs.20{,}000$ for $Rs.25{,}000$.
v) Purchased old vehicle (for office use) for cash $Rs.20{,}000$.
vi) Paid cash $Rs.5{,}000$ for loan and $Rs.3{,}000$ for interest.
vii) With drew cash for personal use $Rs.5{,}000$.
viii) Furniture costing $Rs,10{,}000$ is valued at $Rs.9{,}000$. 08

OR

3 a Journalize the following transaction:

Date	*Particulars*	*Amount (Rs.)*
2007 *Jan* 1	*Started business with cash*	1,00,000
1	*Paid into bank*	5,000
3	*Bought furniture*	900
4	*Purchased goods from Mohan & Co.*	4,000
5	*Sold goods to Omega & Co.*	1,700
6	*Paid telephone rent.*	400
8	*Paid for advertisement by cheque.*	100
12	*With drew cash for personal use*	350
25	*Paid salaries to staff*	600
30	*Received cash from Omega & Co. and allowed him a discount of Rs.100*	1,600

10

b Explain with help of any example the debit and credit rules for personal, real and nominal accounts. 06

4 The following trial balance extracted from the books of merchant Mr. Ramdas on 31/03/2002.

Particulars	*Dr. (Rs)*	*Cr. (Rs)*
Furniture and fittings	640	---
Motor vehicles	6,250	---
Building	7,500	---
Capital	---	12,500
Bad debts	125	---
Provision for Bad Debts	---	200
Sundry Debtors and creditors	3,800	2,500
Stock as on 1/04/2001	3,460	---
Purchase and sales	5,475	15,450
Bank overdraft	---	2,850
Sales and purchase returns	200	125
Advertising	450	---

Particulars	Dr. (Rs)	Cr. (Rs)
Interest on bank overdraft	118	---
Commission	---	375
Cash	650	---
Taxes and insurance premium	782	---
General expenses	1,250	---
Salaries	3,300	---
	34,000	34,000

Adjustments:

a) Stock on hand on 31/03/2002 $Rs.3,250$.
b) Depreciate building at 5% p.a.; Furniture at 10% p.a.; motor vehicles at 20% p.a.
c) $Rs.85$ is due for interest on bank overdraft.
d) Salaries $Rs.300$ and taxes $Rs.200$ are outstanding.
e) Insurance premium amounting $Rs.100$ prepaid.
f) One-third of the commission received is in respect of work to be done next year.
g) Write off a further sum of $Rs.100$ as bad debts from debtors and create a provision for doubtful debts at 5% on debtors.

Construct and present in the proper format:

i) Balance sheet
j) Profit and loss statement. 16

OR

5 a From the following balances extracted from the books of M/s Mythri & Co., pass the necessary closing entries; prepare a trading and profit and loss account and a balance sheet.

Particulars	Rs.	Particulars	Rs.
Opening Stock	1,250	Plant and machinery	6,230
Sale	11,800	Returns outwards	1,380
Depreciation	667	Cash in hand	942
Commission (Cr)	211	Salaries	750
Insurance	380	Debtors	1,905
Carriage inwards	300	Discount (Dr.)	328
Furniture	670	Bills receivable	2,730
Printing charges	481	Wages	1,589
Carriage outwards	200	Return inwards	1,659
Capital	9,228	Bank overdraft	4,000
Creditors	1,780	Purchases	8,679
Bills payable	541	Bad debts	180

The value of stock on 31st December, 1990 was $Rs.3,700$ 14

b Differentiate between Trial balance and balance sheet. 02

6 *XYZ* chemical Ltd. processes a range of products including a detergent, 'Washo', which passes through three processes before completion and transfer to the finished goods warehouse. During April, data relating to this product were as follows:

	Process I	***Process II***	***Process III***	***Total***
Basic Raw material	6,000	-	-	6,000
Direct Raw Material added in process	8,500	9,500	5,500	23,500
Direct Wages	4,000	6,000	12,000	22,000
Direct expenses	1,200	930	1,340	3,470
Production overhead				16,500
Output (units)	9,200	8,700	7,900	
Normal loss in process of input (%)	10	5	10	
Scrap value loss per unit	0.20	0.50	1	

The production overhead is absorbed as a % of direct wages. There was no stock at the start or at the end of any process.
You are required to prepare the following accounts:

a) Process *I*;
b) Process *II*;
c) Process *III*;
d) Abnormal loss;
e) Abnormal gain. 16

OR

7 a From the following data from the sewing machine manufacturer, prepare a statement showing prime cost, work/ factory cost, production cost, total cost and profit.

	Rs.
Value of stock of material as on $1-04-2003$	26,000
Material Purchased	2,74,000
Wages to labor	1,20,000
Rent, taxes and insurance of factory	16,000
General administrative expenses	3,400
Water, power and telephone bills of factory	9,600
Water, lighting and telephone bills of office	2,500
Insurance and rent of office building	2,000
Direct expenses	5,000
Commission and pay of salesman	10,500
Works manager salary	30,000
Repair and maintenance of plant	1,000
Salary of office staff	60,000
Value of stock of material as on $31-03-2004$	36,000
Sale of products	6,36,000

12

b List and explain the objectives of costing. 04

8		The standard cost of a certain chemical mixture is: 40% material A at $Rs.200$ per ton 60% material B at $Rs.300$ per ton. A standard loss of 10% is expected in production. During a period the following materials are used: 90 tons of material A at $Rs.180$ per ton 110 tons of material B at $Rs.340$ per ton The weight of finished product is 180 tons. Calculate: a) Material price variance (MPV), b) Material usage variance (MUV), c) Material mix variance (MMV), d) Material yield variance (MYV).	16
		OR	
9	a	The standard cost card for a product shows, • Material cost 2 kg at $Rs.2.50$ each = $Rs.5.00$ per unit • Labour 2 hours at $Rs.10$ each = $Rs.20$ per unit The actual which have emerged from business operations are as follows: • Production – 8,000 units. • Material Consumed – 16,500 kg at $Rs.2.40$ each = $Rs.39,600$ • Wages paid 18,000 hours at $Rs.8$ each = $Rs.1,44,000$ Calculate appropriate material and labour variances.	12
	b	Explain the significance of the term "variance" relating to standard costing and list the various material variances.	04
10	a	X Ltd. Produces a standard product, the estimated cost of which is given below: • Raw – material – $Rs.10$ per unit • Direct wages – $Rs.8$ per unit • Direct expenses – $Rs.2$ per unit • Variable overheads – $Rs.3$ per unit Semi- variable overheads at 100% activity level (10000 units) are expected to be $Rs.40000$, and these overheads vary in steps of $Rs.2000$ for each change of output of 1000 units. Fixed overheads are estimated at $Rs.50000$. Selling price per unit is expected to be $Rs.40$. Prepare a flexible budget at 50%, 70% and 90% levels of activity.	12
	b	In what respect does the production budget contribute to managerial: i) planning and ii) coordination.	04
		OR	

11	A manufacturing company is currently working at 50% capacity and produces 10,000 units at a cost of $Rs.180$ per unit as per the following details: $Particulars$ | $Amount\ in\ Rs.$ $Materials$ | 100 $Labor$ | 30 $Factory\ overheads$ | 30 [40% $fixed$] $Administrative\ overheads$ | 20 [50% $fixed$] $Total\ cost\ per\ unit$ | 180 The selling price per unit at present is $Rs.200$. At 60% working, material cost per unit increases by 2% and selling price per unit falls by 2%. At 80% working, material cost per unit increases by 5% and selling price per unit falls by 5%. Prepare a flexible budget to show the profits/ losses at 50%, 60% and 80% capacity utilization.	16

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

FINANCIAL ACCOUNTING AND COSTING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
4. Answer FIVE full questions from Part B.

PART-A

	1	Fill in the blanks using debit or credit,	
1	1.1	Increases in the assets are _____ Decreases are _____.	01
	1.2	Increases in the liabilities are _____ Decreases are _____.	01
	1.3	Increases in owners capital are _____ Decreases are _____.	01
	1.4	Increases in expenses are _____ Decreases are _____.	01
	1.5	Increases in revenues are _____ Decreases are _____.	01
		Classify the following accounts	
	1.6	i) Bank account	
		ii) Repairs to machinery account.	01
	1.7	i) Interest account	
		ii) Tools Account.	01
	1.8	i) Loss of goods by fire account	
		ii) Raman account	01
	1.9	i) Capital account	
		ii) Drawings account	01
	1.10	i) Closing stock account	
		ii) Commission received account	01
	1.11	Mention the main three elements of cost in costing.	01
	1.12	The formula for material mix variance is _____.	01
	1.13	If standard hours is 4000, standard rate is $Rs\,3/hr$, actual hours is 5000 and actual rate is $Rs.2.5/hr$, then the labor rate variance is _____.	01
	1.14	The prime cost includes _____.	01
	1.15	The total of all the current assets is termed as _____.	01
	1.16	The difference between current assets and current liabilities is termed as _____.	01
	1.17	Inventory period + accounts receivable period is _____.	01
	1.18	List any two types of balance sheet ratios.	01
	1.19	Enumerate any two differences between single entry and double entry system of book keeping.	01
	1.20	List any two factors influencing working capital management.	01

PART-B

2 a Explain the different types of accounts and their classification. 06

b Journalize the following transactions in the books of Mr. Ganesh.

Date	***Particulars***	***Amount in Rs.***
2104 *Jan*1	*Mr. Ganesh commenced his business*	50000
2014 *Jan*3	*Brought goods for cash*	25000
2014 *Jan*4	*Sold goods for cash*	20000
2014 *Jan*6	*Deposited to Dena Bank*	25000
2014 *Jan*10	*Returned goods to Krishna*	500
2014 *Jan*15	*Purchased machinery from HMT Ltd*	5000
2014 *Jan*25	*Advance paid to Mr. Govind as loan*	2000
2014 *Jan*28	*Withdrew cash for personal use from business*	1500

10

OR

3 a Discuss briefly about accounting and accounting principles. 08

b Explain briefly about trial balance. 08

4 Prepare the trading account and profit and loss account for the following balances for the year ending 31st *March* 2009 and the balance sheet as on that date.

Particulars	***Debit balances***	***Credit balances***
Capital		10000
Machinery	3500	
Debtors	2700	
Drawings	900	
Purchases	9500	
Creditors		1400
Wages	5000	
Cash in bank	1500	
Stock in trade (Opening Stock)	2000	
Rent	450	
Sales		14500
Sundry expenses	200	
Carriage inwards	150	

Additional information is as follows
Stock at the end (closing stock) – $Rs.600$
Rent outstanding – $Rs.50$
Wages prepaid – $Rs.200$
Depreciation on machinery at the rate of 10%. 16

OR

5 a What is trading account? Explain briefly. 04

b From the following information extracted from the books of M/S Prashanth printers, prepare a trading account, profit and loss account and balance sheet. Closing stock on 31 – 03 – 2002 was Rs. 45,000.

Particulars	Amount	Particulars	Amount
Opening stock	12500	Bills receivables	17000
Depreciation	7000	Wages	16000
Carriage inwards	700	Sales returns	14000
Furniture	8000	Purchases	86000
Carriage outwards	500	Sales	189000
Plant & machinery	200000	Commission received	2000
Cash	8900	Capital	171300
Salaries	7500	Creditors	17500
Debtors	19000	Bills payable	5000
Discount	1500	Return outwards	13800

12

6 a Explain the types of process losses and gains in process costing. 04

b The following details are taken from the books of an oil mill for one month ending 31st *March* 1986. Purchase of 100 tons of oil seeds is at $Rs.1000/ton$

Particulars	Crushing	Refining	Finishing
Wages	1000	700	900
Sundry stores	200	600	100
Electricity	400	350	200
Steam	300	250	200
Factory expenses	500	400	300
Containers		-	2350

i) 60 tons of crude oil was produced.
ii) 51 tons of oil was produced in the refinery process.
iii) 50 tons of refined oil was finished for delivery.
iv) Empty bags of oil seeds were sold for $Rs.100$.
v) 35 tons of oil cakes were sold at $Rs.60$ per ton.
vi) Loss in weight in crushing is 5 tons.
vii) 8.5 tons by-products from refinery process were valued at $Rs.2550$.

Prepare the accounts with respect to each of the processes and calculate cost of production at the end of each process. 12

OR

7 a The following figures are extracted from the books of Fashions Ltd. For the year ended 30th *April* 2013. Prepare a cost sheet showing clearly the various elements of cost and also the profit and loss.

Particulars	Amount
Direct material	2400000
Direct labor	500000
Depreciation on factory building	15000
Sales branch office expenses	40000
Depreciation on office buildings	8000
Depreciation on staff cars	12000
Insurance – staff cars	1500
Insurance – office building	1200
Insurance – factory building	1500
Delivery van and other distribution expenses	10000

			Particulars	***Amount***
			Salaries including Sales Manager (25000) *and Factory Chief Engineering* (25000)	300000
			Finished goods warehouse expenses	20000
			Electricity (*including* 4000 *for AOH*)	40000
			Advertisement	20000
			Sundry factory expenses	340000
			Sales promotion	5000
			Office administration expenses	50000
			Expenses for participation in Industrial Exhibition	10000
			Total sales	**4200000**
				16
8	a	Define variance. Give the classification of variances.		04
	b	From the following information calculate: MMV, MCV, MPV and MUV.		

Type of material	***SQ in kg***	***SP in Rs/kg***	***AQ in kg***	***AP in Rs/kg***
A	4	1	2	3.5
B	2	2	1	2
C	2	4	3	3

12

OR

9	a	List all labor variances or wage variance formulae.		04
	b	From the data relating to the manufacture of a standard product during the month of September 1995. Prepare a report showing the cost and the profit per unit.		

Raw materials used	*Rs.* 40000
Direct wages	*Rs.* 24000
Machine hours worked	9500
Machine hour rate	*Rs.* 4/*hr*
Office overheads	20% *on works cost*
Selling overheads	*Rs.* 1/*unit*
Units produced	20000
Units sold	18000@*Rs.* 10/*unit*

12

10		Prepare cash budget for January-June from the following information: i) The estimated sales and expenses are as follows:		

Particulars	***Sales*** (***Rs***)	***Wages*** & *salaries* (*Rs*)	***Misc. expenses*** (***Rs***)
Nov	200000	30000	27000
Dec	220000	30000	27000
Jan	120000	24000	21000
Feb	100000	24000	30000
March	150000	24000	24000
April	240000	30000	27000
May	200000	27000	27000
June	200000	27000	27000

ii) 20 % of the sales are on cash and balance on credit.
iii) The firm has a gross margin of 25% on sales.
iv) 50% of the credit sales are collected in the month following the sales, 30% in the second month and 20% in the third month.
v) Material for the sale of each month is purchased one month in advance on a credit for two months.
vi) The time-lag in the payment of wages and salaries is one-third of a month and of miscellaneous expenses, one month.
vii) Debentures worth $Rs.40000$ were sold in January.
viii) The firm maintains a minimum cash balance of $Rs.40000$. Funds can be borrowed @ 12% per annum in the multiples of Rs 1000, the interest being payable on monthly basis.
ix) Cash balance at the end of December is $Rs.60000$ — 16

OR

11 a Explain the following types of budget:
i) Sales budget
ii) Production Budget. — 04

b From the following particulars presented by Pradeep and Co Ltd you are required to prepare a Master Budget for the year ended 31st March 1985:

Product	*Budgeted Production*	*Budgeted Selling Price/unit*
X	200000	5
Y	100000	3
Z	300000	4

Cost/unit	*X*	*Y*	*Z*
Raw materials	1	0.6	0.8
Labor Cost	0.8	0.4	0.6
Direct Expenses	0.4	0.2	0.3

12

Quality Assurance

V Semester SEE Question Papers

Course codes: IM353IA, 18IM53, 16IM53, 16IM53 / 16IM64

- Feb/Mar 2026 (IM353IA)
- Feb/Mar 2025 (IM353IA)
- Mar/Apr 2023, Statistical Process Control (18IM53)
- Jan/Feb 2021, Quality and Reliability Engineering (16IM53)
- Jan/Feb 2021, Statistical Process Control (18IM53)
- Fast Track Oct 2020, Quality and Reliability Engineering (16IM53)
- Data tables, Fast Track Oct 2020 (16IM53 / 16IM64)
- Nov/Dec 2019, Quality and Reliability Engineering (16IM53)
- Fast Track Jul 2019, Quality and Reliability Engineering (16IM53)
- Nov/Dec 2018 (16IM53)

Question Paper ID: 790679

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Course Code: IM353IA

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026
INDUSTRICAL ENGINEERING AND MANAGEMENT

Quality Assurance

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical table is permitted.
4.Use of statistical process control chart formula table is permitted.

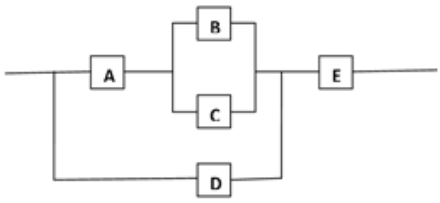
Part A

Question No	Question	M	CO	BT
1.1	The concept of integrating digital technologies with QA practices is called _____.	01	2	2
1.2	What is Quality 4.0? Mention two technologies used.	02	1	2
1.3	A control chart contains a centre line, _____ and _____.	01	1	1
1.4	Define Cpk. How is it different from Cp?	02	2	2
1.5	Define Cp. How is it interpreted	02	2	2
1.6	What is AOQL?	02	2	2
1.7	What is AOQ?	02	2	2
1.8	What is replication and why is it important?	02	2	2
1.9	What is a factorial experiment?	02	3	2
1.10	What is failure rate?	02	2	2
1.11	Differentiate between repairable and non-repairable systems.	02	2	2

Part B

<table>
<tr><th>Question No</th><th>Question</th><th>M</th><th>CO</th><th>BT</th></tr>
<tr><td>2a</td><td>Discuss any four quality dimensions with examples.</td><td>08</td><td>2</td><td>3</td></tr>
<tr><td>2b</td><td>Explain any four statistical concepts used in control charts.</td><td>08</td><td>2</td><td>3</td></tr>
<tr><td>3</td><td>In a foundry the specification on mould hardness is 65 ±15. Twenty-five moulds were chosen at random and mould hardness was estimated at 4 different places/mould. The data obtained are given in the following table:
<table>
<tr><th>Mould No.</th><th>$\bar{X}$</th><th>R</th><th>Mould No.</th><th>$\bar{X}$</th><th>R</th></tr>
<tr><td>1</td><td>68</td><td>5</td><td>13</td><td>70</td><td>10</td></tr>
<tr><td>2</td><td>59</td><td>16</td><td>14</td><td>67</td><td>10</td></tr>
<tr><td>3</td><td>60</td><td>15</td><td>15</td><td>63</td><td>15</td></tr>
<tr><td>4</td><td>57</td><td>12</td><td>16</td><td>69</td><td>10</td></tr>
<tr><td>5</td><td>74</td><td>24</td><td>17</td><td>62</td><td>5</td></tr>
<tr><td>6</td><td>68</td><td>5</td><td>18</td><td>56</td><td>10</td></tr>
<tr><td>7</td><td>66</td><td>10</td><td>19</td><td>66</td><td>10</td></tr>
<tr><td>8</td><td>56</td><td>10</td><td>20</td><td>63</td><td>8</td></tr>
<tr><td>9</td><td>78</td><td>15</td><td>21</td><td>67</td><td>15</td></tr>
<tr><td>10</td><td>55</td><td>10</td><td>22</td><td>75</td><td>10</td></tr>
<tr><td>11</td><td>67</td><td>5</td><td>23</td><td>53</td><td>5</td></tr>
<tr><td>12</td><td>57</td><td>5</td><td>24</td><td>56</td><td>12</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
</table>
i). Construct the control chart for the data
ii).Interpret the state of the manufacturing process
iii).Can the process meet the specification?
iv).What is the current state of affairs regarding non-conformity?
v). What are the different possible ways of reducing the rejection?</td><td>16</td><td>4</td><td>4</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>4a</td><td>Explain the steps in Construct $\bar{X}$–R charts. and interpret $\bar{X}$–R charts.</td><td>10</td><td>3</td><td>2</td></tr>
<tr><td>4b</td><td>What are Process capability index Cp and Cpk? Explain their significance.</td><td>06</td><td>2</td><td>3</td></tr>
<tr><td>5a</td><td>Differenciate between Producer’s Risk vs Consumer’s Risk,AQL and LQL and their Importance</td><td>10</td><td>3</td><td>3</td></tr>
</table>

5b	Explain the following indices used in acceptance sampling plans i) Acceptable Quality Level ii) Rejectable Quality Level	06	3	3
	OR			
6a	What are the characteristics of a good acceptance sampling plan	06	3	2
6b	A double sampling plan is as follows: Inspect a first sample of 25 items randomly form a lot. The lot is accepted if no defectives are found. If there or more found, the lot is rejected. Otherwise a second sample of 100 is drawn and the lot is accepted if the combined number of defectives found does not exceed. i) Determine the producer's risk (a) for a lot fraction defective of 0.01 ii) Determine the consumer's risk (b) for a lot fraction defective of 0.07 iii) What is the ATI of the above plan for a 2% defective lot.	10	2	3
7a	Discuss with examples basic principles of experimental design.	08	2	2
7b	Explicate with illustrations the strategy of experimentation.	08	2	2
	OR			
8a	Define Randomized Block Design RBD. give Example	04	2	1
8b	How do the principles of replication, randomization, and local control contribute to the validity and reliability of experimental results in design of experiments?	10	2	2
8c	What is interaction in factorial design?	02	2	2
9a	A participant in a motor rally is required to complete a mission of 2500 km. His vehicle can be imagined to have 3 subsystems, namely, fuel injection system, ignition system and other systems. The mean distance to repair the 3 subsystems is known to be 6000 kms, 8000 kms & 10000 kms respectively. Find the reliability of motorist completing mission without repair.	08	3	4
9b	The MTBF of a certain unit is 100 hours. Calculate the reliability of the unit for 150 hoursof operation. If the reliability of the units is to be increased by 15%, 25%, 35%,	08	3	4

	45% and55%. Calculate the corresponding % changes in the MTBF that is necessary. Plot % changes in reliability vs % MTBF			
	OR			
10a	Distinguish between: i Repairable and Irrepairable systems; ii Parallel and Series systems.	08	3	3
10b	Consider the following series-parallel configuration:  Determine the reliability of the system for 20 hours of operating period. The configuration is given below. The failure rate / hour: $\lambda_A = 0.01$, $\lambda_B = 0.0125$, $\lambda_C = 0.02$, $\lambda_D = 0.02$, $\lambda_E = 0.025$.	08	3	3

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Regular Examinations Feb/Mar-2025

Industrial Engineering and Management

QUALITY ASSURANCE

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical tables permitted.

PART-A

			M	BT	CO
1	1.1	Differentiate between Quality Control (QC) and Quality Assurance (QA)	02	2	1
	1.2	What is the chance and assignable causes of variation?	02	2	1
	1.3	What is the primary objective of ISO 14000 standards?	02	1	1
	1.4	What are the main objectives of a quality audit?	02	2	1
	1.5	List any four dimensions of quality.	02	1	1
	1.6	Expand AQL and $AOQL$.	02	1	1
	1.7	What is a completely randomized design (CRD) in experimentation?	02	2	1
	1.8	What is the primary advantage of using factorial experiments?	02	2	1
	1.9	A system consists of two components arranged in parallel. The reliabilities of the components are $R1 = 0.80$ and $R2 = 0.85$. What is the overall reliability of the system?	02	3	1
	1.10	What does $MTTF$ stand for and why is it important?	02	1	1

PART-B

2	a	Explain the concept of Rational Subgrouping and its principles. Apply these principles to real-world scenarios in manufacturing or service industries.	08	2	1
	b	Classify the Cost of Good Quality and Cost of Poor Quality in the automotive industry. Provide examples of each cost category and explain how they can be managed effectively to improve the product performance and reduce overall expenses.	08	2	1
3	a	What are the two most common types of control charts for variables? Briefly explain their purpose.	06	2	1
	b	Lots of Bushing screws from a single supplier were inspected 100 percent with the following results.			

Lot No	*Amount in lot*	*Number rejected*
1.	6000	183
2.	6000	131
3.	10000	184
4.	1783	23
5.	3089	171
6.	5774	65
7.	3000	46
8.	2724	85

Draw an appropriate control chart for the data. What can you conclude on the state of control? 10 3 3

OR

4 a What are the two most common types of control charts for attributes? Briefly explain their purpose. 06 2 1

b A manufacturing company measures the diameter of a component in millimeter during a production process. The data are collected in 5 samples, each with 4 measurements. The data are as follows:

Sample	*Measurement* 1	*Measurement* 2	*Measurement* 3	*Measurement* 4
1.	10.2	10.3	10.1	10.4
2.	10.6	10.8	10.4	10.5
3.	10.4	10.2	10.2	10.3
4.	10.1	10.3	10.3	10.2
5.	10.5	10.4	10.6	10.5

i) Calculate the Mean ($\bar{X}$) and Range (R) for each sample.
ii) Determine Control Limits using standard factors for a sample $A2 = 0.729, D3 = 0, D4 = 2.282$.
iii) Interpret the Control Chart. 10 3 3

5 a With the help of a OC curve, define the terms producer's risk and consumer's risk in acceptance sampling plan. 06 2 1

b Draw the OC curve, AOQ curve, ASN curve and ATI curve for the single sampling plan with $N = 5000, n = 100$ and $c = 2$. 10 3 2

OR

6 a Compare single, double and multiple sampling plans. 06 2 1

b Suppose a consumer receives lots of 500 candles from a new supplier. To check the quality of the lot, the consumer draws one sample of size 20 and accepts the lot if the inspected sample contains at most one defective candle. Otherwise he/she rejects the lot. Construct the OC curve for this plan. 10 3 2

7 a Explain the purpose and procedure of a "Single Factor Experiment" in the context of DOE. How is this type of experiment different from a "Multi-Factor Experiment" and when would each be used? 04 2 1

b A chemical engineer has conducted an experiment to study the effect of temperature and pressure on the reaction time of a chemical process. Let the two factors temperature and pressure are denoted by A and B respectively. Suppose the following results are obtained from this study. The number of replications is two.

$Pressure(B)$	$Temperature\ (A)$	
	$Low(60\,℃)$	$High(70\,℃)$
$Low(100MPa)$	40	43
	37	50
$High(150\,MPa)$	59	37
	54	43

Estimate the factor effects. Which effects are large?
Perform an analysis of variance. What are your conclusions? 12 3 4

OR

8 a Explain the concept of "Randomization" in the context of the design of experiments. Why is randomization important and how does it help in eliminating bias and improving the validity of the experimental results? Provide an example of how randomization can be applied in a real-worlds experiment. 08 2 4

	b	A researcher is testing the effect of three different fertilizers (AAA, BBB and CCC) on crop yield (in kg). The researcher conducts an experiment using a randomized block design with 4 replicates per fertilizer type. The observed yields are as follows: $\begin{array}{\|c\|c\|}\hline \boldsymbol{Fertilizer} & \boldsymbol{Yield(kg)} \\ \hline AAA & 22, 24, 20, 23 \\ \hline BBB & 28, 27, 29, 30 \\ \hline CCC & 25,26,24,27 \\ \hline \end{array}$ Perform a one-way $ANOVA$ to determine if there is a significant difference in mean yield among the fertilizers at a 5% significance level.	08	3	3
9	a	Stare the General failure rate curve (Bath tub curve) of an electronic product and identify the causes of failures in the various zones.	08	2	2
	b	In a rerolling process 2 blowers are connected to reheat the furnace using pulverized coal. One of the blowers is standby equipment. The blower system includes: Stator, motor, coupling and blower with the following reliabilities: $0.9, 0.92, 0.85$ and 0.95 respectively. The pulverized coal system has a reliability of 0.99 and the furnace of 0.95 for an operating period of 250 hours. Draw the system configuration and determine the reliability of the total system for 250 hours of operation.	08	3	4
		OR			
10	a	Define the following terms: Reliability, $MTTF, MTBF, MTTR$, Availability and Maintainability.	10	2	1
	b	An amplifier has an exponential time to failure distribution with a failure rate of 10% per 2000 hours. What is the reliability of the amplifier at 8000 hours? Find the mean time to failure.	06	3	3

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations March /April -2023

Industrial Engineering and Management

STATISTICAL PROCESS CONTROL

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of Statistical Tables permitted.

PART-A

1	1.1	Give any one of the definitions of quality.	01
	1.2	What is the broad purview of ISO 9000 standards?	01
	1.3	Consider an operation on a CNC machine. Give an example of a possible assignable cause.	01
	1.4	What is the proportion of the Normal distribution that is covered under the 35 control limits?	01
	1.5	Define the term warning limit.	01
	1.6	What can you conclude when a point falls below the lower control limit on a control chart for fraction non-confirming chart (p-chart)?	01
	1.7	Consider the process of costing. Give an example of a defect.	01
	1.8	Write down the formula for process performance index C_{PK} taxing into account process centering.	01
	1.9	Mention any one of the uses of the chart for individual measurements.	01
	1.10	When do you prefer to use a group control chart for monitoring the process quality?	01
	1.11	Define the term producer's risk.	01
	1.12	Give an example of any one of the published sampling plans.	01
	1.13	Graphically indicate the presence of an interaction effect between two factors.	01
	1.14	What is the meaning of the term randomization?	01
	1.15	How many experimental runs are to be carried out in the case of a 2^5 factorial design?	01
	1.16	What is the meaning of a full factorial design?	01
	1.17	Sketch the time versus failure rate curve for the components failure according to exponential distribution.	01
	1.18	Give a practical example of a series device configuration.	01
	1.19	If there are 'n' components in parallel configuration, all the components are identical. What is the formula for estimating the reliability of the system, given that the reliability of each component in R?	01
	1.20	Why the common failure rate curve is also called as the bath tub curve?	01

PART-B

2	a	Take a simple product and identify the 8 dimensions of quality. Discuss how these dimensions determine the quality of this product.	08
	b	What is the true meaning of a six sigma process? List out the percent inside specifications and the defectives in terms of PPM level for sigma limits of specifications from 1 to 6. Assuming normal distribution and a perfectly centered process. What happens to these two performance measures when there is a shift in the mean of the process to the extent of ± 1.55 from the target? Discuss.	08
3	a	Sample of $n = 6$ items each are taken from a process at regular intervals. A quality characteristic is measured and $\bar{X}$ and R are calculated for each sample. After 50 samples, we have $\sum_{i=1}^{50} \bar{X}_i = 2000$ and $\sum_{i=1}^{50} R_i = 200$. Assume that the quality characteristic is normally distributed. i) Compute the control limits for the $\bar{X}$ and R charts. ii) All points on the control charts fall between the control limits computed in part (i). What are the natural tolerance limits to the process? iii) If the specification limits are 41 ± 5.0, what are your conclusions regarding the ability to the process to produce items within these specifications? iv) Assuming that if an item exceeds the upper specification limit it can be reworked and if it is below the lower specification limit, it must be separated. What percent of scrap and rework is the process producing?	12
	b	Briefly outline any one of the methods of estimating process capability.	04
		OR	
4	a	A process is being controlled with a fraction non-confirming control chart. The process average has been shown to be 0.07. Three-sigma control limits are used and the procedure calls for taking daily samples of 400 items. i) Calculate the upper and lower control limits. ii) If the process average should suddenly shift to 0.10, what is the probability that the shift would be detected on the first subsequent sample?	08
	b	A control chart for non-conformities is maintained on a process producing desk calculators. The inspection unit is defined as two calculators. The average number of non-conformities per machine when the process is in control is estimated to be two. i) Find the appropriate three sigma control limits for this size inspection units. ii) What is the probability of Type-I error for this control chart?	08
5	a	Briefly outline the basic principles of the cusum control chart for monitoring the process mean.	04
	b	Suppose that a supplier ships components in lots of size 5000. A single sampling plan with $n = 50$ and $C = 2$ is being used for receiving inspection. Rejected lots are screened and all defective items are reworked and returned to the lot.	

		i) Draw the OC curve for this plan. ii) Find the level of lot quality that will be rejected 90% of the time. iii) Management has objected to the use of the above sampling procedure and wants to use a plan with an acceptance number $C = 0$ arguing that this is more consistent with their zero defects program. What do you think of this?	12
		OR	
6	a	Give an example of any three situations in which the sample size used for process monitoring is n=1; that is the sample consists of an individual unit is used.	04
	b	Draw the primary and supplementary OC curves for a double sampling plan with $n_1 = 50, C_1 = 2, n_2 = 100, C_2 = 6$. If the incoming lots have fraction non-conforming$p = 0.05$, what is the probability of acceptance on the first sample? What is the probability of final acceptance? Calculate the probability of rejection on the first sample.	12
7	a	An experiment was run to determine whether four specific fusing temperatures affect the density of a certain type of brick. The experiment led to the following data.	

Temperature	Density				
100	21·8	21·9	21·7	21·6	21·7
125	21·7	21·4	21·5	21·4	
150	21·9	21·8	21·8	21·6	21·5
175	21·9	21·7	21·8	21·4	

		i) Does the fusing temperature affect the density of the bricks? Use $\alpha = 0.05$ ii) Analyze the residuals from this experiment.	12
	b	Briefly outline the guidelines for designing experiments.	04
8	a	Define the following terms. i) Reliability ii) Failure rate iii) $MTBF$	03
	b	A component has a failure rate which is according to experimental failure law which is $\lambda = 4 \times 10^{-3}\ failures/hour$. i) Determine the reliability of this component for $t = 10hrs, 100hrs, 1000hrs, 10000hrs$. ii) Plot the time versus reliability graph. Comment on the plot that you have obtained.	05
	c	Three components are in series with failure rate as follows. Component 1 failure rate $\lambda_1 = 4 \times 10^{-6}\ failures/hour$ Component 2 failure rate $\lambda_2 = 4 \times 10^{-4}\ failures/hour$ Component 3 failure rate $\lambda_3 = 6 \times 10^{-6}\ failures/hour$ Determine the reliability of this series system configuration. If the three components were to be arranged in parallel configuration, what would be the system reliability? (Assume $t = 1000hr$).	08

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

QUALITY AND RELIABILITY ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Quality cannot be inspected into product nor can it be advertise in, it must be designed and built in. Comment on the statement.	02
	1.2	A physical or chemical property or any other requirements used to define the nature of the product is termed as _____.	01
	1.3	State Central limit theorem.	01
	1.4	Differentiate between chance and assignable causes of variation.	02
	1.5	The usual purpose of an R-chart is to signal whether there has been a gain or loss in _____.	01
	1.6	_____ plans details the action to be taken once an out of control situation is detected.	01
	1.7	Differentiate between defect and defective.	02
	1.8	The local newspaper receives several complaints per day about typographic errors. Over a seven-day period, the publisher has received calls from readers reporting the following number of errors: 4,3,2,6,7,3 and 9. Based on these data alone,. What type of control chart(s) should the publisher use ?	02
	1.9	The number of invoices that contain errors is an example of _____ data.	01
	1.10	Average total inspection under single sampling plan in which rejected lots are 100% inspected is given by the formula ATI= _____.	01
	1.11	State the three parameters of the single sampling plan.	02
	1.12	The uncontrollable input factors that affect the output process are also termed as _____.	01
	1.13	The experiment in which two factors are varied at two levels is called as _____ experiment.	01
	1.14	The probability that the device/equipment will be restored to its operational effectiveness with the given period is called as _____.	01
	1.15	What is the difference between an experimental and a control group?	01

PART-B

2 a Identify the various dimensions of quality dimensions as proposed by Garvin. Explain their significance with respect to product quality taking suitable examples. 08

b What do you mean by Quality cost? Why is it so important? Differentiate between Failure cost and appraisal cost. 08

3 a In a manufacturing unit of certain transformers the units are required to meet a number of specifications and are subjected to inspection daily. At the end of twenty days 230 units have been rejected out of 4150 produced and inspected. Determine

i. The 3 sigma control limits based on any average daily production of 200 units

ii. Only one point has crossed the control limits. On that day 30 non-conforming items were found in 200 units inspected.

iii. What is the aimed value of p?

iv. What control limits would you recommend for future use? 10

b Distinguish clearly between

(i) P and U charts

(ii) C and U charts. 06

OR

4 a The following $\overline{X}$ – R values of 20 subgroups of 5 readings.

Sub group	$\overline{X}$	R	Subgroup	$\overline{X}$	R
1	34.0	4	11	38.4	4
2	31.6	2	12	34.0	14
3	30.8	3	13	35.0	4
4	33.8	5	14	33.8	7
5	31.6	3	15	31.6	5
6	33.0	5	16	33.0	7
7	28.2	13	17	32.6	3
8	33.8	19	18	31.8	9
9	37.8	6	19	35.6	6
10	35.8	4	20	33.0	4

(i) Determine the control limits of $\overline{X}$ and R chart

(ii) What is the process capability?

(iii) Is the process capable of meeting the specification limits ?

(iv) Determine the percentage of rejection if any. (Note : The specification limits are 33 ± 5)

(v) What values of $\overline{X}$ and R do you recommend for future use? 16

5 a A single sampling plan is as follows : n=80, c=2

(i) Plot type – B OC curve (use distribution which is theoretically correct)

(ii) If AQL is 1.5% find the producer's risk

(iii) If consumer's risk is 15% find LTPD 10

b Sketch the ideal operating characteristic curve of a sampling plan. Explain its significance. 03

c Explain the difference between average sample number and average total inspection. State any assumptions made. 03

		OR	
6	a	A double sampling plan is as follows N=5000 n1=150 c1=3 r1=6 n2=200 c2=8 r2=9 Lot fraction defective is 1.5% . Calculate Pa, AoQ, ATI and ASN	12
	b	Differentiate between type A and type B OC curve.	04
7	a	Explain the utility of "bath tub curve". What are the reasons for failure during the three stages of equipment life?	08
	b	A step down transformer, rectifier, filter comprises a series of systems. The following are the failure rates of these components. Transformer =1.56 percent failures / 1000 hours Amplifier = 2 percent failure / 1000 hours Filter = 1.7 percent failures / 1000 hours The equipment is operated for 1500 hours. What is the probability of survival of the system?	08
8	a	Determine the factors, probable levels of a factor and response for the situations given below. i) The Indian cricket team wants to determine which combination of playing XI will enhance the team's performance in the World Cup matches. ii) A casting company which desires to produce Good Quality Castings.	10
	b	Define the following terms with an example: (i) Factor (ii) Level (iii) Main effect (iv) Interaction effect (v) Treatment Combination	06

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Jan/Feb – 2021

Industrial Engineering and Management

STATISTICAL PROCESS CONTROL

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of Statistical Tables is permitted.

PART-A

1	1.1	What is the first step in Quality Assurance?	01
	1.2	The statistical process control helps identify _____ type of causes that affect the process.	01
	1.3	In Six Sigma, the value of $C_{p=}$ _____.	01
	1.4	_____ is a method to find factor effects and interaction effects.	01
	1.5	Effect of one factor depends on where some other factor is set. This is called as _____.	01
	1.6	The mean of sampling distribution is equal to _____ of process distribution.	01
	1.7	The percent of the sample means will have values that are within ±3 standard deviations of the distribution mean is _____.	01
	1.8	In control charts, the dividing lines between random and nonrandom deviations from mean of the distribution are known as _____.	01
	1.9	Central tendency of a process is monitored in _____ control chart.	01
	1.10	The control chart used for the number of defects per unit is _____.	01
	1.11	The costs related to formal technical reviews and test equipment reviews are related to _____ type of quality cost.	01
	1.12	In double sampling plan, if the number of defects is in between the two cut off numbers C_1 *and* C_2 then the next step is to _____.	01
	1.13	In _____ sampling plan, the samples are accumulated at each stage till the decision of accepting or rejecting a lot is taken.	01
	1.14	In addition to OC curve, _____ and _____ curves are used to select an appropriate sampling pan.	02
	1.15	To find out whether noise affects the capability to solve the assignment of Quality Management, one group solve the assignment in a silent room and another group solve the assignment in a noisy room. The group solving problems in the silent room completes 22 problems in one hour and the group solving problems in the noisy room completed 15 problems in one hour. In this experiment, the independent variable is _____ and the dependent variable is _____.	02

	1.16	What statistical procedure is used to assess the statistical significance of the main effects and the interaction(s) in a factorial design? What is the test statistic used for this?	02
	1.17	One of the concepts of reliability is _____ which is defined as the probability that the system is operating properly when it is requested for use.	01

PART-B

2	a	How is the application of concepts of statistical quality control advantageous over 100% inspection? Explain.	10
	b	Explain the need for the change in the control limits of the control charts.	06
3	a	Identify and explain the cause of i) Trend ii) Cyclic pattern in control charts.	06
	b	Mean values and ranges of data from 20 samples (sample size =4) are shown in the table below: Calculate the control limits for x-bar and R charts. Draw the control charts for the same and give suitable conclusions.	10

S.N.	Mean	Range	S.N.	Mean	Range
1	10	4	11	12	5
2	15	4	12	13	4
3	12	5	13	12	4
4	11	4	14	12	3
5	9	5	15	11	3
6	11	6	16	15	4
7	11	4	17	12	4
8	9	4	18	15	3
9	10	4	19	11	3
10	11	6	20	10	4

OR

4	a	Ten samples of 15 components are tested with results below. Construct p-chart and draw conclusions.	08
	b	A stable normally distributed process has the specification 3.5 ± 0.03, given $\mu = 3.51$ and standard deviation = 0.016, determine $C_p and\ C_{pk}$. What conclusions can you draw about the capability of the process?	04
	c	Discuss any four principles of rational sub-grouping.	04

Sample	No. of defectives
1	2
2	1
3	3
4	2
5	3
6	1
7	0
8	1
9	0
10	1

5 Consider the following data of a double sampling plan. Construct Operating Characteristic Curve, AOQ curve. Also determine ATI and ASN.

N = 5000

n1 = 150 n2 = 200

c1 = 3 c2 = 8

r1 = 6 r2 = 9 — 16

OR

6 a What are the assumptions that you make while designing a single sampling plan? — 05

b Compare and contrast single, double and triple sampling plans. — 05

c How Type A and Type B OC Curves are different? When are they used? — 06

7 a Differentiate between CRD and RBD. — 04

b The data below is from a cattle experiment in which temperatures, protein and gender were evaluated as to efficiency of weight production (feed per pound of gain). Two replicates of each treatment were run according to a Completely Randomized Design. Conduct ANOVA for main effects and interaction effects. Draw suitable conclusions. — 12

House Temp	90°				60°			
Protein Level	High		Low		High		Low	
Gender	M	F	M	F	M	F	M	F
Replication 1	2.96	2.69	2.8	2.6	2.89	2.73	2.94	2.85
Replication 2	3.04	2.75	2.72	2.7	2.75	2.67	3	2.75

8 a Compare and contrast series and parallel configurations of system reliability. Give suitable examples. — 04

b A system has three units with a failure rate of 1.5, 4, 3.8 failures per 10^6 hours respectively. Determine.

i) MTBF of the system.

ii) The reliability of the system for 10 hours if the components are connected in series.

iii) Reliability if these components are connected in parallel. — 08

c What do you think are the causes that lead to high failure during infant mortality period? — 04

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering and Management

QUALITY AND RELIABILTY ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Distinguish between chance assignable cause of variations.	02
	1.2	State any two principles of rational sub grouping.	02
	1.3	Sketch the extreme variation patterns of assignable cause.	02
	1.4	Control unit for R chart are _____ & _____.	02
	1.5	Write any four advantages of sampling technique.	02
	1.6	Write OC curve for single sampling plan.	02
	1.7	Differentiate between consumer risk and producer risk.	02
	1.8	Measure of process performance in terms of the nominal & target value is provided by C_{pk} & the mathematical expression is _____.	02
	1.9	Define level & factors.	02
	1.10	Expand $MTTF$ & $MTBF$.	02

PART-B

2	a	What are the eight dimensions of quality? Discuss with an example any two dimensions.	08
	b	List and explain major quality cost categories.	08
3	a	Sample size $n = 5$ are taken from a manufacturing process every hour. A quality characteristics is measured and $\bar{x}$ and R are computed for each sample after 25 samples have been analyzed we have $\sum_{i=1}^{25} \bar{x} = 66250$ & $\sum_{i=1}^{25} R_i = 900$. The quality characteristics is normally distributed i) Find control unit for $\bar{x}$ and R charts ii) Assume that both chart exhibit control if the specifications are 26.4 ± 0.50. Estimate the fraction non-conforming iii) If the mean of the process were 26.4 what fraction non-conforming would result.	08
	b	Sample of $n = 5$ units are taken from a process every hour. The $\bar{x}$ and R values for a particular quality characteristics are determined. After 25 samples have been collected. We calculate $\bar{x} = 20$ and $R = 4.56$ i) What are the three sigma control limits ii) Both curve exhibit control, Estimate the process standard deviation iii) Assume that the process output is normally distributed, if the specifications 19 ± 5. What are your conclusions regarding the process capability? iv) If the process mean shifts to 24. What is the probability of not detecting this on the first subsequent samples?	08

OR

4 a A process is being controlled with fraction non conforming control chart. The process average has been shown to be 0.07. Three sigma control unit are used and the procedure calls for taking daily samples of 400 items.
i) Calculate the upper & lower control unit.
ii) If the process average should suddenly shift to 0.10. What is the probability that shift would be detected on first subsequent sample
iii) What is the probability that the shift in part (ii) would be detected on the first & second sample takes after the shift? 08

b What are the differences between
i) p chart & np chart
ii) c chart & u chart 08

5 a A single sampling plan of $n = 200, c = 4$ & N is very large when compared to the sample size. Draw an AOQ curve. Find $AOQL$ after computing the probabilities of acceptance of lot of 0.5%, 1%, 2%, 3%, 4% & 5% defectives 08

b Construct an OC curve for a double sampling plan of $N = 2600$, $n_1 = 100$, $c_1 = 1, n_2 = 200$ & $c_2 = 4$. 08

OR

6 a In a single sampling plan $N = 10{,}000$, $n = 300, c = 1$ is used with the stipulation that all rejected lots are to be screened, compute the AOQ, if all lots submitted are 0.50 defective. 08

b Draw the AOQ curve for the data $N = 4000, n = 80, c = 2$. 08

7 a An article in industrial quality control describe an experiment to investigate the effect of glass type and phosphor type of the brightness of a television tube. The response measured in the current necessary (in microamps) to obtain a specific brightness level. The data are shown here. Analyze the data and draw conclusions.

Glass type	*Phosphor type*		
	1	2	3
1	280	300	290
	290	310	285
	285	295	290
2	230	260	220
	235	240	225
	240	235	230

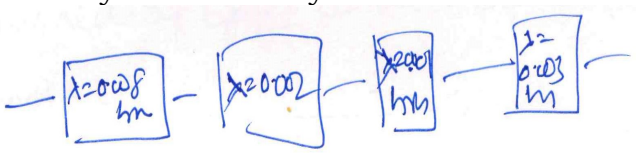
10

b Write the test matrix for the 2^2 factorial design. 06

8 a Suppose that a certain software product has a mean time between failure of 10,000 hours and has a mean time to repair of 20 hrs. If the product is used by 100 customers. What is availability? 06

b Find the reliability of the system with three components A, B & C in parallel. The reliability of A, B & C are 0.95, 0.92 & 0.90 respectively. 06

c The components in the system below are exponentially distributed with the indicated failure rates. Develop an expression for the reliability of the system. What is the system reliability at the time of 100 hrs?



06

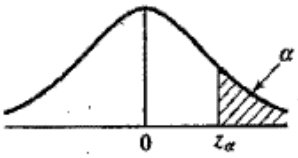
Table A.1 Random Digits

94737	08225	35614	24826	88319	05595	58701	57365	74759
87259	85982	13296	89326	74863	99986	68558	06391	50248
63856	14016	18527	11634	96908	52146	53496	51730	03500
66612	54714	46783	61934	30258	61674	07471	67566	31635
30712	58582	05704	23172	86689	94834	99057	55832	21012
69607	24145	43886	86477	05317	30445	33456	34029	09603
37792	27282	94107	41967	21425	04743	42822	28111	09757
01488	56680	73847	64930	11108	44834	45390	86043	23973
66248	97697	38244	50918	55441	51217	54786	04940	50807
51453	03462	61157	65366	61130	26204	15016	85665	97714
92168	82530	19271	86999	96499	12765	20926	25282	39119
36463	07331	54590	00546	03337	41583	46439	40173	46455
47097	78780	04210	87084	44484	75377	57753	41415	09890
80400	45972	44111	99708	45935	03694	81421	60170	58457
94554	13863	88239	91624	00022	40471	78462	96265	55360
31567	53597	08490	73544	72573	30961	12282	97033	13676
07821	24759	47266	21747	72496	77755	50391	59554	31177
09056	10709	69314	11449	40531	02917	95878	74587	60906
19922	37025	80731	26179	16039	01518	82697	73227	13160
29923	02570	80164	36108	73689	26342	35712	49137	13482
29602	29464	99219	20308	82109	03898	82072	85199	13103
94135	94661	87724	88187	62191	70607	63099	40494	49069
87926	34092	34334	55064	43152	01610	03126	47312	59578
85039	19212	59160	83537	54414	19856	90527	21756	64783
66070	38480	74636	45095	86576	79337	39578	40851	53503
78166	82521	79261	12570	10930	47564	77869	16480	43972
94672	07912	26153	10531	12715	63142	88937	94466	31388
56406	70023	27734	22254	27685	67518	63966	33203	70803
67726	57805	94264	77009	08682	18784	47554	59869	66320
07516	45979	76735	46509	17696	67177	92600	55572	17245
43070	22671	00152	81326	89428	16368	57659	79424	57604
36917	60370	80812	87225	02850	47118	23790	55043	75117
03919	82922	02312	31106	44335	05573	17470	25900	91080
46724	22558	64303	78804	05762	70650	56117	06707	90035
16108	61281	86823	20286	14025	24909	38391	12183	89393
74541	75808	89669	87680	72758	60851	55292	95663	88326
82919	31285	01850	72550	42986	57518	01159	01786	98145
31388	26809	77258	99360	92362	21979	41319	75739	98082
17190	75522	15687	07161	99745	48767	03121	20046	28013
00466	88068	68631	98745	97810	35886	14497	90230	69264

Table A.2 Random Normal Numbers

0.23	−0.17	0.43	2.18	2.13	0.49	2.72	−0.18	0.42
0.24	−1.17	0.02	0.67	−0.59	−0.13	−0.15	−0.46	1.64
−1.16	−0.17	0.36	−1.26	0.91	0.71	−1.00	−1.09	−0.02
−0.02	−0.19	−0.04	1.92	0.71	−0.90	−0.21	−1.40	−0.38
0.39	0.55	0.13	2.55	−0.33	−0.05	−0.34	−1.95	−0.44
0.64	−0.36	0.98	−0.21	−0.52	−0.02	−0.15	−0.43	0.62
−1.90	0.48	−0.54	0.60	−0.35	−1.29	−0.57	0.23	1.41
−1.04	−0.70	−1.69	1.76	0.47	−0.52	−0.73	0.94	−1.63
−.78	0.11	−0.91	−1.13	0.07	0.45	−0.94	1.42	0.75
0.68	1.77	−0.82	−1.68	−2.60	1.59	−0.72	−0.80	0.61
−0.02	0.92	1.76	−0.66	0.18	−1.32	1.26	0.61	0.83
−0.47	1.04	0.83	−2.05	1.00	−0.70	1.12	0.82	0.08
−0.40	1.40	1.20	0.00	0.21	−2.13	−0.22	1.79	0.87
−0.75	0.09	−1.50	0.14	−2.99	−0.41	−0.99	−0.70	0.51
−0.66	−1.97	0.15	−1.16	−0.60	0.50	1.36	1.94	0.11
−0.44	−0.09	−0.59	1.37	0.18	1.44	−0.80	2.11	−1.37
1.41	−2.71	−0.67	1.83	0.97	0.06	−0.28	0.04	−0.21
1.21	−0.52	−0.20	−0.88	−0.78	0.84	−1.08	−0.25	0.17
0.07	0.66	−0.51	−0.04	−0.84	0.04	1.60	−0.92	1.14
−0.08	0.79	−0.09	−1.12	−1.13	0.77	0.40	0.69	−0.12
0.53	−0.36	−2.64	0.22	−0.78	1.92	−0.26	1.04	−1.61
−1.56	1.82	−1.03	1.14	−0.12	−0.78	−0.12	1.42	−0.52
0.03	−1.29	−0.33	2.60	−0.64	1.19	−0.13	0.91	0.78
1.49	1.55	−0.79	1.37	0.97	0.17	0.58	1.43	−1.29
−1.19	1.35	0.16	1.06	−0.17	0.32	−0.28	0.68	0.54
−1.19	−1.03	−0.12	1.07	0.87	−1.40	−0.24	−0.81	0.31
0.11	−1.95	−0.44	−0.39	−0.15	−1.20	−1.98	0.32	2.91
−1.86	0.06	0.19	−1.29	0.33	1.51	−0.36	−0.80	−0.99
0.16	0.28	0.60	−0.78	0.67	0.13	−0.47	−0.18	−0.89
1.21	−1.19	−0.60	−1.22	0.07	−1.13	1.45	0.94	0.54
−0.82	0.54	−0.98	−0.13	1.52	0.77	0.95	−0.84	2.40
0.75	−0.80	−0.28	1.77	−0.16	−0.33	2.43	−1.11	1.63
0.42	0.31	1.56	0.56	0.64	−0.78	0.04	1.34	−0.01
−1.50	−1.78	−0.59	0.16	0.36	1.89	−1.19	0.53	−0.97
−0.89	0.08	0.95	−0.73	1.25	−1.04	−0.47	−0.68	−0.87
0.19	0.85	1.68	−0.57	0.37	−0.48	−0.17	2.36	−0.53
0.49	0.32	−2.08	−1.02	2.59	−0.53	0.15	0.11	0.05
−1.44	0.07	−0.22	−0.93	−1.40	0.54	−1.28	−0.15	0.67
−0.21	−0.48	1.21	0.67	−1.10	−0.75	−0.37	0.68	−0.02
−0.65	−0.12	0.94	−0.44	−1.21	−0.06	−1.28	−1.51	1.39
0.24	−0.83	1.55	0.33	−0.59	−1.24	0.70	0.01	0.15
−0.73	1.24	0.40	−0.61	0.68	0.69	0.07	−0.23	−0.66
−1.93	0.75	−0.32	0.95	1.35	1.51	−0.88	0.10	−1.19
0.08	0.16	0.38	−0.96	1.99	−0.20	0.98	0.16	0.26
−0.47	−1.25	0.32	0.51	−1.04	0.97	2.60	−0.08	1.19

Table A.3 Cumulative Normal Distribution



$$\phi(z_\alpha) = \int_{-\infty}^{z_\alpha} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = 1 - \alpha$$

z_α	0.00	0.01	0.02	0.03	0.04	z_α
0.0	0.500 00	0.503 99	0.507 98	0.511 97	0.515 95	0.0
0.1	0.539 83	0.543 79	0.547 76	0.551 72	0.555 67	0.1
0.2	0.579 26	0.583 17	0.587 06	0.590 95	0.594 83	0.2
0.3	0.617 91	0.621 72	0.625 51	0.629 30	0.633 07	0.3
0.4	0.655 42	0.659 10	0.662 76	0.666 40	0.670 03	0.4
0.5	0.691 46	0.694 97	0.698 47	0.701 94	0.705 40	0.5
0.6	0.725 75	0.729 07	0.732 37	0.735 65	0.738 91	0.6
0.7	0.758 03	0.761 15	0.764 24	0.767 30	0.770 35	0.7
0.8	0.788 14	0.791 03	0.793 89	0.796 73	0.799 54	0.8
0.9	0.815 94	0.818 59	0.821 21	0.823 81	0.826 39	0.9
1.0	0.841 34	0.843 75	0.846 13	0.848 49	0.850 83	1.0
1.1	0.864 33	0.866 50	0.868 64	0.870 76	0.872 85	1.1
1.2	0.884 93	0.886 86	0.888 77	0.890 65	0.892 51	1.2
1.3	0.903 20	0.904 90	0.906 58	0.908 24	0.909 88	1.3
1.4	0.919 24	0.920 73	0.922 19	0.923 64	0.925 06	1.4
1.5	0.933 19	0.934 48	0.935 74	0.936 99	0.938 22	1.5
1.6	0.945 20	0.946 30	0.947 38	0.948 45	0.949 50	1.6
1.7	0.955 43	0.956 37	0.957 28	0.958 18	0.959 07	1.7
1.8	0.964 07	0.964 85	0.965 62	0.966 37	0.967 11	1.8
1.9	0.971 28	0.971 93	0.972 57	0.973 20	0.973 81	1.9
2.0	0.977 25	0.977 78	0.978 31	0.978 82	0.979 32	2.0
2.1	0.982 14	0.982 57	0.983 00	0.983 41	0.983 82	2.1
2.2	0.986 10	0.986 45	0.986 79	0.987 13	0.987 45	2.2
2.3	0.989 28	0.989 56	0.989 83	0.990 10	0.990 36	2.3
2.4	0.991 80	0.992 02	0.992 24	0.992 45	0.992 66	2.4
2.5	0.993 79	0.993 96	0.994 13	0.994 30	0.994 46	2.5
2.6	0.995 34	0.995 47	0.995 60	0.995 73	0.995 85	2.6
2.7	0.996 53	0.996 64	0.996 74	0.996 83	0.996 93	2.7
2.8	0.997 44	0.997 52	0.997 60	0.997 67	0.997 74	2.8
2.9	0.998 13	0.998 19	0.998 25	0.998 31	0.998 36	2.9
3.0	0.998 65	0.998 69	0.998 74	0.998 78	0.998 82	3.0
3.1	0.999 03	0.999 06	0.999 10	0.999 13	0.999 16	3.1
3.2	0.999 31	0.999 34	0.999 36	0.999 38	0.999 40	3.2
3.3	0.999 52	0.999 53	0.999 55	0.999 57	0.999 58	3.3
3.4	0.999 66	0.999 68	0.999 69	0.999 70	0.999 71	3.4
3.5	0.999 77	0.999 78	0.999 78	0.999 79	0.999 80	3.5
3.6	0.999 84	0.999 85	0.999 85	0.999 86	0.999 86	3.6
3.7	0.999 89	0.999 90	0.999 90	0.999 90	0.999 91	3.7
3.8	0999 93	0.999 93	0.999 93	0.999 94	0.999 94	3.8
3.9	0.999 95	0.999 95	0.999 96	0.999 96	0.999 96	3.9

(continued overleaf)

Table A.3 *(continued)*

z_α	0.05	0.06	0.07	0.08	0.09	z_α
0.0	0.519 94	0.523 92	0.527 90	0.531 88	0.535 86	0.0
0.1	0.559 62	0.563 56	0.567 49	0.571 42	0.575 34	0.1
0.2	0.598 71	0.602 57	0.606 42	0.610 26	0.614 09	0.2
0.3	0.636 83	0.640 58	0.644 31	0.648 03	0.651 73	0.3
0.4	0.673 64	0.677 24	0.680 82	0.684 38	0.687 93	0.4
0.5	0.708 84	0.712 26	0.715 66	0.719 04	0.722 40	0.5
0.6	0.742 15	0.745 37	0.748 57	0.751 75	0.754 90	0.6
0.7	0.773 37	0.776 37	0.779 35	0.782 30	0.785 23	0.7
0.8	0.802 34	0.805 10	0.807 85	0.810 57	0.813 27	0.8
0.9	0.824 94	0.831 47	0.833 97	0.836 46	0.838 91	0.9
1.0	0.853 14	0.855 43	0.857 69	0.859 93	0.862 14	1.0
1.1	0.874 93	0.876 97	0.879 00	0.881 00	0.882 97	1.1
1.2	0.894 35	0.896 16	0.897 96	0.899 73	0.901 47	1.2
1.3	0.911 49	0.913 08	0.914 65	0.916 21	0.917 73	1.3
1.4	0.926 47	0.927 85	0.929 22	0.930 56	0.931 89	1.4
1.5	0.939 43	0.940 62	0.941 79	0.942 95	0.944 08	1.5
1.6	0.950 53	0.951 54	0.952 54	0.953 52	0.954 48	1.6
1.7	0.959 94	0.960 80	0.961 64	0.962 46	0.963 27	1.7
1.8	0.967 84	0.968 56	0.969 26	0.969 95	0.970 62	1.8
1.9	0.974 41	0.975 00	0.975 58	0.976 15	0.976 70	1.9
2.0	0.979 82	0.980 30	0.980 77	0.981 24	0.981 69	2.0
2.1	0.984 22	0.984 61	0.985 00	0.985 37	0.985 74	2.1
2.2	0.987 78	0.988 09	0.988 40	0.988 70	0.988 99	2.2
2.3	0.990 61	0.990 86	0.991 11	0.991 34	0.991 58	2.3
2.4	0.992 86	0.993 05	0.993 24	0.993 43	0.993 61	2.4
2.5	0.994 61	0.994 77	0.994 92	0.995 06	0.995 20	2.5
2.6	0.995 98	0.996 09	0.996 21	0.996 32	0.996 43	2.6
2.7	0.997 02	0.997 11	0.997 20	0.997 28	0.997 36	2.7
2.8	0.997 81	0.997 88	0.997 95	0.998 01	0.998 07	2.8
2.9	0.998 41	0.998 46	0.998 51	0.998 56	0.998 61	2.9
3.0	0.998 86	0.998 89	0.998 93	0.998 97	0.999 00	3.0
3.1	0.999 18	0.999 21	0.999 24	0.999 26	0.999 29	3.1
3.2	0.999 42	0.999 44	0.999 46	0.999 48	0.999 50	3.2
3.3	0.999 60	0.999 61	0.999 62	0.999 64	0.999 65	3.3
3.4	0.999 72	0.999 73	0.999 74	0.999 75	0.999 76	3.4
3.5	0.999 81	0.999 81	0.999 82	0.999 83	0.999 83	3.5
3.6	0.999 87	0.999 87	0.999 88	0.999 88	0.999 89	3.6
3.7	0.999 91	0.999 92	0.999 92	0.999 92	0.999 92	3.7
3.8	0.999 94	0.999 94	0.999 95	0.999 95	0.999 95	3.8
3.9	0.999 96	0.999 96	0.999 96	0.999 97	0.999 97	3.9

Source: W. W. Hines and D. C. Montgomery, *Probability and Statistics in Engineering and Management Science*, 2d ed., © 1980, pp. 592–3. Reprinted by permission of John Wiley & Sons, Inc., New York.

Table A.4 Cumulative Poisson Distribution

	α = Mean											
x	.01	.05	.1	.2	.3	.4	.5	.6	.7	.8	.9	x
0	.990	.951	.905	.819	.741	.670	.607	.549	.497	.449	.407	0
1	1.000	.999	.995	.982	.963	.938	.910	.878	.844	.809	.772	1
2		1.000	1.000	.999	.996	.992	.986	.977	.966	.953	.937	2
3				1.000	1.000	.999	.998	.997	.994	.991	.987	3
4						1.000	1.000	1.000	.999	.999	.998	4
5									1.000	1.000	1.000	5

	α = Mean											
x	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0	x
0	.368	.333	.301	.273	.247	.223	.202	.183	.165	.150	.135	0
1	.736	.699	.663	.627	.592	.558	.525	.493	.463	.434	.406	1
2	.920	.900	.879	.857	.833	.809	.783	.757	.731	.704	.677	2
3	.981	.974	.966	.957	.946	.934	.921	.907	.891	.875	.857	3
4	.996	.995	.992	.989	.986	.981	.976	.970	.964	.956	.947	4
5	.999	.999	.998	.998	.997	.996	.994	.992	.990	.987	.983	5
6	1.000	1.000	1.000	1.000	.999	.999	.999	.998	.997	.997	.995	6
7					1.000	1.000	1.000	1.000	.999	.999	.999	7
8									1.000	1.000	1.000	8

	α = Mean											
x	2.2	2.4	2.6	2.8	3.0	3.5	4.0	4.5	5.0	5.5	6.0	x
0	.111	.091	.074	.061	.050	.030	.018	.011	.007	.004	.002	0
1	.355	.308	.267	.231	.199	.136	.092	.061	.040	.027	.017	1
2	.623	.570	.518	.469	.423	.321	.238	.174	.125	.088	.062	2
3	.819	.779	.736	.692	.647	.537	.433	.342	.265	.202	.151	3
4	.928	.904	.877	.848	.815	.725	.629	.532	.440	.358	.285	4
5	.975	.964	.951	.935	.916	.858	.785	.703	.616	.529	.446	5
6	.993	.988	.983	.976	.966	.935	.889	.831	.762	.686	.606	6
7	.998	.997	.995	.992	.988	.973	.949	.913	.867	.809	.744	7
8	1.000	.999	.999	.998	.996	.990	.979	.960	.932	.894	.847	8
9		1.000	1.000	.999	.999	.997	.992	.983	.968	.946	.916	9
10				1.000	1.000	.999	.997	.993	.986	.975	.957	10
11						1.000	.999	.998	.995	.989	.980	11
12							1.000	.999	.998	.996	.991	12
13								1.000	.999	.998	.996	13
14									1.000	.999	.999	14
15										1.000	.999	15
16											1.000	16

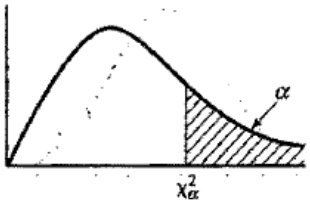
(continued overleaf)

Table A.4 *(continued)*

	α = Mean										
x	6.5	7.0	7.5	8.0	9.0	10.0	12.0	14.0	16.0	18.0	20.0	x
0	.002	.001	.001									0
1	.011	.007	.005	.003	.001							1
2	.043	.030	.020	.014	.006	.003	.001					2
3	.112	.082	.059	.042	.021	.010	.002					3
4	.224	.173	.132	.100	.055	.029	.008	.002				4
5	.369	.301	.241	.191	.116	.067	.020	.006	.001			5
6	.527	.450	.378	.313	.207	.130	.046	.014	.004	.001		6
7	.673	.599	.525	.453	.324	.220	.090	.032	.010	.003	.001	7
8	.792	.729	.662	.593	.456	.333	.155	.062	.022	.007	.002	8
9	.877	.830	.776	.717	.587	.458	.242	.109	.043	.015	.005	9
10	.933	.901	.862	.816	.706	.583	.347	.176	.077	.030	.011	10
11	.966	.947	.921	.888	.803	.697	.462	.260	.127	.055	.021	11
12	.984	.973	.957	.936	.876	.792	.576	.358	.193	.092	.039	12
13	.993	.987	.978	.966	.926	.864	.682	.464	.275	.143	.066	13
14	.997	.994	.990	.983	.959	.917	.772	.570	.368	.208	.105	14
15	.999	.998	.995	.992	.978	.951	.844	.669	.467	.287	.157	15
16	1.000	.999	.998	.996	.989	.973	.899	.756	.566	.375	.221	16
17		1.000	.999	.998	.995	.986	.937	.827	.659	.469	.297	17
18			1.000	.999	.998	.993	.963	.883	.742	.562	.381	18
19				1.000	.999	.997	.979	.923	.812	.651	.470	19
20					1.000	.998	.988	.952	.868	.731	.559	20
21						.999	.994	.971	.911	.799	.644	21
22						1.000	.997	.983	.942	.855	.721	22
23							.999	.991	.963	.899	.787	23
24							.999	.995	.978	.932	.843	24
25							1.000	.997	.987	.955	.888	25
26								.999	.993	.972	.922	26
27								.999	.996	.983	.948	27
28								1.000	.998	.990	.966	28
29									.999	.994	.978	29
30									.999	.997	.987	30
31									1.000	.998	.992	31
32										.999	.995	32
33										1.000	.997	33
34											.999	34
35											.999	35
36											1.000	36

Source: J. Banks and R. G. Heikes, *Handbook of Tables and Graphs for the Industrial Engineer and Manager*, © 1984, pp. 34–35. Reprinted by permission of John Wiley and Sons, Inc., New York.

Table A.6 Percentage Points of The Chi-Square Distribution with ν Degrees of Freedom



ν	$\chi^2_{0.005}$	$\chi^2_{0.01}$	$\chi^2_{0.025}$	$\chi^2_{0.05}$	$\chi^2_{0.10}$
1	7.88	6.63	5.02	3.84	2.71
2	10.60	9.21	7.38	5.99	4.61
3	12.84	11.34	9.35	7.81	6.25
4	14.96	13.28	11.14	9.49	7.78
5	16.7	15.1	12.8	11.1	9.2
6	18.5	16.8	14.4	12.6	10.6
7	20.3	18.5	16.0	14.1	12.0
8	22.0	20.1	17.5	15.5	13.4
9	23.6	21.7	19.0	16.9	14.7
10	25.2	23.2	20.5	18.3	16.0
11	26.8	24.7	21.9	19.7	17.3
12	28.3	26.2	23.3	21.0	18.5
13	29.8	27.7	24.7	22.4	19.8
14	31.3	29.1	26.1	23.7	21.1
15	32.8	30.6	27.5	25.0	22.3
16	34.3	32.0	28.8	26.3	23.5
17	35.7	33.4	30.2	27.6	24.8
18	37.2	34.8	31.5	28.9	26.0
19	38.6	36.2	32.9	30.1	27.2
20	40.0	37.6	34.2	31.4	28.4
21	41.4	38.9	35.5	32.7	29.6
22	42.8	40.3	36.8	33.9	30.8
23	44.2	41.6	38.1	35.2	32.0
24	45.6	43.0	39.4	36.4	33.2
25	49.6	44.3	40.6	37.7	34.4
26	48.3	45.6	41.9	38.9	35.6
27	49.6	47.0	43.2	40.1	36.7
28	51.0	48.3	44.5	41.3	37.9
29	52.3	49.6	45.7	42.6	39.1
30	53.7	50.9	47.0	43.8	40.3
40	66.8	63.7	59.3	55.8	51.8
50	79.5	76.2	71.4	67.5	63.2
60	92.0	88.4	83.3	79.1	74.4
70	104.2	100.4	95.0	90.5	85.5
80	116.3	112.3	106.6	101.9	96.6
90	128.3	124.1	118.1	113.1	107.6
100	140.2	135.8	129.6	124.3	118.5

Source: Robert E. Shannon, *Systems Simulation: The Art and Science*, © 1975, p. 372. Reprinted by permission of Prentice Hall, Upper Saddle River, NJ.

Table A.8 Kolmogorov–Smirnov Critical Values

Degrees of Freedom (N)	$D_{0.10}$	$D_{0.05}$	$D_{0.01}$
1	0.950	0.975	0.995
2	0.776	0.842	0.929
3	0.642	0.708	0.828
4	0.564	0.624	0.733
5	0.510	0.565	0.669
6	0.470	0.521	0.618
7	0.438	0.486	0.577
8	0.411	0.457	0.543
9	0.388	0.432	0.514
10	0.368	0.410	0.490
11	0.352	0.391	0.468
12	0.338	0.375	0.450
13	0.325	0.361	0.433
14	0.314	0.349	0.418
15	0.304	0.338	0.404
16	0.295	0.328	0.392
17	0.286	0.318	0.381
18	0.278	0.309	0.371
19	0.272	0.301	0.363
20	0.264	0.294	0.356
25	0.24	0.27	0.32
30	0.22	0.24	0.29
35	0.21	0.23	0.27
Over 35	$\frac{1.22}{\sqrt{N}}$	$\frac{1.36}{\sqrt{N}}$	$\frac{1.63}{\sqrt{N}}$

Source: F. J. Massey, "The Kolmogorov–Smirnov Test for Goodness of Fit," *The Journal of the American Statistical Association*, Vol. 46. © 1951, p. 70. Adapted with permission of the American Statistical Association.

Table A.9 Maximum Likelihood Estimates of the Gamma Distribution

$1/M$	β	$1/M$	β	$1/M$	β
0.020	0.0187	2.700	1.494	10.300	5.311
0.030	0.0275	2.800	1.545	10.600	5.461
0.040	0.0360	2.900	1.596	10.900	5.611
0.050	0.0442	3.000	1.646	11.200	5.761
0.060	0.0523	3.200	1.748	11.500	5.911
0.070	0.0602	3.400	1.849	11.800	6.061
0.080	0.0679	3.600	1.950	12.100	6.211
0.090	0.0756	3.800	2.051	12.400	6.362
0.100	0.0831	4.000	2.151	12.700	6.512
0.200	0.1532	4.200	2.252	13.000	6.662
0.300	0.2178	4.400	2.353	13.300	6.812
0.400	0.2790	4.600	2.453	13.600	6.962
0.500	0.3381	4.800	2.554	13.900	7.112
0.600	0.3955	5.000	2.654	14.200	7.262
0.700	0.4517	5.200	2.755	14.500	7.412
0.800	0.5070	5.400	2.855	14.800	7.562
0.900	0.5615	5.600	2.956	15.100	7.712
1.000	0.6155	5.800	3.056	15.400	7.862
1.100	0.6690	6.000	3.156	15.700	8.013
1.200	0.7220	6.200	3.257	16.000	8.163
1.300	0.7748	6.400	3.357	16.300	8.313
1.400	0.8272	6.600	3.457	16.600	8.463
1.500	0.8794	6.800	3.558	16.900	8.613
1.600	0.9314	7.000	3.658	17.200	8.763
1.700	0.9832	7.300	3.808	17.500	8.913
1.800	1.034	7.600	3.958	17.800	9.063
1.900	1.086	7.900	4.109	18.100	9.213
2.000	1.137	8.200	4.259	18.400	9.363
2.100	1.188	8.500	4.409	18.700	9.513
2.200	1.240	8.800	4.560	19.000	9.663
2.300	1.291	9.100	4.710	19.300	9.813
2.400	1.342	9.400	4.860	19.600	9.963
2.500	1.393	9.700	5.010	20.000	10.16
2.600	1.444	10.000	5.160		

Source: S. C. Choi and R. Wette, "Maximum Likelihood Estimates of the Gamma Distribution and Their Bias," *Technometrics*, Vol. 11, No. 4, Nov. © 1969, pp. 688–9. Adapted with permission of the American Statistical Association.

Table A.7 Percentage Points of The F Distribution with $\alpha = 0.05$

ν_2 \ ν_1	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
	Degrees of Freedom for the Numerator (ν_1)																		
Degrees of Freedom for Denominator (ν_2)																			
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	245.9	248.0	249.1	250.1	251.1	252.2	253.3	254.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.55	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

Source: W. W. Hines and D. C. Montgomery, *Probability and Statistics in Engineering and Management Science*, 2d ed., © 1980, p. 599. Reprinted by permission of John Wiley & Sons, Inc., New York.

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

QUALITY AND RELIABILITY ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of statistical table is permitted.

PART-A

1	1.1	Write different definitions for quality.	02
	1.2	Define reliability.	02
	1.3	Write any four benefits of ISO 9000.	02
	1.4	List 7 QC tools.	02
	1.5	When is type-A OC curve used?	02
	1.6	Define average outgoing quality.	02
	1.7	When do you use attribute control charts?	02
	1.8	When do you use Cusum chart?	02
	1.9	List two applications of statistical experimental design in engineering design.	02
	1.10	Consider a system consisting of three components in parallel, each with a reliability of 0.65. Find the reliability of the system.	02

PART-B

2	a	List and explain the eight dimensions of quality.	08
	b	List and explain four categories of quality cost.	08
3	a	What is variability? Distinguish clearly chance causes and assignable causes of variation.	08
	b	Explain the principle of rational subgrouping.	04
	c	Write a note on process capability study.	04
		OR	

4	a	Subgroup of 5 items each are taken from a manufacturing process at regular intervals. A certain quality characteristic is measured and $\bar{X}$ and R values compared for each subgroup. After 25 subgroups, $\sum \bar{X} = 357.5 \qquad \sum R = 8.8$ Assume that all the points are within the control limits on both the charts. The specifications are 14.4 ± 0.4 i) Compute the control limits for $\bar{X}$ and R chart ii) What is the process capability? iii) Determine the percentage of rejections if any. iv) What can you conclude regarding its ability to meet the specifications?	12
	b	When do we use moving range chart? Explain.	04
5	a	The inspection results in a machine shop based on sampling size of 50 are given below.	

Serial Number	*Number of defectives*	*Serial Number*	*Number of defectives*
1	6	11	3
2	3	12	7
3	1	13	1
4	2	14	15
5	12	15	4
6	6	16	18
7	4	17	3
8	7	18	2
9	1	19	6
10	8	20	7

		i) Calculate the control limits for the P-chart using 3σ limits. ii) Plot the data and offer your comments on the behavior of the process. iii) What standard fraction defectives you recommend for the future period.	12
	b	Distinguish between P-chart and C-chart.	04
		OR	
6	a	A single sampling plan is as follows: $N = 5000 \qquad n = 100 \qquad C = 3$ i) Plot the OC curve. ii) Determine the AQL and $LTPD$ for $P_r = 10\,\%$ and $C_r = 15\,\%$ respectively. iii) What is the ATI of the above plan for 1.5 % defectives of the incoming lots?	10
	b	Write the flowchart for double sampling plan and explain with an example.	06

7	a	Discuss with examples the principles of experimental design.	04
	b	The quality control department of an automobile company is studying the effects of several factors on the surface finish of a spindle that is manufactured using a Lathe. Three operators and two machines were selected and two spindles were produced under each combination of the experimental conditions. The results are shown in table. Assume a significance level of 0.05 i) Write the model of $ANOVA$ of this reality ii) Check the significance of each of the components of the model $operator$ / $machine$: 1: 23, 27; 25, 26 — 2: 36, 34; 36, 39 — 3: 24, 38; 37, 34	12
8	a	Explain failure rate curve/bath tub curve with a neat sketch.	09
	b	A step down transformer, rectifier and filter constitutes a series system. The following are the failure rate of these components. Transformer =1.56 % $failure/$ 1000 $hours$ Rectifier =2 % $failure/$ 1000 $hours$ Filter = 1.7 % $failure/$ 1000 $hours$ The equipment is to operate for 1500 $hours$. What is the probability of survival of the system?	07

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. FastTrack Examinations July-19

Industrial Engineering and Management

QUALITY AND RELIABILITY ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of statistical and control chart tables permitted.

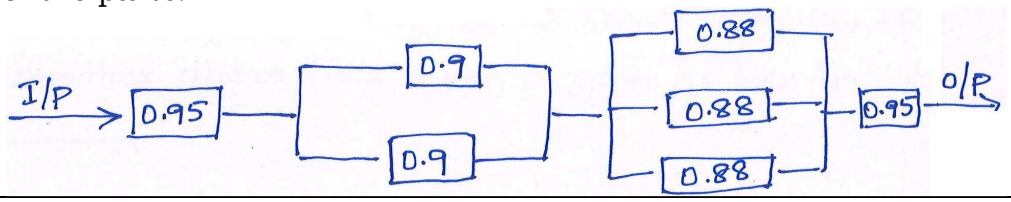
PART-A

1	1.1	The _____ are the means of translating the term 'fitness for use' into the language of technologists.	01
	1.2	The cost associated with measuring, evaluating or auditing the product comes under the costs of _____.	01
	1.3	Statethe central limit theorem	01
	1.4	Differences among materials, workers and machines is _____ type of cause of variation	01
	1.5	The usual purpose of an R-chat is to signal whether there has been a gain or loss in _____.	01
	1.6	Jars of pickles are sampled and weighed. Sample measures are plotted on control charts. The ideal weight should be precisely 11oz. which type of chart(s) would you recommend?	01
	1.7	The factors A_1, A_2, D_1, D_2, D_3 and D_4 etc. to calculate control limits depends upon _____.	01
	1.8	The local newspaper receives several complaints per day about typographic errors. Over a seven-day period, the publisher has received calls from readers reporting the following number of errors: $4, 3, 2, 6, 7, 3$ and 9. Based on these data alone, what type of control chart(s) should the publisher use?	01
	1.9	In a lot of 50 parts, 4 parts were rejected because of 12 non-conformities found in them. The proportion defectives is _____ and the average number of non-conformities per part in the lot is _____.	02
	1.10	The three parameters used for a single sampling plan are _____, _____ and _____.	02
	1.11	$MIL-STD-105E$ plans are based on _____.	01
	1.12	The uncontrollable input factors that affect the output of a process are termed as _____.	01
	1.13	In the experimental designs, the process in which every experimental unit has the same chance of being allocated to treatments s known as _____.	01

	1.14	If there are 'm' levels for a factor 'A' and 'n' levels for factor 'B' in a factorial design of experiments, then there shall be _____ combinations	01
	1.15	If 'θ' is $MTBF$, then failure rate λ is equal to _____ and reliability of the unit over the time 't' is equal to _____.	02
	1.16	Stand-by redundancy produces _____ reliability compared to parallel redundancy.	01
	1.17	Early failure of component is also called as _____.	01

PART-B

2	a	Define 'quality'. Giving two examples each, explain the five quality characteristics	06
	b	"Spending on prevention of poor quality leads to decrements in appraisal costs". Do you agree with this statement why or why not? Discuss this with the focus on components of these quality costs.	04
	c	Define and explain the type I and type II errors in the context of control charts. Are they related? How does the choice of control limits influence these two errors?	06
3	a	Subgroup of 4 items each are taken from a manufacturing process at regular intervals. A certain quality characteristic is measured and $\bar{X}$, R values are computed for each subgroup. After 25 subgroup, $\sum \overline{X} = 15350, \sum R = 411.4$. i) Compute the control limits for $\bar{X} - R$ chart ii) Assume all the points are falling within the control limits on both the charts. The specification limits are 610 ± 15. If the quality characteristic is normally distributed what % of product would fail to meet specifications iii) Any product that falls below LSL will be scrapped and above USL must be reworked. It is suggested that the process can be centered at a level so that not more than 0.1% of the product will be scrapped. What should be the aimed value of $\bar{X}'$ to make the scrap exactly 0.1%	12
	b	A cloth manufacturing company is maintaining 100% inspection of all the finished fabric. The past 8 months data from the inspection indicated that the average number of defects/unit of fabric is 8. i) Determine the 3σ control limits ii) With these control limits, what is the probability that the process is out of control when it is really not.	04
		OR	
4	a	An item is made in large lots and a sample of 200 items are drawn and given 100% inspection. The record sheet for the first 25 samples showed that a total of 75 items were defective. i) Determine the trial control limits for np-chart ii) Assuming that all the points fall within control limits, what is the process average fraction defective? iii) If the 'p' remains constant, what is the probability that the 26th sample will contain exactly 7 defectives. iv) What is the probability that 26th sample will contain 7 or more defectives.	10

	b	What is process capability? Explain the procedure of computing process cpacnbilty using average range method?	06
5		A single sampling plan is as follows: $N = 4000;\ n = 75;\ c = 2$ a) Construct the OC curve of this plan b) If AQL is 1.5%, find producers risk and if consumers risk is 10%, find $LTPD$ c) Construct the AOQ curve and determine $AOQL$ d) Find the ATI of the above plan at 1.5% defective of the incoming lot.	16
		OR	
6		The following data refers to a 'double sampling plan': $N = 50000$, $n_1 = 150, c_1 = 0, n_2 = 100, c_2 = 1$. If 1% defective lots are submitted compute a) P_a –probability of acceptance b) AOQ- Average Outgoing Quality c) ASN d) ATI- Average Total Inspection e) AFI.	16
7	a	A single factor experiment is to be designed for estimating the tensile strength of a composite material. One of the constituents of the composite material varies in the combination at 5%, 10% and 20%. Six specimens are prepared for each combination. Explain the process of data tabulation for a completely randomized design	08
	b	Explain the application of a "linear statistical model" and $ANOVA$ for the above case	08
8	a	Sketch and explain stages of 'bath tub' curve	08
	b	Calculate the reliability of the system given below from the reliability of the parts. 	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-18

Industrial Engineering and Management

QUALITY AND RELIABILITY ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

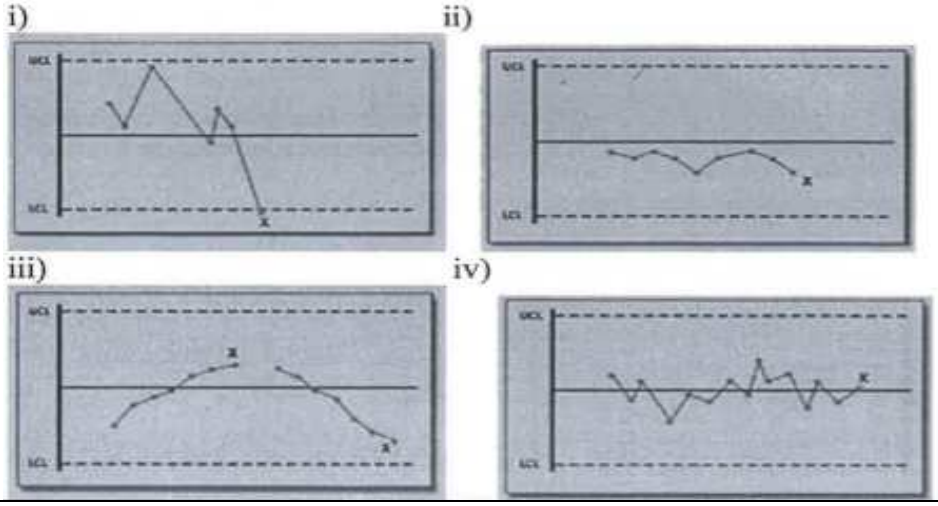
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Differentiate between quality control and quantity assurance.	02
	1.2	What are ISO 9000 and ISO14000?	02
	1.3	State any two principles of rational subgrouping.	02
	1.4	State three requisites for group control chart.	02
	1.5	Differentiate between cp and cpk.	02
	1.6	Draw the ideal OC curve.	02
	1.7	What is rectifying sampling plan?	02
	1.8	State the principles of experimental design.	02
	1.9	Differentiate between $MTTR$ and $MTBF$.	02
	1.10	What is balanced design? Give an example.	02

PART-B

2	a	How are external failure costs different from internal failure costs?	04
	b	Provide a list of chance and assignable causes.	04
	c	Identify the patterns that exist in the following control charts. What do you think are the reasons for such patterns? 	08

3 a A stable normally distributed process has the specifications 3.5 ± 0.03. Given $\mu = 3.51$ and standard deviation= 0.016, determine Cpand Cpk. What conclusions can you draw about the capability of the process? 05

b The following are the X bar $-R$ values for 20 subgroups with 5 readings each. The specification limits are 40.37 ± 0.10. Establish the control limits for X bar and R charts. State the control limits for which the process will be able to meet the specifications.

Sub group No	*X bar*	*R*	*Sub group No*	*X bar*	*R*
1	34	4	11	35.8	4
2	31.6	4	12	38.4	4
3	30.8	2	13	34	14
4	33	3	14	35	4
5	35	5	15	33.8	7
6	33.2	2	16	31.6	5
7	33	5	17	33	5
8	32.6	13	18	28.2	3
9	33.8	19	19	31.8	9
10	37.8	6	20	35.6	6

11

OR

4 a A company manufactures valves for industrial use. Ten samples if 15 valves each were taken from the production line and tested. The results are reported below:

Number	*Sample defective*	*Number*	*Sample defective*
1	3	6	2
2	1	7	0
3	0	8	3
4	0	9	1
5	0	10	0

i) Compute the fraction defective for each sample
ii) Construct a p-chart for this process using control limits that include 95.5 percent of the sample means.
iii) What comments can you make about the results? Is the process in-control? What action should be taken? 10

b What are the advantages of:
i) Control charts for attributes?
ii) Control charts for variables? 06

5 a Consider the following data of a double sampling plan. Construct Operating Characteristic curve, AOQ curve. Also determine ATI and ASN.
$N = 5000$
$n1 = 150 \quad n2 = 200$
$c1 = 3 \quad c2 = 8$
$r1 = 6 \quad r2 = 9$ 12

	b	How Type-A and Type-B OC curves are different? When are they used?	04
		OR	
6	a	What are the assumptions that you make while designing a single sampling plan?	05
	b	Compare and contrast single, double and triple sampling plans.	05
	c	Define the following: i) AQL ii) $LTPD$ iii) Producer's risk.	06
7	a	Three different washing solutions are being compared to study their effectiveness in retarding bacteria growth in 5-gallon milk containers. The analysis is done in a laboratory, and only three trials can be run on any day. As days could represent a potential source of variability, the experimenter decides to use a randomized block design. Observations are taken for four days, and the data are shown here. $Days$ / $Solution$ / 1 / 2 / 3 / 4 1 / 13 / 22 / 18 / 39 2 / 16 / 24 / 17 / 44 3 / 5 / 4 / 1 / 22 Construct $ANOVA$ table and determine if different washing solutions have significant effect in retarding bacteria growth.	10
	b	How is completely randomized design different from a randomized block design?	06
8	a	Consider a system with 5 kinds of component in series, with the following reliabilities • Component 1: 0.95, • Component 2: 0.95, • Component 3: 0.70, • Component 4: 0.75, • Component 5: 0.90 Because of the low reliability of the third and fourth components, they are replicated (parallel to each other); the system contains 3 of the third component and 2 of the fourth component. Find the reliability of the system.	08
	b	Prove that for a series system, the failure rates of the system is the sum of the failure rates of the individual component comprising the system.	08

Digital Metrology

V Semester SEE Question Papers

Course codes: 21IM52

- Apr 2024 (21IM52)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

V Semester B. E. Examinations Apr-2024

Industrial Engineering and Management

DIGITAL METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	List any two static and dynamic errors in the context of measurements?	02	1	1
	1.2	Differentiate between resistive and capacitive transducers in the context of sensing technology.	02	2	2
	1.3	Explain the roles of A/D and D/A converters in a data acquisition system	02	2	2
	1.4	List the applications of a Capacitive Transducer in Proximity Sensing.	02	2	2
	1.5	Differentiate between hole and shaft basis systems in engineering fits.	02	2	3
	1.6	Define the concept of limits, and fits	02	1	3
	1.7	Describe the Working Principle of an Autocollimator.	02	2	2
	1.8	Enumerate on the concept: i) Interchangeability ii) selective assembly	02	1	2
	1.9	What are the advantages of using interferometry in metrology?	02	2	4
	1.10	What is the application of multiple beam interferometer?	02	2	4

PART-B

2	a	Enumerate briefly on different types of Standards and Calibration. Explain the role of calibration in ensuring accuracy and reliability in measurements, and how do standards contribute to the traceability of measurements in various industries.	08	3	1
	b	State the advantages of electrical transducers elements over mechanical transducers.	08	2	1
3	a	Discuss the specifications of sensors and their importance in metrology. Elucidate various types of sensors employed in measurement procedures.	08	3	2
	b	Explain the Velocity and motion sensors and their application in metrology for the detection of infrared radiation and temperature variations.	08	2	2
		OR			
4	a	Describe the factors influencing the selection of sensors for a particular application. How can one ensure the compatibility between sensors and data acquisition systems?	08	3	2

	b	Compare and contrast position sensors and proximity sensors, highlighting their working principles and applications.	08	3	2
5	a	Design the 'General' type 'GO' and '$NOGO$' gauges for the components having $50H\,8\,c\,8$ fit. Given the following with usual notations. i) Upper deviation for Shaft $= (95 + 0.8D)$ microns ii) $50mm$ falls in the diameter step of $50 - 80mm$ iii) $IT8 = 25i$ Take wear allowance as 20% of the gauge Tolerance. Also mention the type of fit. Represent the dimensions graphically.	08	4	4
	b	What is the difference between unilateral and bilateral tolerances? why unilateral is preferred over bilateral tolerance?	08	2	3
		OR			
6	a	Define Fit. Explain briefly the various types of fits with examples.	08	2	3
	b	In a limit system the following limits are specified to give a clearance fit between a shaft and a hole: $Shaft\ 30\ ^{-0.005}_{-0.018}\ mm\phi$ $Hole\ 30\ ^{+0.020}_{-0.000}\ mm\phi$ Determine: i) Basic size ii) shaft and hole tolerances iii) the shaft and hole limits iv) the maximum and minimum clearance.	08	4	4
7	a	With the help of a neat sketch briefly explain the working of a tool maker's microscope.	08	3	2
	b	With the help of a neat sketch briefly explain the method of determining the chordal thickness of a gear tooth using gear tooth vernier caliper.	08	2	2
		OR			
8	a	With the help of a neat sketch briefly explain the working of a floating carriage micrometer.	08	3	2
	b	With the help of a neat sketch, explain the terminologies used to represent the surface roughness.	08	3	2
9	a	List and explain the components of a Coordinate Measuring Machines (CMM) system and how they work together to measure the geometric characteristics of objects	08	2	4
	b	Explain the principle of laser metrology and its significance in precision measurements. Discuss the advantages of using lasers over traditional measurement methods. Provide examples of applications where laser metrology is commonly used.	08	2	3
		OR			
10	a	With the help of sketches, explain any two types of Coordinate Measuring Machines (CMM)	08	3	4
	b	Describe the working principle of a Michelson Interferometer in detail. How is it used for interferometric measurement of angles? Discuss the advantages and limitations of using a Michelson Interferometer for angle measurements compared to other methods.	08	2	3

Operations Research

V Semester SEE Question Papers

Course codes: 21IM54, 12IM53, 12IM53 / 10IM54

- Apr 2024 (21IM54)
- Fast Track Jul 2018 (12IM53)
- Nov/Dec 2017 (12IM53)
- Fast Track Jul 2017 (12IM53)
- Nov/Dec 2016 (12IM53)
- Fast Track Jul 2016 (12IM53 / 10IM54)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

V Semester B. E. Examinations Apr-2024

Industrial Engineering and Management

OPERATIONS RESEARCH

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of Statistical table is permitted.

PART-A

			M	BT	CO
1	1.1	For Maximize $Z = 3x_1 + 4x_2$ subject to the constraints: $x_1 + x_2 \leq 4, x_1 \geq 0\ x_2 \geq 0$, what will be the value of Z.	01	1	1
	1.2	Decision variables in optimization problems represent the _____ factors or parameters whose values are to be determined in order to optimize a certain objective.	01	1	1
	1.3	The simplex method efficiently identifies the optimal solution by traversing through the _____ of the feasible region.	01	2	2
	1.4	The primary objective of the two-phase simplex method is to find a feasible solution before transitioning to the _____ phase.	01	2	2
	1.5	Sensitivity analysis in linear programming examines the impact of changes in _____ coefficients, constraints, or other parameters on the optimal solution..	01	1	1
	1.6	Sensitivity analysis in linear programming assesses the impact of changes in _____ on the optimal solution and associated variables.	01	1	1
	1.7	In linear programming, the concept of duality establishes a relationship between a primal problem and its corresponding _____ problem.	01	2	2
	1.8	In the dual simplex method, each iteration involves pivoting on a non-basic variable to improve feasibility and bring the _____ variable into the basis.	01	2	2
	1.9	In the transportation problem, the objective is to minimize the total _____ of transporting goods from suppliers to destinations.	01	1	1
	1.10	The transportation problem can be formulated as a _____ programming model.	01	1	1
	1.11	The assignment problem can be solved using _____ algorithms such as the Hungarian algorithm.	01	3	3
	1.12	The Travelling Salesman Problem can be formulated as a _____ optimization problem.	01	3	3
	1.13	In CPM (Critical Path Method), the _____ path represents the longest sequence of activities that determines the minimum duration of the project.	01	4	4
	1.14	In graphical sensitivity analysis, the _____ is the line representing the objective function value.	01	1	1

	1.15	*PERT* (Program Evaluation and Review Technique) is a project management tool used to estimate the _____ time for completing each task.	01	2	2
	1.16	In the context of project management, floats represent the _____ time available for an activity without delaying the project's completion.	01	2	2
	1.17	In a zero-sum game, the total gains of the participants are equal to _____.	01	3	3
	1.18	In game theory, the dominance rule states that if one strategy of a player is always better than another strategy, regardless of the other player's actions, then the dominated strategy can be _____ from consideration.	01	3	3
	1.19	Game theory is a branch of mathematics and economics that studies _____ interactions where the outcome of one participant's decision depends on the decisions of others.	01	1	1
	1.20	The graphical method in game theory is a visual tool used to analyze and illustrate the possible _____ of strategies for each player in a game.	01	1	1

PART-B

2	a	Explain the seven step model building process with an example for each step.	06	2	2
	b	Solve the following Linear Programming Problem by graphical method: Max $Z = 100x_1 + 40x_2$ Subject to $5x_1 + 2x_2 \leq 1000; 3x_1 + 2x_2 \leq 900; x_1 + 2x_2 \leq 500$ and $x_1, x_2 \geq 0$.	10	2	2
3	a	Discuss the following with respect to linear program: i) Slack variable ii) Surplus variable	04	3	3
	b	Solve the following Linear Programming Problem by simplex method: $Maximize\ Z = 3x_1 + 5x_2 + 4x_3$ Subject to the constraints $2x_1 + 3x_2 \leq 8$ $2x_2 + 5x_3 \leq 10$ $3x_1 + 2x_2 + 4x_3 \leq 15\ and\ x_1, x_2, x_3 \geq 0.$	12	3	3
		OR			
4	a	List and explain the characteristics of duality.	06	2	2
	b	Use dual simplex method to solve the following linear programming problem: Minimize $Z = 2x_1 + x_2$ Subject to $3x_1 + x_2 \geq 3$ $4x_1 + 3x_2 \geq 6$ $x_1 + 2x_2 \geq 3\ and\ x_1 \geq 0, x_2 \geq 0.$	10	4	4

5	a	Find the initial basic feasible solution for the following transportation problem, using North-West Corner Rule method. $Destination$ $D1$ $D2$ $D3$ $Supply$ $S1$ 3 8 5 7 $S2$ 4 4 2 8 $Sources$ $S3$ 6 5 8 10 $S4$ 2 6 3 15 $Demand$ 8 10 22	04	3	3
	b	A potato chip manufacturer has three plants and four warehouses. Transportation cost for shipping from plants to warehouses, the plant availability and warehouses requirements as follows: $Plants$ / $Warehouses$: $W1$ $W2$ $W3$ $W4$ / $Pant\ Availability\ (quintals)$ F_1 7 4 3 5 — 235 F_2 6 8 7 4 — 280 F_3 5 6 9 10 — 110 $Requirements(quintals)$ 125 160 110 230 Find optimum shipping schedule.	12	3	3
		OR			
6	a	Solve the following assignment problem shown in the following Table using Hungarian method. The matrix entries are processing time of each job to each machine in hours. J/M I II III IV V 1 9 22 58 11 19 2 43 78 72 50 63 3 41 28 91 37 45 4 74 42 27 49 39 5 36 11 57 22 25	08	2	2
	b	A raveling salesman has to visit five cities. He wishes to start from a particular city, visit each city once and then return to his starting point. The travelling cost of each city from a particular city is given below: A B C D E A ∞ 2 5 7 1 B 6 ∞ 3 8 2 C 8 7 ∞ 4 7 D 12 4 6 ∞ 5 E 1 3 2 8 ∞ What is the sequence of visit of the salesman, so that the cost is minimum?	08	2	2
7	a	Distinguish between CPM and $PERT$ of network analysis based in orientation, evolution, model and estimates.	08	3	3
	b	Draw the network diagram, identify critical path and total project duration for the following: $Activity$ $1-2$ $1-3$ $1-4$ $2-5$ $3-7$ $4-6$ $5-7$ $Duration$ 10 8 9 8 16 7 7 $Activity$ $5-8$ $6-7$ $6-9$ $7-10$ $8-10$ $9-10$ $Duration$ 6 7 5 12 13 15	08	3	3
		OR			

8	a	Distinguish between CPM and $PERT$ of network analysis	06	2	2
	b	A project schedule has the following characteristics as shown in following table:			

$Activity$	$Name$	$Time$	$Activity$	$Name$	$Time$
1 – 2	A	4	5 – 6	G	4
1 – 3	B	1	5 – 7	H	8
2 – 4	C	1	6 – 8	I	1
3 – 4	D	1	7 – 8	J	2
3 – 5	E	6	8 – 10	K	5
4 – 9	F	3	9 – 10	L	7

i) Construct a network diagram.
ii) Find the critical path.

Compute early start & finish, late start & finish for each activity. | 12 | 4 | 4

9	a	A company management and the labour union are negotiating a new three-year settlement. Each of these has 4 strategies i) Hard and aggressive bargaining ii) Reasoning and logical approach iii) Legalistic strategy iv) Conciliatory approach. The costs to the company are given for every pair of strategy choice.			

		$Company\ Strategies$			
		I	II	III	IV
$Union\ Strategies$	I	20	15	12	35
	II	25	14	8	10
	III	40	2	10	5
	IV	−5	4	11	0

Find the saddle point and value of the game. | 06 | 3 | 3

	b	Use the graphical method for solving the following game and find the value of the game.			

		$Player\ B$			
		$B1$	$B2$	$B3$	$B4$
$Player\ A$	$A1$	2	2	3	−2
	$A2$	4	3	2	6

10 | 4 | 4

OR

10	a	Use the principle of dominance to find the optimal strategy for both the players and also value of the game for the following pay – off matrix:			

		$Player\ B$			
		I	II	III	IV
$Player\ A$	1	4	2	3	6
	2	3	4	7	5
	3	6	3	5	4

08 | 2 | 2

	b	Use graphical method in solving the following game and find the value of the game.			

		$Player\ B$	
		$B1$	$B2$
$Player\ A$	$A1$	−2	4
	$A2$	8	3
	$A3$	9	0

08 | 3 | 3

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Fast Track Examinations July-18

Industrial Engineering and Management

OPERATIONS RESEARCH

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
4. Answer FIVE full questions from Part B.
5. Use of statistical tables permitted.

PART-A

1	1.1	Any two isoprofit or isocost lines for a given linear programming problem are _____ to each other.	01
	1.2	Distinguish between Convex and Non Convex solution spaces.	02
	1.3	State all the assumptions that form the basis for an LPP model.	02
	1.4	The solution to a transportation problem with –m-rows (supplies) and n-columns (destinations) is feasible if number of positive allocations are _____.	01
	1.5	Mention any two differences between transportation problem and assignment problem.	02
	1.6	Distinguish between balking, reneging and jockeying.	02
	1.7	Explain two person zero sum game.	02
	1.8	The dual of a dual is a _____.	01
	1.9	Give an example for an infinite server model.	01
	1.10	When 2 or more moves are considered by a player for optimal strategy pay offs, then the game has a _____ strategy.	01
	1.11	List any 4 characteristics of queuing model.	02
	1.12	Activities K, M and S immediately follow activity H, and their latest start times are $14, 18$, and 11. The latest finish time for activity H is _____.	01
	1.13	For a particular project, eight activities are to be carried out. Their relationships with other activities and expected durations are mentioned in the table below.	

$Activity$	$Predecessors$	$Duration\ (days)$
a	–	3
b	a	4
c	a	5
d	a	4
e	d	2
f	b	9
g	c, e	6
h	f, g	2

Draw the project network diagram. 02

PART-B

2 a Define Operations Research and briefly discuss its industrial applications. 08

b A company is planning to determine its product mix out of three different products; $P1, P2$ and $P3$. The monthly sales of the product $P1$ is limited to a maximum of 500 units. For every two units $P2$ produced, there will be one unit of by-product which can be sold at the rate of $Rs.20$ per unit. The highest monthly demand for this by-product is 200 units. The contribution per unit of the products $P1, P2$, and $P3$ are $Rs.50, Rs.70$ and $Rs.60$, respectively. The processing requirements of these products are shown in the table.

	Hours / Unit			
Process	$P1$	$P2$	$P3$	*Available Hours*
I	3	5	2	1000
II	4	–	3	700
III	4	3	2	1300

Formulate a linear programming model of this problem. 08

OR

3 a Define "Degeneracy"? How do you solve a Degenerate $L.P.P$? 06

b The owner of Metro Sports wishes to determine how many advertisements to place in the selected three monthly magazines A, B, C. His objective is to advertise in such a way that total exposure to principal buyers to expensive sports goods is maximized. The percentage of readers for each magazine is known. Exposure to any particular magazine is the number of advertisements placed multiplied by the number of principal buyers. The following data may be used:

	Magazines		
	A	B	C
Readers	1 *Lakh*	0.6 *lakh*	0.4 *lakh*
Principal buyers	10%	15%	7%
Cost per advertisements $(Rs.)$	5,000	4,500	4,250

The budget amount is at the most $Rs.1$ lakh for the advertisements. The owner has already decided that magazine A should have no more than six advertisements and that B and C each at least two advertisements. Formulate this problem as an LP model to determine the number of advertisements to be placed in each magazine. 10

4 a Consider the following LP Model:

Maximize $Z = 5x_1 + 2x_2 + 3x_3$

Subjected to $x_1 + 5x_2 + 2x_3 \leq b1$

$x_1 - 5x_2 - 6x_3 \leq b2$

$x_1, x_2, x_3 \geq 0$

The following optimal table corresponds to specific values of $b1$ and $b2$:

Basic	x_1	x_2	x_3	x_4	x_5	*Solution*
Z	0	a	7	d	e	150
x_1	1	b	2	1	0	30
x_5	0	c	−8	−1	1	10

Determine the following:

i) The RHS values, $b1$ and $b2$.

ii) The elements a, b, c, d, e.

iii) The optimal dual solution. 08

b Solve the following LP problem using simplex method.

Maximize $Z = 10x_1 + 15x_2 + 20x_3$

Subject to $2x_1 + 4x_2 + 6x_3 \leq 24$

$3x_1 + 9x_2 + 6x_3 \leq 30$

x_1, x_2 and $x_3 \leq 0$ 08

OR

5 a Apply the dual simplex method to solve the following problem:

Minimize $Z = 10x_1 + 6x_2 + 2x_3$

Subject to $-x_1 + x_2 + x_3 \geq 1$

$3x_1 + x_2 - x_3 \geq 2$

$x_1, x_2, x_3 \geq 0$ 10

b Differentiate between slack and surplus variables with suitable examples. 06

6 Consider 3 sources and 4 destinations as shown below. The cell entries represent the cost of transportation per unit.

		Destination				
		1	2	3	4	*Supply*
Source	1	3	1	7	4	300
	2	2	6	5	9	400
	3	8	3	3	2	500
Demand		250	350	400	200	

Obtain the initial basic feasible solution using the northwest –corner cell method and then optimize the solution using $U - V$ method. 16

OR

7 a What is meant by unbalanced transportation problem? Explain the method for solving such a problem 04

b A company has received a contract to supply gravel for three new construction projects located in towns A, B and C. Construction engineers have estimated the required amounts of gravel which will be needed at these construction projects:

Project location	A	B	C
Weekly requirement (truckloads)	72	102	41

The company has three gravel pits located in towns X, Y and Z. The gravel required by the construction projects can be three pits. The amount of gravel which can be supplied by each pit is as follows:

Plant	X	Y	Z
Amount available (truckloads)	76	82	77

The company has computed the delivery cost from each pit to each project site. These costs (in Rs.) are shown in the following table:

	A	B	C
X	4	8	8
Y	16	24	16
Z	8	16	35

		Schedule the shipment from each pit to each project in such a manner so as to minimize the total transportation cost within the constraints imposed by the capabilities and project requirements. Also find the minimum cost.	12
8	a	Formulate an assignment problem as a square matrix.	04
	b	A college is having degree programme for which the effective semester time available is very less and the programme requires field work. Hence, a few hours can be saved from the total number of class hours, and can be utilized for the field work. Based on past experience the college has estimated the number of hours required to teach each subject by each faculty. The course in its present semester has 5 subjects and the college has considered 6 existing faculty members to teach these courses. The objective is to assign the best 5 teachers out of these 6 faculty members to teach 5 different subjects so that the number of class hours required is minimized. The data is summarized below. Solve this assignment problem optimally.	

Subject				
1	2	3	4	5
30	39	31	38	40
43	37	32	35	38
34	41	33	41	34
39	36	43	32	36
32	49	35	40	37
36	42	35	44	42

12

OR

9	a	The arrival rate of the customer at a banking counter follows Poisson distribution with a mean of 45 per hour. The service rate of the counter clerk also follows Poisson distribution with a mean of 60 per hour. i) What is the probability of having 0 customer in the system (po)? ii) What is the probability of having 5 customers in the system $(p5)$? iii) What is the probability of having 10 customers in the system $(p10)$? iv) Find Ls, Lq, Ws, Wq.	09
	b	At a central warehouse, vehicles arrive at the rate of 18 per hour and the arrival rate follows Poisson distribution. The unloading time of the vehicles follows exponential distribution and the unloading rate is 6 vehicles per hour. There are 4 unloading crews. Find the following: i) po and $p3$ ii) Lq, Ls, Wq and Ws.	07
10	a	Game theory provides a systematic quantitative approach for analyzing competitive situations in which the competitors make use of logical processes and techniques in order to determine an optimal strategy for winning: Comment.	06

b Using the Dominance property reduce the matrix and solve it.

Player 'A' \ Players 'B'	I	II	III	IV
I	3	2	4	0
II	3	4	2	4
III	4	2	4	0
IV	0	4	0	8

10

OR

11 a State the circumstances where CPM is a better techniques of project management than $PERT$. 04

b A project consists of 10 activities, lettered A through J. For each activity, the deterministic time estimate in weeks is shown in the following table.

Activity	Precedence	Time	Activity	Precedence	Time
A	–	2	F	E, D	3
B	A	4	G	D, E	4
C	B	3	H	G	4
D	A	5	I	C, F	3
E	–	6	J	I, H	1

i) List all of the paths through the network.
ii) What is the duration of each of the paths?
iii) What is the critical path?
iv) What is the second-most critical path?
v) Calculate the ES, EF, LS and LF times for each activity. 12

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-17

Industrial Engineering and Management

OPERATIONS RESEARCH

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

13. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
14. Answer FIVE full questions from Part B.

PART-A

1	1.1	What is an Iconic and physical model in Operations Research?	02
	1.2	Mention three basic requirements for formulating a linear programming problem.	02
	1.3	Out of the algorithms for initial basic feasible solution in a transportation model which one yields the nearer optimal solution and why?	02
	1.4	State the condition for applying initial basic feasible solution to any transportation problem.	02
	1.5	How will you convert an unbalanced assignment problem to a balanced assignment problem, justify with appropriate example.	02
	1.6	State the objective of travelling salesman problem.	02
	1.7	Given, expected duration of project is 47 days and variance is 9 days. What is the probability of not completing the project in 50 days?	

z	0.33	1.0
$Area$	0.1293	0.3413

02

	1.8	What do you mean by ideal activity plan?	02
	1.9	List the basic elements of queuing system.	02
	1.10	What is two person Zero-sum game?	02

PART-B

2	a	Explain the different phases of operations research.	06
	b	The manager of an oil refinery must decide on the optimal mix of two possible blending processes of which the inputs and outputs per production run are as follows;	

$Process$	$Inputs(Units)$		$Outputs(units)$	
	$Crude1$	$Crude\ 2$	$Petrol\ (superior)$	$Petrol\ (ordinary)$
A	10	06	10	16
B	12	15	12	12

The availability of two varieties of crude is limited to the extent of 400 units and 450 units respectively per day. The market demand indicates that at least 200 units and 240 units of superior and ordinary quality petrol are required every day. The profitability analysis indicates that process A contributes Rs. 480 per run while the process B contributes Rs. 400 per run. The manager is interested in determining an optimal product mix for maximizing the company's profit. 10

OR

3 a Solve the following Linear Programming problem by graphical method:

$Z_{max} = 450x_1 + 480\,x_2$

Subject to constraints,

$2x_1 + 3\,x_2 \leq 1500$

$3x_1 + 2\,x_2 \leq 1500$

$x_1 \leq 400$

$x_2 \leq 400$

x_1 and $x_2 \geq 0$ 08

b Solve the following linear programming problem by simplex method.

Maximize $Z = x_1 + 2x_2 - x_3$ subject to

$2x_1 + x_2 + x_3 \leq 14$

$4x_1 + 2x_2 + 3x_3 \leq 28$

$2x_1 + 5x_2 + 5x_3 \leq 30$

$x_1, x_2, x_3\ all \geq 0$ 08

4 a Find the initial basic feasible solution to the following Transportation Problem using Vogel's Approximation Method(VAM):

		Destination Centers				*Supply*
		D_1	D_2	D_3	D_4	
Plants	P_1	2	3	11	7	6
	P_2	1	0	6	1	1
	P_3	5	8	15	9	10
Demand		7	5	3	2	

06

b The Hardrock Concrete Company has plants in three locations and is currently working on three major construction projects, each located at a different site. The shipping cost per truckload of concrete, daily plant capacities and daily project requirements are provided in the accompanying table.

		A	B	C	*Plant Capacity*
Plant	P_1	10	4	11	70
	P_2	12	5	8	50
	P_3	9	7	6	30
Project Requirements		40	50	60	

i) Formulate an initial basic feasible solution to Hardrock's transportation problem using VAM.

ii) Then solve using the $MODI$ method.

iii) Was the initial solution optimal? 10

OR

5

A company has four manufacturing plants and five warehouses. Each plant manufactures the same product, which is sold at different prices in each warehouse area. The cost of manufacturing and cost of raw materials are different in each plant due to various factors. The capacities of the plant are also different. The relevant data is in the following table.

Item	*Plant*			
	1	2	3	4
Manufacturing cost (Rs) per unit	12	10	8	8
Raw material cost (Rs) per unit	8	7	7	5
Capacity per unit time	100	200	120	80

The company has five warehouses. The sales price, transportation costs and demands are given in the following table:

Warehouse	*Transportation Costs (Rs) per unit*				*Sales price*	*Demand per unit(Rs)*
	1	2	3	4		
A	4	7	4	3	30	80
B	8	9	7	8	32	120
C	2	7	6	10	28	150
D	10	7	5	8	34	70
E	2	5	8	9	30	90

a) Formulate this problem as a transportation problem in order to maximize profit.

b) Find *IBFs* using *VAM*.

c) Test for optimality and find the optimal solution. 16

6 a

A machine tool company decides to make four sub-assemblies through four contractors. Each contractor is to receive only one sub-assembly. The cost of each sub-assembly is determined by the bids submitted by each contractor and is shown in the table in hundreds of rupees:

		Contractors			
		1	2	3	4
Sub − assemblies	1	15	13	14	17
	2	11	12	15	13
	3	13	12	10	11
	4	15	17	14	16

i) Formulate the mathematical model for the problem.

ii) Show that the assignment model is special case of the transportation model. 08

b

In a job shop operation, four jobs may be performed on any of four machines. The hours required for each job on each machine are presented in the following table. The plant supervisor would like to assign jobs so that total time is minimized. Use the assignment method to find the best solution.

		Machines			
		W	*X*	*Y*	*Z*
Job	*A*	10	14	16	13
	B	12	13	15	12
	C	9	12	12	11
	D	14	16	18	16

08

OR

7 a In the modification of a plant layout of a factory four new machines M_1, M_2, M_3 and M_4 are to be installed in a machine shop. There are five vacant places A, B, C, D and E available. Because of limited space, machine M_2 cannot be placed at A. the cost of placing of machine I at place j (in rupees) is shown below.

		Location				
		A	B	C	D	E
Machine	M_1	9	11	15	10	11
	M_2	12	9	-	10	9
	M_3	-	11	14	11	7
	M_4	14	8	12	7	8

08

b A travelling salesman has to visit five cities. He wishes to start from a particular city, visit each city once and then return to his starting point. The travelling cost of each city from a particular city is given below:

	A	B	C	D	E
A	∞	2	5	7	1
B	6	∞	3	8	2
C	8	7	∞	4	7
D	12	4	6	∞	5
E	1	3	2	8	∞

What is the sequence of visit of the salesman, so that the cost is minimum? 08

8 a Differentiate between critical path method and project evaluation review technique with respect to the following characteristics:

i) Type of project
ii) Time estimates
iii) Nature of time estimates. 06

b Characteristics of a project schedule are given below:

Activity	Preceding Activity	Optimistic Time(Days)	Most Likely Time(Days)	Pessimistic Time(Days)
A	-	6	9	12
B	-	3	5	9
C	-	10	14	18
D	A	7	10	13
E	B	3	4	8
F	C	4	10	12
G	A	8	11	14
H	C	5	10	15
I	D, E, F	3	4	5
J	D, E, F	11	15	17
K	G, I	7	9	12
L	H	4	8	10
M	J, L	6	7	9

i) Construct network diagram
ii) Calculate the expected time and variance.
iii) What is the expected project length and critical path? 10

OR

9 The following table gives details of a project:

$Activity$	$Immediate\ Predecessor$	$Normal$		$Crash$	
		$Time(weeks)$	$Cost(Rs)$	$Time(weeks)$	$Cost(Rs)$
A	-	10	20	7	30
B	-	8	15	6	20
C	B	5	8	4	14
D	B	6	11	4	15
E	B	8	9	5	15
F	E	5	5	4	8
G	A, D, C	12	3	8	4

The indirect cost is Rs.400 per day. Find the optimum duration and the associated minimum project cost. 16

10 a List and explain different types of queuing models. 06

b A self-help canteen employs one cashier at its counter. 8 customers arrive in every 10 minutes on an average. The cashier can serve at the rate of 1 customer per minute. Assume Poisson's distribution for arrival and exponential distribution for service patterns. Determine:

i) Average number of customers in the system
ii) Average queue length
iii) Average time a customer spends in the system. 10

OR

11 a Calculate the saddle point for the following game:

		$Player\ B$	
		P	Q
$Player\ A$	L	-3	3
	M	-5	2
	N	6	2

06

b Solve the following game and find optimal strategy for player B by graphical method.

		$Player\ B$		
		1	2	3
$Player\ A$	1	-2	6	3
	2	2	-3	4
	3	1	2	2

10

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

OPERATIONS RESEARCH

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

5. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
6. Answer FIVE full questions from Part B.
7. Use of statistical tables permitted.

PART-A

1	1.1	A constraint in an LP model restricts the use of _____ variable.	01
	1.2	The objective function for a LP model is $3x_1 + 2x_2$, if $x_1 = 20$ and $x_2 = 30$, what is the value of the objective function?	01
	1.3	Before formulating a formal LP model, it is better to verbally _____ decision variables.	01
	1.4	A feasible solution to an LP problem must satisfy all of the problem's _____ simultaneously.	01
	1.5	The occurrence of degeneracy while solving a transportation problem means that the solution so obtained is not _____.	01
	1.6	One disadvantage of using North-west corner rule to find initial solution to the transportation problem is that it does not take into account _____ of transportation.	01
	1.7	The degeneracy in the transportation problem indicates that the optimal solution exist.	01
	1.8	In a transportation problem, there are 4 rows and 5 columns, the number of non-allocated cells are _____.	01
	1.9	The method used for solving an assignment problem is called _____ method.	01
	1.10	To proceed with the $MODI$ algorithm for solving an assignment problem, the number of dummy allocations need to be added are _____.	01
	1.11	In a 6×6 assignment problem, the number of allocated basic cells are _____.	01
	1.12	During the process of assignment when we do not find a single zero in any row or column, then it is an indication of _____ optimal solution.	01
	1.13	The term _____ is used to refer to nodes in a $PERT$ using AOA.	01
	1.14	Of all paths through the network, the critical path has the _____ expected time.	01
	1.15	The $PERT$/Cost model assumes that the _____ are uniformly distributed over the life of the activity.	01
	1.16	In the CPM time cost trade off function, cost _____ linearly as time increases.	01

	1.17	Service mechanism in a queuing system is characterized by _____.	01
	1.18	The calling population is assumed to be infinite when, arrivals are _____ of each other.	01
	1.19	The size of the payoff matrix of a game can be reduced by using the principle of _____.	01
	1.20	When the sum of gains of one player is equal to the sum of losses to another player in a game, this situation is known as _____.	01

PART-B

2	a	Operations research is an ongoing process. Comment on this statement with examples.	06
	b	The owner of Metro Sports wishes to determine how many advertisements to place in the selected three monthly magazines A, B and C. His objectives is to advertise in such a way that total exposure to principal buyers to expensive sports goods is maximized. The percentage of readers for each magazine is known. Exposure to any particular magazine is the number of advertisements placed multiplied by the number of principal buyers. The following data may be used	

	Magazines		
	A	B	C
Readers	1 *lakh*	0.6 *lakh*	0.4 *lakh*
Principal buyers	10%	15%	07%
Cost per advertisements (rs)	5,000	4,500	4,250

		The budget amount is at the most Rs.1 lakh for the advertisements. The owner has already decided that magazine A should have no more than six advertisements and that B and C each at least two advertisements. Formulate this problem as an LP model to determine the number of advertisements to be placed in each magazine.	10
		OR	
3	a	Solve the following LPP by graphical method: $Z_{max} = 60x_1 + 40x_2$ Subject to constraints $x_1 < 25, x_2 \leq 35, 2x_1 + x_2 \leq 60, x_1 \& x_2 \geq 0$	08
	b	Solve the following linear programming problem by simplex method: Maximize $Z = 5x_1 + x_2$ Subject to $5x_1 + 2x_2 \leq 20$ $x_1 \geq 3, x_2 \leq 5, x_1, x_2, all \geq 0$	08
4	a	Find the initial solution to the following transportation problem using Vogel's Approximation Method (VAM)	

		Destination				*Supply*
		D_1	D_2	D_3	D_4	
Factories	F_1	3	3	4	1	100
	F_2	4	2	4	2	125
	F_3	1	5	3	2	75
Demand		120	80	75	25	

			06

	b	Solve the transportation problem for minimization, when the unit transportation costs, demands and supplies are as given $Destination$ / $Origin$ / $Supply$ / $Demand$ O_1: 6, 1, 9, 3 — 70 O_2: 11, 5, 2, 8 — 55 O_3: 10, 12, 4, 7 — 70 $Demand$: 85, 35, 50, 45	10
		OR	
5	a	There are three factories A, B and C which supply goods to four dealers $D_1, D_2, D_3, \& D_4$. The production capacities of these factories are $1000, 700 \& 900$ units per month respectively. The requirements from the dealers are $900, 800, 500 \& 400$ units per month respectively.The per unit return (excluding transportation cost) are $Rs. 8, 7, \& 9$ at the three factories. The following table gives the unit transportation costs from the factories. The following table gives the unit transportation costs from the factories to the dealers. $Destination$ / $Factories$ A: 2, 2, 2, 4 B: 3, 5, 3, 2 C: 4, 3, 2, 1 Determine the optimum solution to maximize the total returns.	16
6	a	Differentiate between transportation problem and assignment problem.	06
	b	A dairy plant has five milk tankers I, II, III, IV & V. these milk tankers are to be used on five delivery routes A, B, C, D and E. The distances (in km) between dairy plant and the delivery routes are given in the following distance matrix: $Milk\ tankers$ / $Delivery\ Routes$ A: 160, 130, 175, 190, 200 B: 135, 120, 130, 160, 175 C: 140, 110, 155, 170, 185 D: 50, 50, 80, 80, 110 E: 55, 35, 70, 80, 105 How the milk tankers should be assigned to the chilling so as to minimize the distance travelled?	10
		OR	
7	a	The cost matrix of assigning four operators on four existing machine is given below. Note that the operator one cannot be assigned to machine four. A new machine is available whose respective cost assignment are $2, 1, 2 \& 8$. Determine whether it is economical to replace an existing machine with new machine. If the answer is yes, which machine is to be replaced? What should be the savings?	

				$Machines$				
		$Operators$		M_1	M_2	M_3	M_4	
			1	5	5	-	2	
			2	7	4	2	3	
			3	9	3	5	-	
			4	7	2	6	7	12
	b	What is the objective of travelling salesmen problem?						04
8	a	State the applications of CPM & $PERT$.						06
	b	A project consists of the following activities with the duration in days						

$Activity$	A	B	C	D	E	F	G	H	I
$Precedence$	-	A	A	B,C	A	D,E	C	F,G	H
$Duration$	10	4	12	10	9	8	7	9	6

i) Draw the network of the above project
ii) Identify the critical path and determine the project duration
iii) Calculate early and late start, early and late finish, independent float, free float for all the activities. 10

OR

9	a	Discuss the importance of float and critical path in a project network.						04
	b	Consider the following project						

$Activity$	A	B	C	D	E	F	G
$Precedence$	-	-	A	B	B	C,D	E
$Optimistic\ Time$	3	2	2	2	1	4	1
$Most\ Likely\ Time$	6	5	4	3	3	6	5
$Pessimistic\ Time$	9	8	6	10	11	8	15

i) Draw the network of the above project, calculate the expected time and variance for all the activities.
ii) Identify the critical path and determine the project duration
iii) Find the probability that the project will be completing in 18 weeks. 12

10	a	Discuss the following with respect to a customer's behavior in a queue: i) Balking ii) Collusion iii) Jockeying.	06
	b	A person repairing radios finds that the time spent on the radio sets has exponential distribution with mean 20 minutes. If the radios are repaired in the order in which they come in and their arrival is approximately Poisson distributed with an average rate of 15 for 8 hour day, what is the repairman's expected idle time each day? How many jobs are ahead of the average set just bought in?	10
		OR	
11	a	Explain the following with respect to game theory: i) Two person zero sum game ii) Pay-off matrix iii) Saddle point.	06

b	In a certain game, A has three possible choices L, M and N while player B has two choices P and Q. Payments are to be made according to the following arrangements LP : $A\ pays\ B\ Rs.3$ LQ : $B\ pays\ A\ Rs.3$ MP : $A\ pays\ B\ Rs.5$ MQ : $B\ pays\ A\ Rs.2$ NP : $B\ pays\ A\ Rs.2$ NQ : $B\ pays\ A\ Rs.2$ What are the best strategies for players A and B in the game? What is the value of game for A and B?	10

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

OPERATIONS RESEARCH

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

12. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
13. Answer FIVE full questions from Part B.

PART-A

1	1.1	The person who coined the name Operations Research is _____.	01
	1.2	Slack variables in LPP indicates _____.	01
	1.3	Mention any two characteristics of operations research.	02
	1.4	Mention any two assumptions made while formulating a linear programming problem.	02
	1.5	A line on the graph drawn as per the objective function, assuming certain profit is called as _____.	01
	1.6	Mention any two differences between graphical and simplex methods.	02
	1.7	Alternate solution in LPP arises when _____.	02
	1.8	Big-M-method is also called as _____.	01
	1.9	What is the condition for degeneracy in transportation model?	01
	1.10	Mention the condition on which assignment problem can be converted to transportation problem.	01
	1.11	Queuing model is also called as _____.	01
	1.12	The person who coined Games theory was _____.	01
	1.13	Mention any two differences between $PERT$ and CPM.	02
	1.14	Arrival → Service → Service → out. This sequence represents what type of queuing system?	02

PART-B

2	a	What are the methods of solving operations research problems?	06
	b	In a manufacturing process, the final product has a requirement that it must weigh exactly 150 pounds. The two raw materials used are A, with a cost of $Rs.4$ per unit and B with a cost of $Rs.8$ per unit. At least 14 units of B and not more than 20 units of A must be used. Each unit of A weighs 5 pounds; each unit of B weighs 10 pounds. Setup an LPP to minimize the cost, do not solve. Write down the mathematical structure of the problem. Also mention which generally stated problem structure does this exercise on formulation does this situation belong?	10
		OR	

3 a Solve the following LP problem graphically:

Minimize $z = -2x_1 + 3x_2$

subject to $-x_1 + 2x_2 \leq 2;$

$2x_1 - x_2 \leq 3;$

$x_2 \geq 4;$

$x_1, x_2 \geq 0$ 08

b Solve the following LP problem by simplex method:

Maximize $z = 3x_1 + 2x_2$

subject to $x_1 + x_2 \leq 4;$

$x_1 - x_2 \leq 2;$

$x_1, x_2 \geq 0$ 08

4 A company has 3 factories manufacturing the same product at 4 sole agencies in different parts of the country. The production cost differs from factory to factory and selling price differs from agency to agency. The shipping cost/unit is known. Given the following data, determine the production and distribution schedule most profitable to the company.

	S_1	S_2	S_3	S_4	*Requirement*
F_1	3	9	5	4	180
F_2	1	7	4	5	120
F_3	5	8	3	6	150
Production capacity	10	180	50	100	

Production cost/unit	15	18	14	13
Raw material cost/unit	10	9	12	9

16

OR

5 Three refineries with maximum daily capacities of $6,5$ and 8 million gallons of gasoline supply three distribution areas with daily demand of $4,8$ and 4 million gallons, through a network of pipeline. The transportation costs per 100 gallons are given below:

	To	*Distribution Areas*		
From		1	2	3
Refineries	1	120	180	–
	2	300	100	80
	3	200	250	120

Note: refinery 1 is not connected to distribution area 3. Any surplus production at refinery 1 and 2 must be diverted to other areas by truck. The resulting transportation cost per 100 gallons is $Rs.150$ from refinery 1 and $Rs.220$ from refinery 2. The refinery 3 can direct its surplus to a chemical process within its own plant at no cost. Formulate the problem as transportation model and find the optimal solution. 16

6 a Give a mathematical formulation of the assignment problem. 06

b A college athletic conference has 6 basketball officials. It must assign to 3 conference games. Two officials must be assigned to each game. The conference officer desires to assign the officials such that the total distance travelled by all the 6 officials must be minimized. The distances each official would have to travel to each game are given below. The conference office has decided not to assign the official 4 to game A because of previous conflicts with one of the coaches. Determine the optimal assignment.

$Official$	$Games$		
	A	B	C
1	20	45	10
2	40	90	70
3	60	70	30
4	20	60	40
5	70	15	50
6	80	25	35

10

OR

7 a Give a mathematical formulation of travelling sales person problem. 06

b Solve the travelling salesman problem given by the following data: $C_{12} = 20, \quad C_{13} = 4, \quad C_{14} = 10, \quad C_{23} = 5, C_{34} = 6, \quad C_{25} = 10, \quad C_{35} = 6, C_{45} = 20,$ where $C_{ij} = C_{ji}$, there is no route between cities i & j, if the values are not shown. 10

8 a The following table gives the data about duration and cost of various activities.

$Activities$	$Normal$		$Crash$	
	$Time\ (weeks)$	$Cost\ (Rs)$	$Time\ (weeks)$	$Cost\ (Rs)$
1 – 2	2	500	1	800
1 – 4	5	900	3	1300
1 – 3	4	800	3	1000
2 – 4	1	400	1	400
3 – 4	3	1200	2	1800
4 – 5	6	700	4	900
3 – 5	7	600	4	1200

$Project\ Duration$	9	10	11	12	13
$Indirect\ Cost$	6000	6150	6200	6500	7100

i) Draw the network and determine the normal duration of the project.
ii) Find the optimum duration and its least cost(after crashing)
iii) Plot a graph of duration versus various costs. 12

b Mention the applications for $PERT$ models and crashing using CPM models (any 2 each). 04

OR

9 a List some of the limitations of network techniques. 06

b A project consists of 9 jobs with the following precedence relationship

$Activity$	$Precedence$	$Pessimistic\ time\ (Days)$	$Optimistic\ time\ (days)$	$Most\ likely\ time\ (days)$
A	-	8	2	5
B	A	12	6	9
C	A	8	6	7
D	B, C	7	1	4
E	A	8	8	8
F	D, E	17	5	14
G	C	21	3	12
H	F, G	9	3	6
I	H	11	5	8

i) Draw the network and determine the critical path

ii) What is the probability of completing the project 2 days sooner and 3 days later than expected? 10

10 a Why studying of Queue are considered important in one field of industrial engineering. 06

b In a bank there is only one window. A solitary employer performs all the series required and the window remain continuously open from $7am$ to $1pm$. It was discovered that an average number of clients is 54 during the day and the average service time is $5min/$person. Find:

i) Average number of clients in the system

ii) Average waiting time

iii) The probability that a client has to spend more than 10 minutes in a system. 10

OR

11 a A and B play a game as follows: They simultaneously and independently write one of the three numbers as $1, 2$ and 3. If the sum of the number written is even, B pays to A this sum in rupees. If it is odd, A pays the sum to B in rupees. Formulate the payoff matrix with respect to player A. 04

b Solve the following game whose pay off matrix is as given below:

$PLAYER\ A$	$PLAYER\ B$				
		B_1	B_2	B_3	B_4
	A_1	8	15	-4	-2
	A_2	19	15	17	16
	A_3	0	20	15	5

By motion of dominance reduce the game to 2×4 and then solve graphically. 12

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

OPERATIONS RESEARCH

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

5. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
6. Answer FIVE full questions from Part B.

PART-A

1		Explain briefly the following:	
	1.1	Mathematical problem	02
	1.2	Phases of $O.R.$	02
	1.3	Artificial variable	02
	1.4	Degeneracy in Transportation problem	02
	1.5	Travelling salesman problem	02
	1.6	Crashing the project	02
	1.7	Characteristics of Queuing model(only listing)	02
	1.8	Critical path	02
	1.9	Mixed strategy	02
	1.10	Saddle Point	02

PART-B

2	a	Define $O.R.$ and list the applications.	06
	b	A company produces 3 models of a certain product A, B and C. he uses 2 types of raw materials X and Y of which 4000 and 6000 units are available. Their requirements are given below: $Raw\ Material$ | A | B | C X | 2 | 3 | 5 Y | 4 | 2 | 7 The labor time for each unit of A is twice that of model B and 3 times of C. The entire labor force of the factory can produce an equivalent of 700 units of B. there is a commitment that 200 units of A must be produced. The profit of A, B and C are $Rs.30, Rs.20$ and $Rs.50$ respectively. Formulate the problem for optimum mix.	10
		OR	
3	a	Differentiate between unbounded and infeasible solution. Also show it with the help of a sketch.	06

b	Solve the following LPP: Maximize $z = -x_1 + 2x_2$ Subjected to $-x_1 + 3x_2 \leq 10$ $3x_1 + 8x_2 \leq 24$ $10x_1 + 7x_2 \leq 35\,, \quad x_1,\ x_2 \geq 0$	10
4	A company has 3 production shops supplying a product to 5 warehouses. The cost of production varies from shop to shop and also transportation cost. Each shop has a specific production capacity and the cost of transportation are given below:	

		Warehouse					*Supply*
		I	*II*	*III*	*IV*	*V*	
shop	*A*	6	4	4	7	5	100
	B	5	6	7	4	8	125
	C	3	4	6	3	4	175
		60	80	85	105	70	

The cost of manufacturing the product at different shops are as follows:

Shop	*Variable Cost*
A	14
B	16
C	15

Find the optimum quantity to be supplied from each shop to different warehouses at minimum cost. 16

OR

5 Given a transportation problems supply, demand and unit profit (Rs.) between factory and warehouse, find the optimal solution.

Factory	*Warehouse*			*supply*
	I	*II*	*III*	
*F*1	5	8	3	50
*F*2	5	10	9	40
*F*3	6	4	7	10
Demand	50	30	20	

16

6	A process can be carried out on any of the 6 machines. The average time taken by any of the operator is given below:	

operator	*Existing m/cs*						*New m/c*
	1	2	3	4	5	6	
A	10	12	8	10	8	12	11
B	9	10	8	7	8	9	10
C	8	7	8	8	8	6	8
D	12	13	14	14	15	14	11
E	9	9	9	8	8	10	9
F	7	8	9	9	9	8	8

It is proposed to buy a new machine to replace the existing one and its estimated time is given in the table. Is it advantageous to use the new machine? If so, which of the original machine should be replaced and how many hours can be saved? 16

OR

7 a Explain the differences between a transportation and Assignment Problem. 06

b A salesman travels from one place to another and distances between pairs of cities are given below:

		To			
		P	Q	R	S
From	P	10	15	25	20
	Q	22	∞	45	55
	R	40	30	∞	25
	S	20	26	38	∞

Checkout a route which enables him to visit each of the cities only once, so that the total distance covered is minimum. 10

8 A Project is composed of 11 activities:

Activity	*Least time*	*Most likely*	*Greatest*
$1-2$	3	5	7
$1-3$	2	6	10
$3-5$	6	7	8
$2-5$	3	4	5
$2-4$	7	8	9
$3-6$	6	8	10
$4-7$	4	7	10
$5-7$	5	6	7
$5-8$	3	4	5
$6-8$	7	8	9
$7-8$	7	9	11

a) Draw the network, calculate earliest and latest occurrences.
b) Find the critical path and expected project length.
c) Compute the probability of completing the project one day later than expected. 16

OR

9 a State the applications of CPM and $PERT$. 04

b The utility data for a network is given below:

	Normal		*Crash*	
Activity	*Time*	*Cost*	*Time*	*Cost*
$1-2$	8	100	6	200
$1-3$	4	150	2	350
$2-4$	2	50	1	90
$3-4$	5	100	1	200

Indirect cost is $Rs.100$/day. Crash systematically and determine the optimum project duration with cost. 12

10		A road transport company has one reservation clerk on duty a time. She handles information of bus schedules and makes reservation. Customers arrive at a rate of 8 per hour and the clerk can service 12 customers on an average. Answer the following: a) Average number of customers waiting. b) Average time a customer has to wait before getting service. c) The management is contemplating to install a computer system to handle information and reservation. This is expected to reduce the service time from 5 to 3 mins. The additional cost of having the new system works out to be 12 paise per minute spent waiting before being served. Should the company install the computer system? Assume 8 hours working day.	16
		OR	
11	a	In a game of matching coins, player A wins $Rs.8$, if both coins match head and $Rs.1$ if both are tail. Player B wins $Rs.3$ when coins do not match. Given the choice of being player A or B, which would you choose and what would be your strategy?	06
	b	Use the concept of dominance to reduce the size of the matrix to 2×3 matrix and solve graphically. The payoff matrix is: $Player\ B$: 1, 8, 3 $Player\ A$: 6, 4, 5 0, 1, 2	10

Marketing Management (and Research)

V Semester SEE Question Papers

Course codes: 18IM55, 16IM5A4

- Mar/Apr 2023 (18IM55)
- Jan/Feb 2021 (16IM5A4)
- Jan/Feb 2021 (18IM55)
- Fast Track Oct 2020 (16IM5A4)
- Nov/Dec 2019 (16IM5A4)
- Nov/Dec 2018 (16IM5A4)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Mar/Apr 2023

Industrial Engineering and Management

MARKETING MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	The key to winning and keeping customers is to understand their needs and buying processes better than competitors do and _____.	01
	1.2	The way the product is defined by consumers on important attributes is called _____.	01
	1.3	The strategy of choosing one attribute to excel at to create competitive advantage is known as _____.	01
	1.4	A brand's _____ is the full positioning of the brand- the full mix of benefits upon which it is positioned.	01
	1.5	Comment on importance of demographic segmentation.	02
	1.6	Identify the 4 *P's* of Marketing.	02
	1.7	List any two functions of packaging.	02
	1.8	Identify two types of research data.	02
	1.9	Provide your own Psychographic and demographic profile when you are being assessed by any marketing personnel.	02
	1.10	Highlight one advantage and one limitation of psychographic segmentation.	02
	1.11	You want to test an app you have developed which addresses some needs of Engineering college students. How can you use digital marketing to get initial quantitative feedback?	02
	1.12	List any four online marketing tools that can be useful for beginners in the field of marketing.	02

PART-B

2	a	Explain the core concepts of Marketing. Support the explanation with an illustration.	08
	b	Examine how Globalization, Technology and social responsibility has changed marketing for Good in the last decade.	08
3	a	Choose a company of your choice and identify the product variants they produce. Place the variants in product Portfolio matrix (*BCG* matrix) and provide rationale.	06
	b	*AICTE* activity points requires engineering students from 2018 batch onwards to involve in community services in order to develop in to individuals of higher human values. But, current scenario prevailing from past two years due to corona virus has hindered students from indulging in outreach programs organized by institutions and *NGO's*. Develop a marketing plan for digital product/online portal which can solve this issue for students.	10

		OR	
4	a	Discuss the importance of Research in marketing with special emphasis on new product development.	10
	b	Explain the three key management questions that the holistic management framework is designed to address.	06
5	a	A lot of students now have their own YouTube channel where they publish good content sometimes but are unable to get the required audience to appreciate. You as a student of Marketing, showcase 10 ways in which these YouTube videos can be made to reach the right audience.	10
	b	Explain any three steps a company can take to improve the quality of its marketing intelligence.	06
		OR	
6	a	Similar to $4P's$ of marketing, There are $4\,A's$ of Rural marketing ie., affordability, awareness, availability, acceptability. Now, imagine you have a product which you think can find a decent user base in rural India. So, take an example of a product that which you think addresses the needs of the rural market (Clearly mention the product characteristics) and design a questionnaire to test the $4A's$ of rural market. Questionnaire to contain 4 heads corresponding to $4A's$ of rural marketing and each head to contain minimum 4 questions with appropriate unit of measure.	10
	b	Explain any three components of Modern Marketing Information System.	06
7	a	Discuss the 4 scales of measurement with respect to marketing.	08
	b	Explain the steps in marketing research.	08
8	a	It is interesting to know that pandemic has pushed several people to take a plunge in to entrepreneurship in the past two years. Many entrepreneurs though have a product that functions really well and has all the characteristics of being a winner in the market but still not reflecting in the sales figure. The reason lies in differentiation aspect of marketing. So, as a student of marketing, what would your 5 suggestions be for such entrepreneurs with regard to product or service differentiation?	10
	b	As part of "Sustainable Development Goals- SDG", every sector of economy should strive towards sustainable practices to reduce carbon footprint and pollution. Marketing division also has found a new niche called "Green Customers" and efforts to cater to their needs is being met positively. The fashion and lifestyle choices of top engineering college students from metro cities like Bengaluru have huge influence on students from across the state of Karnataka. Being a responsible student of the institution, if you were given an opportunity to make changes in daily utility products and promote them, which 6 products would you choose and why?	06

16IM5A4

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

MARKETING MANAGEMENT AND RESEARCH (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Explain the essence of Marketing.	02
	1.2	What is Industrial Marketing?	02
	1.3	List out the necessary skills for a Marketing Manager.	02
	1.4	What are the characteristics of Service Marketing?	02
	1.5	What is the Marketing Mix?	02
	1.6	What is Marketing Segmentation?	02
	1.7	Define Marketing Research.	02
	1.8	What is Customer Retention?	02
	1.9	What is online Marketing?	02
	1.10	What is Retail Research?	02

PART-B

2	a	Define Marketing Management and explain the core concepts of Marketing.	08
	b	What are the various factors in Indian Marketing environment? And what is the need to analyze the marketing environment.	08
3	a	What is Marketing Strategy? Explain the key drivers of Marketing Strategies.	08
	b	Explain the various stages of product life cycle.	08
		OR	
4	a	What do you understand by Product Planning? Discuss briefly the problems of introducing new products.	08
	b	Enumerate the Marketing Mix Strategies for Services.	08
5	a	Explain the Micro Environmental Elements in detail.	08
	b	Give the process of carrying out a Marketing Research.	08

		OR	
6	a	What do you understand by Marketing Information System (MIS)? Explain the various elements of MIS in detail.	08
	b	Explain the various problems in Channel Management in rural marketing.	08
7	a	What is attitude measurement scale? Explain its types.	08
	b	Explain the eight characteristics of qualitative research.	08
8	a	What is Retailing? What are the challenges faced by retail marketing?	08
	b	Discuss the purpose of packaging?	08

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Jan/Feb–21

Industrial Engineering and Management

MARKETING MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	The way a product is defined by consumers on important attributes is called _____.	01
	1.2	When Coca-Cola produced only one drink for the whole market, it was practicing _____ type of marketing strategy.	01
	1.3	Dividing buyers into groups based on their knowledge, attitude, uses or responses to a product is called _____.	01
	1.4	The consumer does not know whether to buy toothpaste, mouthwash, or an antiseptic gargle, or some mint. It suggests that he want to seek _____ benefits.	01
	1.5	Manufacturer and Consumer is an example for _____level channel.	01
	1.6	Segmentation on the basis of values, attitudes and behavioral patterns is an example of _____.	01
	1.7	Targeting affluent customers with luxurious goods is an example of _____ segmentation.	01
	1.8	Stock exchange of Mumbai is an for _____market.	01
	1.9	The research which is "unstructured, qualitative, highly flexible" is called as _____.	01
	1.10	Improving existing business is called as _____.	01
	1.11	When an organization acquires wholesaler and retailers, it is called as _____ integration.	01
	1.12	_____ & _____ are two types of research data.	01
	1.13	In marketing research, the _____phase is generally the most expensive and most subject to error.	01
	1.14	When firms use symbols, colors, or characters to convey their personalities, they are using _____ differentiation.	01
	1.15	A company is practicing _____ if it focuses on sub segments with distinctive traits that may seek a special combination of benefits.	01
	1.16	The stage of product life cycle that focuses on expanding market and creating product awareness and trial is the _____stage.	01

	1.17	Mr. X has discovered that effective programs can be designed for attracting and serving your chosen segments. This segmenting requirements is called _____	01
	1.18	Costs that do not vary with production or sales levels are called _____.	01
	1.19	_____ is called pre-assumption of the expected result of the research.	01
	1.20	_____ is a part of the product, which carries verbal information about the product.	01

PART-B

2	a	Explain the relationships between customer value, satisfaction and quality.	06
	b	Discuss the differences between the marketing concept and the production, product, and selling concepts. Which concepts are easier to adopt in the short run? Which concept offers that the best chance for long-run success? Why?	10
3	a	Explain the steps involved in business unit strategic-planning process.	08
	b	Examine the Boston Consulting Group's growth-share matrix. Which one of the cells provide the primary revenues for the organization's SBUs? How can an organization use the BCG growth-share matrix to plan its future?	08
		OR	
4	a	Define Strategic Planning. List and briefly describe the four steps that lead managers and the firm through the strategic planning process.	08
	b	What is a mission statement? List and briefly discuss the four characteristics of effective mission statements.	08
5	a	With suitable examples explain the marketing intelligent systems.	08
	b	List and explain the challenges faced when researching Rural Markets.	08
		OR	
6	a	Identify the barriers in using Marketing Research.	08
	b	Imagine that you are the marketing research manager for a small electronics firm that has recently changed management. You are concerned when you receive a memo stating that "marketing research is not really necessary as long as you have a good marketing information system (MIS)". Write a brief response that outlines the role of marketing research and clarifies how it differs from MIS.	08
7	a	Differentiate between explanatory and descriptive techniques in marketing research with suitable examples.	08
	b	Write the short note on the following rating scales i) Itemized Non-comparative ii) Graphic Non-comparative	08
8	a	Explain the various factors which have contributed to the growing use of packaging as a marketing tool.	06
	b	In product-mix pricing, the firm searches for a set of prices that maximizes profits on the total mix. Explain the six types of product-mix pricing.	10

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RV COLLEGE OF ENGINEERING®

(Autonomous Institution affiliated to VTU)

V Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering & Management

MARKETING MANAGEMENT & RESEARCH (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	In marketing , identifying and profiling distinct groups of buyers who might prefer or require varying product and service mixes is termed as _____.	01
	1.2	All the activities in gathering and acting upon information about the market is termed as _____.	01
	1.3	If Videocon engineers its washing machines to spin the clothes faster regardless of whether or not customers want the speed, then Videocon has characteristics associated with _____ concept.	01
	1.4	Demands are _____ for specific products backed by an ability to pay.	01
	1.5	At which level of the organization is marketing plan formulated?	01
	1.6	_____ is a tool for identifying ways to create more customer value.	01
	1.7	_____ provide diagnostic information about how and why we observe certain effects in the market place, and what that means to marketers.	01
	1.8	_____ scales are carefully constructed sets of rating scales designed to measure one or more aspects of an individual's attitude towards some object.	01
	1.9	What is the difference between a laboratory and a field experiment?	01
	1.10	Examination of archives, or secondary sources can be termed as _____.	02
	1.11	Non attribute based techniques in marketing research are generally referred to as _____.	01
	1.12	_____ analysis offers the potential of evaluating the interest generated by a commercial or package and the nature-emotional or rational of the interest.	01
	1.13	_____ offers a variety of merchandise, including impulse goods such as soft drinks, coffee, candy, newspaper, magazines and other products such as hosiery, cosmetics, hot food and paperbacks	01
	1.14	A _____ is a brand that retailers and wholesalers develop.	01
	1.15	_____ include materials management, material flow systems and physical distribution, aided by information technology.	01
	1.16	_____ are unbranded, plainly packaged, less expensive versions of common products.	01
	1.17	_____ refers to the elapsed time between an order's receipt, delivery and payment.	01

	1.18	_____ is a type of error, one that occurs in a manner each time something is measured.	01
	1.19	Projective techniques and physiological measures can be used to measure _____ responses to ads, products or product use.	01

PART-B

2	a	Explain the 4 *P*'s of Marketing.	10
	b	New terms are now being used to describe where marketing is done. Using the automobile market, describe automotive buying behavior for a: i) Market place ii) Market space iii) Meta market	06
3	a	A locally well known kannada book publication house in state of Karnataka, used to sell books through physical retailing outlets stretched across Bengaluru, Mysore and Mangalore. It is now planning to take their business completely online, thereby reaching out to Kannada book readers stretched across the globe. In this scenario, discuss the nature and contents of a marketing plan to successfully accomplish their agenda.	10
	b	How important is corporate and divisional strategic planning for a business firm to establish its business; grow, and have durable competitive stand in the industry?	06
		OR	
4	a	A well established Media company specially associated with sports telecast, now would want to establish an SBU in restaurant industry with a theme of sports. With respect to this scenario discuss the business unit strategic planning.	08
	b	Googles' Mission statement is stated as "To organize the world's information and make it universally accessible and useful." With this as a case example discuss the features of a good mission statement, philosophy and the competitiveness reflected to capture the major market share.	08
5	a	By considering a suitable case example, elucidate the six-phased marketing research process.	08
	b	Discuss all the demographic factors to be considered in an urban environment of a developing country such as India, for establishing organic food chain outlets.	08
		OR	
6	a	"Marketing intelligence is very much important to innovate and grow, which helps a company to sustain its competitive position for a long duration", with a suitable case example justify this statement.	08
	b	Explain briefly the following modules of *MIS*: i) Internal Record system ii) Marketing Research.	08

7	a	Write the short notes on the following approaches to conjoint analysis i) Trade off matrices ii) Full profile.	10
	b	"Facial expressions have been used as an indicator of baby's preference for various food flavorings". Give reason for preferring observational data over alternative methods in marketing research.	06
8	a	Why are wholesalers used at all? Why don't manufactures sell directly to retailers or final consumers? Discuss the functions wholesalers perform to make the selling process more efficient.	08
	b	Retailers are facing increasing competition from other manufacturer to consumer distribution channels. A retailer must develop a product differentiation strategy. List and explain these product differentiation strategies.	08

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

MARKETING MANAGEMENT AND RESEARCH (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	_____ is a management philosophy which states that achieving organizational goals depends on determining the needs and wants of target markets and delivering the desired satisfaction more effectively and efficiently than competitors.	01
	1.2	Fitting company's core competencies and distinctive capabilities into tight interlocking is termed as _____.	01
	1.3	If Videocon company engineers its washing machines to spin the clothes faster regardless of whether or not customers want the speed, then Videocon has characteristics associated with _____ concept.	01
	1.4	Demands are _____ for specific products backed by an ability to pay.	01
	1.5	What are the different types of rating scales?	02
	1.6	Differentiate between a laboratory and a field experiment.	02
	1.7	Holistic marketers succeed by managing a _____ that delivers a high level of product quality, service and speed.	01
	1.8	_____ provide diagnostic information about how and why we observe certain effects in the market place and what that means to marketers.	01
	1.9	At which level of the organization is marketing plan formulated?	01
	1.10	In marketing, identifying and profiling distinct groups of buyers who might prefer or require varying product and service mixes is termed as _____.	01
	1.11	Companies address customer needs by putting forth _____ and _____ that satisfy those needs.	02
	1.12	_____ scales are carefully constructed sets of rating scales designed to measure one or more aspects of an individual's attitude towards some object.	01
	1.13	Examination of archives or secondary sources can be termed as _____.	01
	1.14	The GE Matrix classifies each SBU by the extent of its _____ and the attractiveness of its industry.	01
	1.15	Projective techniques and physiological measures can be used to measure _____ responses to ads, products or product use.	01
	1.16	A _____ is a brand that retailers and wholesalers develop.	01
	1.17	With respect to measurement accuracy under real world conditions: $M = C + E$, Where M is measurement, what does C and E stand for in the equation?	01

PART-B

2	a	Elucidate the 4*P's* of Marketing.	10
	b	New terms are now being used to describe where marketing is done. Considering the Electronic goods market, describe electronic goods buying behavior for: i) Market place ii) Market space iii) Meta market.	06
3	a	A locally well-known Kannada book publication house in state of Karnataka, which used to sell books through physical retailing outlets stretched across Bengaluru, Mysore and Mangalore, is now planning to take their business completely online, there by reaching out to Kannada book readers stretched across the globe. In this scenario, discuss the nature and contents of a marketing plan to successfully accomplish their agenda.	10
	b	How important is corporate and divisional strategic planning for a business firm to establish its business, grow and have durable competitive stand in the industry?	06
		OR	
4	a	A well-established Media company specially associated with sports telecast, now would want to establish an *SBU* in restaurant industry with a theme of sports. With respect to this scenario discuss the business unit strategic planning.	08
	b	Company Google's Mission Statement is stated as "To organize the world's information and make it universally accessible and useful." With this as a case example discuss the features of a good mission statement, philosophy and the competitiveness reflected to capture the major market share.	08
5	a	Describe the six-phased marketing research process by considering a suitable case example.	08
	b	Discuss all the demographic factors to be considered in an urban environment of a developing country such as India, for establishing organic food chain outlets.	08
		OR	
6	a	"To spot important opportunities and potential problems, marketing managers rely on internal reports." Elucidate the components of internal records in relevance to the above statement.	08
	b	Explain briefly the following modules of *MIS*: i) Marketing Research ii) Marketing Intelligence.	08
7	a	Write the short notes on the following approaches to conjoint analysis i) Trade-off matrices ii) Full profile.	10
	b	"Facial expressions have been used as an indicator of baby's preference for various food flavorings". Give reason for preferring observational data over alternative methods in marketing research.	06
8	a	Explain the reason why wholesalers are preferred for selling the products and why don't manufactures sell directly to retailers or final consumers.	08
	b	Identify and explain the various product-differentiation strategies.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-18

Industrial Engineering and Management

MARKETING MANAGEMENT AND RESEARCH (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	In the context of marketing management, distinguish between the terms 'market place' and 'market space'.	02
	1.2	What is the key objective of 'Relationship marketing'?	02
	1.3	The essence of successful business is to focus on the 'core competencies' by organizations. Identify the three characteristics of core competency.	02
	1.4	Holistic marketers address three key management questions. What are they?	02
	1.5	A large social, economic, political and technological change is slow to form, and once in place, influences us for some time-between seven and ten years or longer. What is this phenomenon termed as?	02
	1.6	One of the measures preferred by marketers for gauging consumer opinion is 'Laddering'. Describe the term.	02
	1.7	A sample of experts may be asked to rank preferences for several brands. Identify the appropriate scale of measurement in such context.	02
	1.8	One of the approaches to assessing reliability is 'Test Retest reliability'. Mention how this measurement helps in assessing the reliability in marketing research.	02
	1.9	Store retailers offer services at four different types of levels. Identify the four levels.	02
	1.10	Differentiate between the two prominent players 'Wholesaler' and 'Retailer'.	02

PART-B

2	a	Illustrate a simple marketing system depicting the relationship between the industry and the market. Also categorize and explain the key customer markets.	08
	b	Holistic marketing recognizes and reconciles the scope and complexities of marketing activities. Provide a schematic overview of four broad components characterizing holistic marketing. Also provide a brief review on each of these categories.	08
3	a	Good mission statements have five major characteristics. Elaborate. Also write the mission statement of the institution you are currently associated with?	08
	b	The overall evaluation of a company's strengths, weaknesses, opportunities and threats is called as *SWOT* analysis. Conduct the SWOT analysis for the 'program of engineering' you are associated with.	08
		OR	
4	a	Summarize the key constituents of a typical marketing plan.	08
	b	The value chain identifies nine strategically relevant activities that create value and cost in a specific business. Describe the same.	08
5	a	To identify important opportunities and potential problems, marketing managers rely on various categories of internal reports. Analyze the purpose of these reports and bring out their relevance to the context.	08
	b	Identify different ways marketers can research their competitor's product strengths and weaknesses online.	08
		OR	
6	a	As part of analyzing the macro environment, identify and explain the major forces to be considered by the organizations.	08
	b	Analyze the typical steps followed for effective marketing research.	08
7	a	Analyze the five major issues, or problem areas, involved with question content.	08
	b	Evaluate the basic approaches to generating perceptual maps.	08
8	a	Today, the market comprises of different types of retailers. One such format is 'Non store retailing'. Classify and characterize different types of non-store retailing.	08
	b	Classify and characterize different types of major wholesaler formats prevailing today.	08

Simulation Modeling and Analysis

V Semester SEE Question Papers

Course codes: 16IM54, 16IM54 / 12IM64

- Jan/Feb 2021 (16IM54)
- Fast Track Oct 2020 (16IM54 / 12IM64)
- Nov/Dec 2019 (16IM54)
- Fast Track Jul 2019 (16IM54 / 12IM64)
- Nov/Dec 2018 (16IM54)

16IM54

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

SIMULATION MODELING AND ANALYSIS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	The collection of variables necessary to describe the system at any time, relative to the objectives of study is known as _____.	01
	1.2	Static models are simulated when there is existence of _____ randomness and _____ interaction among variables.	01
	1.3	Q type inventory is one in which _____ is fixed.	01
	1.4	The two types of documentation employed in simulation study are _____ and _____.	01
	1.5	_____ distribution is related to a sequence of Bernoulli trials.	01
	1.6	In simulation, iid with respect to random number represents _____.	01
	1.7	_____ distribution is often overused when there are no data.	01
	1.8	What techniques are used for parameters estimation, once the distribution is identified from the collected data?	01
	1.9	For the triangular distribution with end points as 0.2, and the mode at 1, the random variate corresponding to the random number 0.35 is _____.	02
	1.10	Given a standard normal variate $z_1 = (-1.3)$ the corresponding chi-square variate with 1 degree of freedom is _____.	01
	1.11	The sales is uniformly distributed between 300 and 500. If r is a random number between 0 and 1, then the proper expression for sales is _____.	01
	1.12	Building model correctly as per specification is _____.	01
	1.13	Whether a simulation in a state that is more representative of long run conditions is called _____.	01
	1.14	Initializing simulation in a state that is more representative of long run conditions is called _____.	01
	1.15	_____ and _____ are two prominent examples of robust heuristics which can be applied to problems with mixed types of decision variables.	01
	1.16	_____ software has manufacturing oriented modeling and rule based decision logic.	01

	1.17	_____ is the tweaking / tuning of existing parameters and usually does not involve introduction of new ones, changing the model structure.	01
	1.18	_____ estimation typically requires a methods of estimating the variance of point estimator.	01
	1.19	Analysis of simulation data and drawing inferences about the real life situation is _____.	01

PART-B

2	a	Explain the different types of simulation models.	08
	b	The following patterns of arrivals are observed in each counter of a certain back in 1 minute interval. No of persons arriving: 0 1 2 3 4 5 Frequency: 3 8 16 12 6 5 If 2 persons can be served in 1 minute interval, simulate the activities of the cash counter for 15 minute period using the following 2 digit random numbers : 14,23,92,07,37,12,97,62,34,21,09,42,59,67,39. Find the average number of persons waiting and percentage Idle time of the server (counter).	08
3	a	Enumerate the steps involved in the development of useful model of input data.	04
	b	Apply the following tests to the sequence, identify which property is being tested and offer your comments: 0.16, 0.27, 0.58, 0.63, 0.45, 0.21, 0.72, 0.87, 0.27, 0.15 0.92, 0.85, 0.78, 0.59, 0.43, 0.67, 0.58, 0.26, 0.06, 0.84 (i) Autocorrelation Test with log 4 starting at 2nd Number (ii) Runs up and down test.	12
		OR	
4	a	Inter arrival times of parts (in minutes) is claimed to be exponential with a mean of 30 seconds. Perform a chi-square goodness of fit to see if the following data fits the claim. Use 3 classes of equal expected values. Follow standard hypothesis testing procedure. 0.0, 0.1, 0.1, 0.3, 0.8, 1.1, 1.2, 1.4, 1.5, 1.6, 1.8, 1.8, 1.9, 2.1, 2.1, 2.6, 2.9, 3.7, 4.8	10
	b	Use multiplicative congruential method to generate a sequence of four three digit random integers. Let $X_0 = 117, a = 43$ *and* $m = 1000$.	02
	c	Perform test for uniformity using Kolmogorov-Smirnov test with level of significance α = 0.05. Consider the following random number: 0.44, 0.81, 0.14, 0.05 and 0.95.	04
5	a	Lead times have been found to be exponentially distributed with mean 3.7 days. Generate 4 random lead times for this distribution. R_i = 0.931, 0.417, 0.612, 0.079, 0.0826.	03
	b	Develop a generator for a triangular distribution with range (0,2) and mode at 1. Use R_i = 0.13, 0.14, 0.65, 0.79, 0.96	09

	c	Explain the procedure to generate random variates for Poisson distribution. **OR**	04
6	a	What is random variate? Explain the inverse transform technique for generation of a random variable, sampling from weibull distribution.	08
	b	Apply Acceptance-Rejection method for a gamma distribution with mean 2.2 minutes and variance 2.10 minutes^2. Generate random variates using the random numbers R_1 = 0.832,0.434 R_2 = 0.021,0.716	08
7	a	Explain the difficulty of optimization via simulation.	08
	b	Explain the 3-step approach formulated by Naylor and Finger to aid in the validation process.	08
8	a	List and explain the types of simulation with respect to output analysis.	08
	b	Explain briefly trends in simulation software	04
	C	Highlight the various features of ARENA simulation software.	04

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Fast track Examinations Oct-2020

Industrial Engineering and Management

SIMULATION MODELING AND ANALYSIS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of statistical tables are permitted.

PART-A

1	1.1	An architectural model of a building is an example of _____ type of model	01
	1.2	The reason for delay in components arrival may be considered as system _____ for a production system	01
	1.3	Provide two examples of events in a baggage handling operation in an airport	02
	1.4	When the data is available, the sample _____ and sample _____ are used to estimate the parameters of a hypothesized distribution	01
	1.5	If the range space Rx of the random variable X is an interval, then X is called a _____ Variable	01
	1.6	What are the statistical properties of random numbers	02
	1.7	The _____ method generates random variates by adding together two or more random variables to obtain new random variable with the desired distribution.	01
	1.8	Consider a random variable X that is uniformly distributed on the interval $[a, b]$. Then pdf for uniform distribution is given by _____.	01
	1.9	What do you mean by empirical distribution?	02
	1.10	An example of robust heuristics applied to problems with mixed type of decision variables is _____.	01
	1.11	_____ proposed a three step approach for validation in 1967.	01
	1.12	Why are expected values in chi-squared goodness of fit test sometimes pooled?	02
	1.13	'Setting the inventory levels, number of back orders, number of items on order and their arrival dates in an inventory simulation ' etc are some examples for intelligent initialization. This reduces ……. In steady state simulation.	01
	1.14	A model used to analyze time series data is _____.	01
	1.15	What is embedded simulation?	02

PART-B

2	a	Classify the simulation models. What is discrete event system simulation? Explain.	06
	b	Customers arrive at a hotel to order a specialty dish which is made at the rate of 2 per hour. Being a fresh dish it cannot be stored. Each customer orders 1 to 5 numbers of the dish. The dishes produced but left over are sold to an agency at half of the regular price of $Rs.42/-$ per unit. The production cost is $Rs.40/-$ per unit. Construct a simulation table and calculate the total profit which consists of revenues (Sales to customers, sales of left over unordered units) and costs (Production and lost sales) over 4 hours of simulation. Customer arrival and order probabilities are given below:	10

#Customers per hour	0	1	2	3
Probability	0.05	0.25	0.45	0.25

#Units ordered by Customers	0	1	2	3
Probability	0.3	0.3	0.2	0.2

Use the following random numbers 12,10,47,77,16,43,94,7,26 for both arrivals and orders.

3	a	How do you develop useful model of input data? Explain the stepwise procedure	08
	b	Why random numbers are used in simulation? Why random numbers generated by computers are called pseudo-random numbers? What problems are faced when generating pseudo random numbers?	08
		OR	
4	a	Explain how chi-square test can be used as a test for goodness of fit?	08
	b	Consider the following sequence of 40 random numbers	08

0.90	0.89	0.44	0.21	0.67	9.17	0.46	0.83	0.79	0.40
0.94	0.22	0.66	0.42	0.99	0.67	0.41	0.73	0.02	0.72
0.43	0.47	0.17	0.56	0.45	0.78	0.56	0.30	0.71	0.19
0.93	0.37	0.42	0.96	0.73	0.47	0.60	0.29	0.78	0.26

Based on the runs ups and downs, determine whether the hypothesis of independence can be rejected.

5	a	Corresponding to the two random numbers 0.432 and 0.87 , generate two random variates of: i) Triangular distribution with minimum -2 , mode 0 and maximum 2 ii) Exponential distribution with a mean of ¼ .	12
	b	Time of failure of a machine is distributed as a Weibull random variable with shape parameter $\beta = 0.75$ and scale parameter $\alpha = 8$. Generate two time to failure values using random numbers 0.947 and 0.872.	04
		OR	

6	a	Processing of certain application is done in 5 sequential steps, with each step clearing 6 applications in an hour on the average, as a Poisson process. In order to simulate application processing as a single process, derive a method to generate application processing time. Generate a random variate using the required numbers sequentially from the following numbers: .947,.872,.638,.307,.696,.377,.014	08
	b	When acceptance – rejection technique of generating random variate is useful? Write steps in acceptance-rejection technique to generate random variates X uniformly distributed between ¼ and 1.	08
7	a	What is the meaning of optimization via simulation of a process? What are the difficulties encountered in this? Explain	08
	b	Explain with a block diagram model building, verification and validation	08
8	a	What are transient and steady state simulations with respect to output analysis? Explain with an example each	10
	b	Identify the trends in simulation packages	06

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

SIMULATION MODELING AND ANALYSIS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Statistical tables are permitted.

PART-A

1	1.1	The collection of variables used to describe the system at any time are called _____.	01
	1.2	_____ is the determination that a model is an accurate representation of reality.	01
	1.3	Two important properties of a random number sequence are _____ and _____.	01
	1.4	In a linear congruential generator, the seed, multiplier, increment and the modulus are $27, 17, 43$ and 40 respectievly. The first number generated, x_1 = _____.	01
	1.5	Frequency tests are used to test the property of _____ in a random number sequence.	01
	1.6	An example of a probability distribution with memorylessness property is _____ distribution.	01
	1.7	Which test can be used to check if all the intervals have equal representation, in a random number sequence?	01
	1.8	How can you confirm the distribution and parameters of a data set?	01
	1.9	A process has the processing time distribution _____, when it has 3 substages, each consuming exponentially distributed time with an average of $4\ minutes$?	01
	1.10	The process (or technique) of generating a random variate by adding together several other random variates is known as _____.	01
	1.11	Random numbers R_i in (0,1) can be transformed to x_i in uniform (10,30) distribution with relation x_i = _____.	01
	1.12	Initialization bias is the phenomenon that _____.	01
	1.13	Use of traces (values of state variables at each instance of time) is most useful in _____ stage of simulation study.	01
	1.14	The iterative process of comparing a model to the real system and refining the model is _____.	01
	1.15	Two measures of performance for a parameter estimate are _____ and _____.	01

	1.16	Optimization of systems without regular problem structure and with mixed types of decision variables are routinely undertaken via simulation using techniques such as _____.	01
	1.17	In a simulation run in time (0,1000) *in hours*, the period (0,100) *hours*, if ignored while collecting output statistics, then the period (0,100) is called as _____, in order to get a better output estimate.	01
	1.18	What is covariance-stationary time series?	01
	1.19	How is the precision in an interval estimate of an output measure increased by simulation?	01
	1.20	A Turing test involves comparison of outputs of _____ and _____, undertaken by experts.	01

PART-B

2	a	Name the entities, attributes, activities, events and state variables for the following systems: i) Check-in process at an airport ii) Communication network system in a small office iii) Assembly line of an automobile manufacturer iv) An engineering college.	08
	b	The inter arrival times of a job shop are distributed as follows. $IAT\ (hours)$ | 0 | 1 | 2 | 3 $probability$ | 0.23 | 0.37 | 0.27 | 0.13 Processing times for jobs are uniformly distributed in interval $(0.8, 1.4)$ in hours. Assume that when simulation begins, there is one job being processed with 30 *minutes* of processing left. Simulate the operation for 6 new jobs processed. i) What is the average waiting time in queue for a job? ii) What is the average queue length of jobs? iii) What is the proportion of time the job shop is free of parts (Job)? Use the stream 0.94, 0.73, 0.78, 0.26, 0.86, 0.68 for arrival times and $0.8, 0.3, 0.2, 0.5, 0.4, 0.4$ for processing.	08
3	a	When regular data collection is not possible, how are input models built for simulation? Explain in the context of a manufacturing system.	08
	b	The sequence $0.83, 0.31, 0.19, 0.95, 0.04, 0.38, 0.81, 0.64, 0.80, 0.69$ when sorted is seen to have a gap between 0.38 and 0.64. Compare the observation with KS test as well as the x^2 test for uniformity, using two intervals. Assume 5% significance level. n | 8 | 9 | 10 | 11 | 12 $D_{0.5}$ | 0.457 | 0.432 | 0.41 | 0.391 | 0.375	08
		OR	
4	a	What are the steps in input modeling? Explain briefly, listing the guidelines/techniques in each step.	06
	b	Test if the following lifetime measurements on a component are exponentially distributed with a mean of 14. Use 3 intervals and 5% significance. $80, 3, 0.1, 1.9, 6, 3, 6.5, 0.2, 0.1, 6.8, 60, 1.2, 35, 5.1, 18.4, 0.1, 0.4$	10

5	a	Generate one random variate each for the following distributions: i) Uniform in the interval (15,30) ii) Exponential with a mean of 0.4 iii) Binomial with 20 trials and success probability of 18 % iv) Poisson with a mean of 3.	06
	b	i) A server has 4 stages of service, each taking a duration of exponentially distributed time with a mean of 1.2 $minutes$. Generate one random variate for the total time the server takes for all the stages. What is the distribution of the overall service time? ii) Amount of money spent for formal training, and the profit a company earns are both normally distributed, with a mean and standard deviation of 12 and 2 lakhs of rupees for training expenses and 35 and 4 lakhs of Rupees for profit respectively. Also both the variables are positively correlated with a correlation coefficient of 0.72 . Generate one random variate each for training expense and profit. For each of the subparts of part (a) and part (b), use the same random number sequence as input, starting from the first number for each sub part. $0.9, 0.1, 0.4, 0.2, 0.9, 0.1, 0.6, 0.7, 0.8, 0.1, 0.9, 0.7, 0.9, 0.5, 0.7, 0.3, 0.2, 0.8, 0.2, 0.6$	10
		OR	
6	a	Develop a variate generator for X which has a piecewise linear density function, connecting coordinates$(0,0), (2, 0.2)$ and (4,0.6). Generate two variates using $R_i = 0.832,\ 0.021$	08
	b	Downtime of machines is found to be Gamma distributed with mean 2.2 $minutes$ and a standard deviation of 1.45 $minutes$. Generate a variate, using $R_i = 0.832,\ 0.021,\ 0.434, 0.716$	08
7	a	Why is optimization via simulation very difficult?	06
	b	Explain briefly the 3 step approach for validating models.	06
	c	What are some of the guidelines to follow during verification?	04
8	a	Differentiate between terminating and non-terminating simulations. Provide an example each of such simulations.	08
	b	What are some of the features to look for during model building and during output analysis in software? Name atleast 4 in each of these 2 stages.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Fast Track Examinations July - 19

Industrial Engineering and Management

SIMULATION MODELING AND ANALYSIS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Statistical tables are permitted.

PART-A

1	1.1	In simulation, iid with respect to random numbers represents _____.	01
	1.2	Building model correctly as per specification is _____.	01
	1.3	_____ distribution models different processes when only the minimum, most likely and maximum values of the distribution are known.	01
	1.4	Time to failure for a component is modeled by _____.	01
	1.5	_____ is the 1st step in input modeling.	01
	1.6	Auto correlation tests are used to test the _____ of the random numbers.	01
	1.7	Frequency test used are _____ and _____ tests.	01
	1.8	Two important statistical properties of random numbers are _____.	01
	1.9	In the formula $x_i + 1 = (ax_i + c)\ mod\ m$ for linear congruential method, what is X_0 called _____.	01
	1.10	Acceptance – Rejection technique is used to generate _____.	01
	1.11	Whether a simulation is terminating or not depends on _____ and _____.	01
	1.12	A continuous system that runs continuously for long period of time is called _____.	01
	1.13	A simulation system that runs continuously for long period of time is called _____.	01
	1.14	Kolmogorov – Smirnov test is used when sample size is _____ than 40.	01
	1.15	Activity is also called a _____ wait.	01
	1.16	Simulation models that contain no random variables are classified as _____.	01
	1.17	Discrete models are used to simulate only _____ events.	01
	1.18	Simulation of a bank would involve Random _____ and random service time.	01
	1.19	Exogenous describes _____ outside the system.	01
	1.20	Instantaneous occurrence that may change the state of the system is called as _____.	01

PART-B

2 a What are the parameters to be considered while simulating a queuing system? Explain. 08

b Consider a technical support call center where call center executives take calls and provide service over phone. The time between calls ranges from 1 to 4 minutes with the distribution shown as follows:

Time between arrivals (min)	*Probability*
1	0.25
2	0.40
3	0.20
4	0.15

There are two executives Johnny and David with the service time distribution given below:

Probability distribution of Johnny		*Probability distribution of David*	
Service time (min)	*Probability*	*Service time (min)*	*Probability*
2	0.30	3	0.35
3	0.28	4	0.25
4	0.25	5	0.20
5	0.17	6	0.20

Simulate the call center system for 10 customers assuming that if both are idle Johnny gets the call. Find the average delay. Use the following random digits: 08

Customer no.	1	2	3	4	5	6	7	8	9	10
Random digits for service	95	21	51	92	89	38	13	61	50	49
Random digits for arrival	–	26	98	90	26	42	74	80	68	22

3 a Explain how chi-square test can be used as a test for goodness of fit. 08

b Based on the runs up and down test, determine whether the following sequence is such that the hypothesis of independence with $\alpha = 0.05$ 0.41, 0.68, 0.89, 0.94, 0.74, 0.74, 0.91, 0.55, 0.62, 0.36, 0.27, 0.29, 0.72, 0.75, 0.08, 0.54, 0.02, 0.01, 0.36, 0.16, 0.28, 0.18, 0.01, 0.62, 0.73, 0.04, 0.83, 0.45, 0.13, 0.51, 0.63, 0.29. 08

OR

4 a Enumerate the steps involved in the development of useful model of i/p data. 08

b Perform test for uniformity using KS Test with level of significance $\alpha = 0.05$. Consider the following $R.No$ 0.44, 0.81, 0.14, 0.05 and 0.95. 08

5 a Lead times have been found to be exponentially distributed with mean 3.7 days. Generate 5 random mean time with the distribution 0.931, 0.417, 0.612, 0.079, 0.826. 08

b Explain the step by step procedure for random variate generation using inverse transform technique. 08

OR

6	a	Explain the step by step procedure of acceptance rejection technique.	08
	b	Generate three Poisson varieties with mean $\alpha = 0.2$ and the random numbers being $0.4357, 0.8353, 0.9952, 0.8004$.	08
7	a	Compare and contrast between verification and validation.	08
	b	Briefly explain the iterative process of calibrating a model.	08
8		Write short notes on the following: a) Trends in simulation softwares b) Batch means for interval estimation c) Replication method d) Optimization via simulation.	 04 04 04 04

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-18

Industrial Engineering and Management

SIMULATION MODELING AND ANALYSIS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	With the help of sketch show the variation in state variables change with respect to time in a discrete system and in a continuous system.	02
	1.2	Consider the choice between the analytical approach and the simulation. Mention one of the advantage of the simulation model versus the analytical model	01
	1.3	Consider a machine in a factory, the machine is subject to failures. The status of the machine is traced by defining the random variable $V(t)$ as follows $V(t) = \begin{cases} 0 & Idle \\ 1 & Busy \\ 2 & down \end{cases}$ A job that is to be taken up will wait for the machine for what values of $V(t)$.	01
	1.4	How do you overcome the problem related to model building when data are lacking? Give any one work around to simulate the model.	01
	1.5	Identify the input distribution which models the processes that can be viewed as the sum of several exponentially distributed process.	01
	1.6	Write down the set of Hypothesis null hypothesis and alternate hypothesis for the kolmognor-smirnov Goodness of fit test. Assume parameters of interest is exponentially distributed.	01
	1.7	Mention the recursive relationship for the linear congruential method of generating random numbers.	01
	1.8	Why are the Inverse-transform technique said to be most straight forward but not always the most efficient technique? Give one reason.	01
	1.9	What is the meaning of an empirical distribution?	01
	1.10	Mention the technique used for generating the random variate that is expected to follow the binomial distribution	01
	1.11	Random variate generation techniques that exploit the special features of a particular family of probability distribution, rather than being general purpose technique like the inverse transform or the acceptance rejection technique. Mention any one distribution that uses this technique for random variate generation.	01

	1.12	Mention any one distinguishing feature of models that leads to optimization of system performance via the simulation model.	01
	1.13	Define the term random search algorithm.	01
	1.14	Is the conceptual model assumptions about system components and system structure parameter valves abstractions and simplification) accurately represented by the model? This question does it pertain to verification or validation. Justify.	01
	1.15	Graphical representation of the model is essentially a form of self-documentation. How?	01
	1.16	Give a situational example for terminating simulations.	01
	1.17	Give an example for the use of intelligent initialization method for reducing bias in estimation.	01
	1.18	If R denotes the number of replications. Write down the formulae for estimating the minimum number of replications at a certain significance level and desired precision of $\pm t$ to estimate parameter θ	01
	1.19	"The purpose of the statistical analysis is to acquire same assurance that the estimates are sufficiently precise for the purpose of the model". Justify the statement.	01

PART-B

2	a	Estimate by simulation the average number of lost sales per week for an inventory system that functions as follows i) Whenever an inventory level falls to below 10 units an order is placed. Only one order can be outstanding at a time. ii) The size of each order is equal to $20 - I_1$, where I_1 be the inventory level when the order is placed. iii) If a demand occurs during a period when the inventory level is zero the sale is lost. iv) Daily demand is normally distributed with a mean of 5 units and a standard deviation of 1.5 units (round off demands to the nearest integer during the simulation) and (if negative value results , give it a demand of zero) v) Lead time is distributed uniformly between zero and 5 days – integers only vi) The simulation will start with 18 units in inventory. vii) For simplicity assume that orders are placed at the close of the business day and received after the lead time has occurred. That the lead time is one day the order is available for distribution on the morning of the second day of the business following the placement of the order. viii) Let the simulation run for 5 weeks.	10
	b	Name entities, attributes, activities, events and state variables for the following systems: i) Course registration online website portal ii) Fire service station iii) Airport terminal.	06

3	a	Consider the following data for the monthly number of stoppages (due to failures or any other reason) in the assembly line of an automotive assembly plant.	

12	14	26	13	19
15	18	17	14	10
11	11	14	18	13
9	20	19	9	12
17	20	25	18	12
16	20	20	14	11

		Apply the Chi-square test to the sample data of stoppages to test the hypothesis that the underlying distribution is Poisson at a significance level of $\alpha = 0.05$.	12
	b	Generate four digit random numbers by linear congruential method $X_0 = 117, a = 43, m = 1000$.	04
		OR	
4	a	For a preliminary version of a simulation model, the number of pallets X to be loaded on to a Truck at a loading dock was uniformly distributed between 8 to 24. Devise a method for generating X, assuming that the loads on successive trucks are independent. Use the discrete uniform distribution. Finally generate 10 successive trucks by using four digit random numbers.	08
	b	Following data represent the time to perform transactions in a bank measured in minutes 0.740,1.28,1.46,2.36,0.354,0.750,0.912,4.44,0.114,3.08,3.24,1.10,1.59,1.47,1.17, 1.27,9.12,11.5,2.42,1.77 Develop an input model for these data.	08
5	a	The time to attend a breakdown call is found to follow exponential distribution with a mean of 2 hours. Generate five(5) service times(use random number 94737,89326,86689,61674,07471)	05
	b	Service time of a bank teller is found to follow normal with $\mu = 5\ minutes$ and $\sigma = 1\ minute$. Generate five service times. (Use random number 8556,9437,8210,4315,3567)	05
	c	Generate three values from a geometric distribution on the range ($x \geq 1$) with mean 2. Such a geometric distribution has $p(x) = p(1-p)^{x-1}$ $(x = 1,2 \ldots)$ with mean $1/p = 2$, $p = 1/2$. thus X can be generated using $X = Q + \left\{\frac{ln\,(1-R)}{ln\,(1-p)} - 1\right\}$ $R_1 = 0.932, R_2 = 0.687, R_3 = 0.105$	06
		OR	
6	a	Downtimes for a high production candy making machine has found to be gamma distributed with mean 2.2 minutes and variance 2.10 minutes^2. Thus $\frac{1}{\theta} = 2.2$ and $\frac{1}{\beta\theta^2} = 2.10$ implying $\beta = 2.30$ and $\theta = 0.4545$. Carryout the steps to generate random variate over 2 trials based on gamma distribution.	06

	b	Develop an acceptance- rejection technique for generating a geometric random variable X, with parameters p on the range $\{0,1,2 \ldots\ldots\}$[Hint $-X$ can be thought of as the number of trials before the first success occurs in a sequence of independent Bernoulli's trails]	06
	c	Outline the procedure for generating empirical discrete distribution for discrete uniform distribution.	04
7	a	Compare validation in simulation to the validation of theories in physical sciences.	04
	b	Contrast the difficulties and compare the techniques in validating a model of a manually operated warehouse with Fork trucks and other manually operated vehicles versus a model of facility with automated guided vehicles, conveyers and on automated storage and retrieval system.	06
	c	Outline the process of calibrating a model through iteratively comparing the real system and the model generated system using simulation.	06
8	a	Bring out the relevance of point estimation and confidence interval estimation statistical procedures, in the context of simulating the production system to derive performance measures of cycle time and throughput time as outputs from the simulation model.	06
	b	Clearly bring out the distinctions between terminating simulations and steady state simulations.	04
	c	"The purpose of the simulation experiment is to obtain estimates of the performance measures of the system under study" Explain this in the context of the following systems. i) Customers arriving into the ATM kiosu ii) Batch manufacturing process in which a machine process jobs in batches of three units. iii) Repairs stop with two repairmen.	06

Industrial Ergonomics

V Semester SEE Question Papers

Course codes: 16IM52

- Nov/Dec 2019 (16IM52) - Set 1 [DSpace: UG JAN 21 FEB 21 PDF]
- Nov/Dec 2019 (16IM52) - Set 2 [DSpace: NOV-DEC-2019]
- Nov/Dec 2018 (16IM52)

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

INDUSTRIAL ERGONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Identify two reasons for the occurrence of ergonomic entropy in work systems.	02
	1.2	List the three primary effectors of human body.	02
	1.3	Identify the design issue in the interaction between processing and Environment components of a work system.	02
	1.4	Which is considered as a key factor in any assessment of Risk of musculoskeletal disorder in the workplace?	01
	1.5	What is meant by the term 'plasticity' in human ergonomics?	02
	1.6	Mention the method of measurement for the variable, 'Hand length', as per ISO/DIS7250.	02
	1.7	Identify the two important issues in psycho-anthropometry.	02
	1.8	Mention the method in which engineers and designers can avoid anthropometric mismatches in the absence of anthropometric data.	01
	1.9	Static trunk inclination greater than a particular value is not permitted as per ISO guidelines. Mention the value.	01
	1.10	Distinguish between 'sensation' and 'perception' in the context of cognitive Ergonomics.	02
	1.11	What are the two characteristics that define a pure tone?	02
	1.12	Identify the common auditory effect of exposure to land noise both in short term and long term.	01

PART-B

2	a	Examine the possible interactions in a basic work system. Also explain the interactions in which 'human' is involved.	08
	b	Depict a typical human-machine model. Also elaborate on the human and machine components.	08
3	a	Analyze the broad categories of anthropometric data necessary in the design of products and services.	08

	b	Describe the application of the following anthropometric variables in Ergonomic design of products. (i) Standing Eye Height (ii) Sitting Height (iii) Hip Breadth (iv) Vertical Reach (Sitting/Standing)	08
		OR	
4	a	Examine the approaches followed in solving the issue of designing products to suit a wide range of users.	08
	b	Discuss the philosophies of 'Minimum dimensions' and 'Maximum Dimensions' in the application of ergonomic principles in product designs.	08
5	a	Categorize the three classes of variables influencing the ergonomic work postures of human workers. Analyze any two of them.	08
	b	All the tasks have three sets of requirements that influence workspace design. What are they? Review any two of them in detail.	08
		OR	
6	a	Summarize the workplace design faults that increase postural stress in standing workers. Also write a note on how the postural constraint in standing workers can be relieved.	08
	b	Review the product features and identify suitable accessories for enhancing VDU workstation flexibility for the following items. (i) Source documents (ii) Seating (iii) VDU screen (iv) Desk	08
7	a	Examine the relevance of cognitive models of human operator for problem solving in organizations.	08
	b	Illustrate a composite, general model of human information processing as advocated by Wickens. Also explain how a stimuli is transformed into a Response.	08
8	a	Examine the Non-auditory effects of noise on human health in industrial environment.	08
	b	Review the principles of designing a sociotechnical job in organizations.	08

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

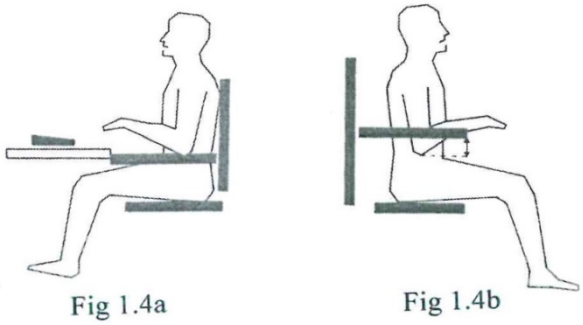
INDUSTRIAL ERGONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Identify the three primary effectors in Human body.	02
	1.2	List the basic interactions in a typical work system.	02
	1.3	When it comes to interaction between human and environment, the evaluation should be _____.	01
	1.4	State the common anthropometric mismatches in the following figures.  Fig 1.4a Fig 1.4b	02
	1.5	In the tent analogy, the _____ is the tent pole and the _____ are the guy ropes.	02
	1.6	_____ type of Anthropometric data is used in mechanical analysis of the loads on the body.	01
	1.7	Mention the three components of a postural triangle.	02
	1.8	In context to human operator as a decision maker, _____ models attempt to specify the optimum way of making a decision under a particular set of conditions.	01
	1.9	List the verbal methods of capturing data for modeling human operator.	02
	1.10	State the hygiene factors according to Herzberg.	02
	1.11	Distinguish between the job centered and person centered approaches to systems design.	02
	1.12	_____ principle of socio-technical job design states that information should go where it is needed in the first instance rather than via senior management to subordinate.	01

PART-B

2	a	Discuss with help of an example, the various interactions between different components of simple work system. Also mention the corresponding evaluation methods.	12
	b	Provide a note on Ergonomic entropy.	04
3	a	Explain the below mentioned common anthropometric variables and their influence in the design of chair for seated worker. i) Knee height and thigh depth ii) Buttock- popliteal length iii) Hip breadth iv) Sitting elbow height v) Sitting height vi) Popliteal height.	12
	b	Illustrate with help of an example, the role of mannequins in work space and equipment design.	04
		OR	
4	a	Write a report that explains to a design engineer, why anthropometric data should be used in the design of Human Machine systems. Give examples from everyday life to support your arguments in the report.	08
	b	Explain the common approaches adopted to achieve appropriate match between the product dimensions and the dimensions of a range of users. Also supplement your answer with relevant examples.	08
5	a	Discuss with help of an example, the factors that influence the working posture.	10
	b	List the relative advantages of standing work position over sitting.	06
		OR	
6	a	Explain with an example, the task requirements that influence the workspace design.	08
	b	Explain the workspace design faults which increase postural stress in standing workers.	08
7	a	Describe the four stages involved in reviewing strategies and biases in human decision making.	10
	b	Explain the three main classes of heuristic techniques for human decision making.	06
8	a	Explain with a block diagram, the steps involved in systems approach for design of a system.	12
	b	Explain the following principles of socio-technical system theory: i) Compatibility ii) Boundary location.	04

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-18

Industrial Engineering and Management

INDUSTRIAL ERGONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Identify the common reasons for the occurrence of "incompatibility' in a human machine system.	02
	1.2	Categorize the main risk factors for occurrence of musculo skeletal disorders.	02
	1.3	*ISO/DIS* 7250 lists different anthropometric variables and their measurement methods. Mention the method of measurement for "eye height" and "shoulder breadth".	02
	1.4	What is meant by a "workspace envelope"?	02
	1.5	Identify an appropriate anthropometric variable for the following work related problem: a) Reaching for work objects. b) Work surface too high.	02
	1.6	Identify the method in which engineers and designers can avoid anthropometric mismatches in the absence of anthropometric data.	02
	1.7	Recognize the three sets of tasks requirements that influence workspace design.	02
	1.8	Mention the two aspects that define a pure tone.	02
	1.9	Distinguish between 'Sensation' and 'Perception' in the context of cognitive ergonomics.	02
	1.10	Differentiate between "Job Enlargement" and "Job enrichment" in the context of understanding employees emotional responses to job design.	02

PART-B

2	a	Describe different components of a typical human machine system. Support your answer with an appropriate illustration.	08
	b	The skeletal and muscular system can be likened to the scaffolding to which all other parts are attached. In the light of above statement, describe the functions of the human skeletal and muscular systems.	08
3	a	Human variability is considered to be one of the important factors in designing products for a population of users. In this context anthropologists distinguish four types of human adaptations. Analyze the same.	08

	b	Different sets/types of anthropometric date is required in the design of products and services. In this context, analyze the broad categories of anthropometric data.	08
		OR	
4	a	Examine the approaches followed in solving the problem of designing products to suit a range of users.	08
	b	Describe and provide examples of use of the following anthropometric variables: i) Standing eye height ii) Sitting height iii) Standing knuckle height iv) Sitting elbow height.	08
5	a	Ergonomic workstation design encourages good posture. Three classes of variables influence the ergonomic work postures. In the light of above statement, analyze the three classes of variables.	08
	b	Anthropometric mismatches can have serious consequences for health and efficiency of the workers. Substantiate the above statement with suitable illustrations.	08
		OR	
6	a	Summarize the work space design faults increase postural stress in standing posture adapted by the workers.	08
	b	With the help of a suitable illustration, list the essential furniture considerations in visual Display Unit (VDU) work.	08
7	a	Illustrate and describe Wicken's composite and generic model of human information processing.	08
	b	Analyze the growing importance of cognitive Ergonomics in the context of industries embracing internet of things philosophy. Support your answer with a suitable example.	08
8	a	Evaluate the common auditory and Non- auditory effects of noise on health.	08
	b	Elaborate the steps needed for an ergonomic approach to design as drafted by ISO/DIS 6385.	08

Statistical Quality Control

V Semester SEE Question Papers

Course codes: 12IM54

- Nov/Dec 2017 (12IM54)
- Fast Track Jul 2017 (12IM54)
- Nov/Dec 2016 (12IM54)
- Fast Track Jul 2016 (12IM54)

12IM54

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-17

Industrial Engineering and Management

STATISTICAL QUALITY CONTROL

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

15. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
16. Answer FIVE full questions from Part B.
17. Use of statistical and control chart tables permitted.

PART-A

1	1.1	Mention any two critical-to-quality characteristics (CTQ).	02
	1.2	Differentiate between change and assignable cause of quality variation.	02
	1.3	What are Rational subgroups?	02
	1.4	What are the differences between control limits and specification limits?	02
	1.5	Give the expression for defect per million opportunity ($DPMO$).	02
	1.6	Write some statistics that can be used to measure the capability of a process.	02
	1.7	Mention some important considerations in forming lots for inspection.	02
	1.8	How do you measure the performance of an acceptance sampling plan?	02
	1.9	Draw a general model of a process.	02
	1.10	Mention the applications of statistical experimental design in engineering design.	02

PART-B

2	a	Select a specific product or service and discuss how the eight dimensions of quality impact its overall acceptance by consumer.	08
	b	Identify critical to quality to improve service quality for a banking process using the $DMAIC$ problems solving process.	08
		OR	
3	a	State a set of decision rules for detecting nonrandom patterns on control charts as suggested by the Western Electric Handbook(1956).	08
	b	Mention the various reasons for the popularity of control charts.	08
4		The following table contains the inside diameter measurements of forged automobile engine piston rings. Each sample as subgroup consists of five piston rings. Construct $\bar{x}$ and r charts using the data.	

Sample Number	*Sample Measurements in mm*				
1	36	35	34	33	32
2	31	31	34	32	30
3	30	30	32	30	32
4	32	33	33	32	35
5	32	34	37	37	35
6	32	32	31	33	33
7	33	33	36	32	31
8	23	33	36	35	36
9	43	36	35	24	31
10	36	35	36	41	41
11	34	38	35	34	38
12	36	38	39	39	40
13	36	40	35	26	33
14	36	35	37	34	34
15	30	37	33	34	35
16	28	31	33	33	33
17	33	30	34	33	35
18	27	28	29	27	30
19	35	36	29	27	32
20	33	35	35	39	36

16

OR

5 The following table contains the flow width measurements(microns) for the hard bake process. Construct $\bar{x}$ and R chart for the given data.

Sample Number	Wafers				
1	1.3235	1.4128	1.6744	1.4573	1.6914
2	1.4314	1.3592	1.6075	1.4666	1.6109
3	1.4284	1.4871	1.4932	1.4324	1.5674
4	1.5028	1.6352	1.3841	1.2831	1.5507
5	1.5604	1.2735	1.5265	1.4363	1.6441
6	1.5955	1.5451	1.3574	1.3281	1.4198
7	1.6274	1.5064	1.8366	1.4177	1.5144
8	1.419	1.4303	1.6637	1.6067	1.5519
9	1.3884	1.7277	1.5355	1.5176	1.3688
10	1.4039	1.6697	1.5089	1.4627	1.522
11	1.4158	1.7667	1.4278	1.5928	1.4181
12	1.5821	1.3355	1.5777	1.3908	1.7559
13	1.2856	1.4106	1.4447	1.6398	1.1928
14	1.4951	1.4036	1.5893	1.6458	1.4969
15	1.3589	1.2863	1.5996	1.2497	1.5471
16	1.5747	1.5301	1.5171	1.1839	1.8662
17	1.368	1.7269	1.3957	1.5014	1.4449
18	1.4163	1.3864	1.3057	1.621	1.5573
19	1.5796	1.4185	1.6541	1.5116	1.7247
20	1.7106	1.4412	1.2361	1.382	1.7601
21	1.4371	1.5051	1.3485	1.567	1.488
22	1.4738	1.5936	1.6583	1.4973	1.472
23	1.5417	1.4333	1.5551	1.5295	1.6866
24	1.6399	1.5243	1.5705	1.5563	1.553
25	1.5797	1.3663	1.624	1.3732	1.6887

16

6 The following table shows the data collected in manufacturing unit. Compute the trial limits for P chart and draw the control chart for fraction defective.

Subgroup Numbers	*Number inspected*	*Number of defective*	*Subgroup Number*	*Number inspected*	*Number of defective*
1	300	12	13	300	9
2	300	3	14	300	3
3	300	9	15	300	0
4	300	4	16	300	5
5	300	0	17	300	7
6	300	6	18	300	8
7	300	6	19	300	**18**
8	300	1	20	300	2
9	300	8	21	300	5
10	300	11	22	300	6
11	300	2	23	300	0
12	300	10	24	300	3
			25	300	2

16

OR

7 a A subgroup of 5 items each are taken from a production process at a regular interval. A certain quality characteristic is measured and $\bar{x}$ and R values are computed. After 25 subgroups it is found that $\Sigma\bar{x} = 357.50$ and $\Sigma R = 8.80$. If the specification limits are 14.40 ± 0.40; If the process is in statistical control what conclusion can you draw about the capability(or ability) of the process to produce items within specification?(For subgroup size 5 $d_2 = 2.326$) 10

b Bringout different types of attribute charts. 06

8 a In a double sampling plan $N = 5000, n_1 = 100,\ c_1 = 0, n_1 + n_2 = 200,\ c_2 = 2$

i) Compute P_a for 1% defective lot

ii) What will be AOQ if the lot submitted are 1% defective. 10

b Differentiate between different types of sampling plane. 06

OR

9 a Explain the different considerations in forming lots for inspection to influence the effectiveness of the acceptance-sampling plan. 06

b Three single sampling plans are given. Identify which one of the three sampling plans are the best at fraction defective of 0.005.
Sample 1. $N = 3000, n = 400, c = 4$
Sample 2. $N = 2500, n = 300, c = 1$
Sample 1. $N = 2000, n = 100, c = 2$ 10

10 a Define 'Reliability.' Derive an expression for reliability of a component for time 't' $R(t)$. 08

	b	Define the terms: i) $MTTF$ ii) $MTBF$ iii) $MTTR$ iv) $MTBM$.	08
		OR	
11	a	Recommend a procedure for designing an experiment.	08
	b	With the help of suitable example explain the objectives and applications of experimental design.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

STATISTICAL QUALITY CONTROL

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

8. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
9. Answer FIVE full questions from Part B.
10. Use of statistical and control chart tables permitted.

PART-A

Clarify the meaning of the following statements in the context of statistical quality control processes, procedures, policies and programs. Reason out the need to follow these percepts, principles and practices to manage the quality of the product and process and continuously improve the quality aspects.			
1	1.1	In many cases customers rely on the past reputation of the company concerning quality aspects of its products. This reputation is directly influenced by failures of the product that are highly visible to the public or that require product recalls and by how the customer is treated when a quality related problem with the product is reported. This dimension Is known as perceived quality.	01
	1.2	Quality improvement is the reduction of variability in process and products. Excessive variability in process performance results in waste.	01
	1.3	One of the Deming's 14 points: Do not rely on mass inspection to "control" quality. All inspection can do is sort out defectives and at that point it is too late-the organization already has paid to produce those defectives.	01
	1.4	A process that is operating with only chance causes of variation present is said to be "In statistical control". In other words, the chance causes are an inherent part of the process.	01
	1.5	If the process is in control all the plotted points on the control chart should have an essentially random pattern. Methods for looking for sequence or non-random patterns can be applied to control charts as an aid in detecting out-of –control conditions.	01
	1.6	Preliminary samples are used to construct $\overline{x}$ and R control charts. It is customary to treat these limits obtained on the basis of these preliminary samples as "Trial control limits".	01
	1.7	Control limits are driven by the natural variability of the process (measured by the process standard deviation σ) that is by the natural tolerance limits of the process.	01

1.8	Samples for constructing the control charts for variables namely $\overline{X} - R$ control charts should be drawn as rational subgroups. $\overline{X}$ Charts monitor between–sample variability (variability in the process overtime) and the R chart measures within-sample variability (the instantaneous process variability at a given time).	01
1.9	Cyclic patterns on the $\overline{X}$ chart may result from systematic environmental changes such as temperature, operator fatigue, regular rotation of operators and/or machines or fluctuations in voltage or pressure or some other variable in the production equipment.	01
1.10	In interpreting patterns on $\overline{X}$ and R charts one should consider the two charts jointly if the underlying distribution is normal, then the random variables $\overline{X}$ and R computed from the same sample are statistically independent.	01
1.11	The fraction non-conforming control chart has three parameters that must be specified; the sample size, the frequency of sampling and the width of control limits.	01
1.12	Care must be exercised in interpreting points that plot below the lower control limit on the control chart for fraction nonconforming. These points often do not represent a real improvement in process quality.	01
1.13	Defect or nonconformity data are always more informative than fraction nonconforming because there will usually be several different types of non-conformities.	01
1.14	The units selected for the inspection from the lot should be chosen at random and they should ne representative of all the items in the lot. The random sampling concept is extremely important in acceptance sampling.	01
1.15	A consumer often establishes a sampling plan for a continuing supply of components or raw material with reference to an acceptable quality level (AQL).	01
1.16	The principal advantage of a double sampling plan with respect to single plan is that it may reduce the total amount of required inspection.	01
1.17	The important measure of the performance of an acceptance sampling is the operating characteristic curve (OC) curve. This curve plots the probability of accepting the lot versus the lot fraction defective.	01
1.18	Experimental design methods to be used at times in process development or process troubleshooting to improve process performance or to obtain a process that is robust or intensive to external sources of variability.	01
1.19	A factorial design of experiments is the one in which the trials or replicate of the experiments all possible combinations of the levels of the factors are investigated.	01
1.20	The main effect of a factor is defined as the change in response produced by the change in the level of the factor.	01

PART-B

2	a	Garvin (1987) has identified eight components or dimensions of quality. Summarize the key points concerning these eight dimensions of quality. The two other definitions of Quality are Quality means fitness for use and Quality is inversely proportional to variability. Discuss these definitions and their applicability to product and service quality.	08
	b	Discuss the chance and assignable causes of Quality variation. How does the distinction between chance causes and assignable causes provide the statistical basis for control chart? Elaborate on the linkages between causes of variation and the statistical control of dimensional quality.	08
		OR	
3	a	Bring out the relationship between $SIPOC$ and $DMAIC$. Take an example of the machining process and outline how you would address the quality issue that arises due to dimensional quality problems using $DAMIC$ process (outline the tools that you would use at each stage).	08
	b	What is the difference between control chart for averages ($\overline{X}$) and control chartfor individual measurements (X)? Why is control chart for averages preferred to control chart for individuals for quality control procedures in mass production environments? (Give illustrations on patterns on both the type of control charts).	08
4	a	Samples of size $n = 5$ are taken from a manufacturing process every hour. A quality characteristic is measured and $\overline{X}$ and R are computed for each sample. After 25 samples have been analyzed we have, $\sum_{i=1}^{25} \overline{X}_i = 662.50$ and $\sum_{i=1}^{25} \overline{R}_i = 9.00$ The quality characteristic is normally distributed. i) Find the control limits for the $\overline{X}$ and R charts. ii) Assume that both charts exhibit control if the specifications are 26.40 ± 0.50, estimate the fraction nonconforming. iii) If the mean of the process were 26.80, what fraction from conforming would result?	08
	b	A normally distributed quality characteristic is monitored through the use of $\overline{X}$ and R chart. These charts have the following parameters: $\underline{\overline{X} - chart}$ $\quad$ $\underline{R - chart}$ $UCL = 626.0$ $\quad$ $UCL = 18.795$ $Central\ line = 620.0$ $\quad$ $Central\ line = 8.236$ $LCL = 614.0$ $\quad$ $LCL = 0$	

Both charts exhibit control

i) What is the estimated standard deviation of the process?

ii) Suppose an 'S' chart were to be substituted for the R-chart, what would be the appropriate parameters of the s chart.

iii) If specifications of the product were 610 ± 15, what would be your estimate of fraction nonconforming?

iv) What would be done to reduce fraction non-conforming?

v) What is the probability of detecting the shift in the process mean to 610 on the first sample following the shift(σ remains constant) 08

OR

5 a The data in the table below pertain to molecular weight measurements made every two hours on a polymer.

Molecular weight measurements on a polymer				
2048	2039	2051	2002	2029
2025	2015	2056	1967	2019
2017	2021	2018	1994	2016
1995	2010	2020	2001	2010
1983	2012	2023	2013	2000
1943	2003	2036	2016	2009
1940	1979	2019	2019	1900
1947	2006	2000	2036	1986
1972	2042	1986	2015	1947
1983	2000	1952	2032	1958
1935	2002	1988	2016	1983
1948	2010	2016	2000	2010
1966	1975	2002	1988	2000
1954	1983	2004	2010	2015
1970	2021	2018	2015	2012

i) Construct an individual control chart with the standard deviation estimated using the moving range control chart. How do you interpret the chart? Are you comfortable with the interpretation?

ii) Construct a $CUSUM$ control chart on the residuals by fitting a model $x_i = \xi + \phi x_{i-1} + \tau_i$. 12

b What is a pre control chart? What is its utility? 04

6 a The data in the table below gives the number of non-conforming bearing and seal assemblies in samples of size 100. Construct a fraction non-conforming control chart for these data. If any point plot out of control, assume that assignable cause can be found and determine revised control limits.

Sample number	*Number of non conforming assemblies*	*Sample number*	*Number of non − conforming assemblies*
1	7	11	6
2	4	12	15
3	1	13	0
4	3	14	9
5	6	15	5
6	8	16	1

$Sample\ number$	$Number\ of\ \ non\ conforming\ assemblies$	$Sample\ number$	$Number\ of\ non-conforming\ assemblies$
7	10	17	4
8	5	18	5
9	2	19	7
10	7	20	12

12

b Why is that in a fraction nonconforming chart a point plotting below the lower limit indicates the presence of a favorable assignable cause and that it must be traced? 04

OR

7 a What factors are taken into consideration to make a choice between using a variable control chart and an attribute control chart? 04

b A control chart for non conformities is to be constructed with $c = 2.0$, $LCL = 0$ and UCL such that the probability of a point plotting outside the control limits when $c = 2.0$ is only 0.005.

i) Find the UCL (Upper control limit)

ii) What is the type I error probability if the process is assumed to be out of control only when two consecutive points fall outside control limits? 08

c Consider the machining of a cast work piece. What are some of the casting defects which can influence the machinability of work pieces? 04

8 a Sketch the ideal OC curve. Explain the critical points on the ideal OC curve. 04

b Consider a sampling plan with $n = 150$ and $c = 2$ being used for receiving inspection where the supplier ships the product in lots of size $N = 3000$.

i) Draw the OC curve for this plan

ii) Draw the AOQ curve and find the $AOQL$. 12

OR

9 a Consider a Double sampling plan with the following parameters $n_1 = 50 c_1 = 1, n_2 = 100, c_2 = 3$. Draw the operating characteristic curve by building up the probability of acceptance equations, based on P_a^I and P_a^{II} denoting the probability of acceptance based on first and second sample respectively. 12

b Outline the mechanics of working of the published sampling plans. Give any one example of a published sampling plan. 04

10 a What is experimental design? Explain the role of experimental design with the help of a general model of a process and the concepts related to quality improvement of a process. 08

b Develop an $ANOVA$ table for the following data with respect to experiment on paint adhesion-

$Primer\ type$	$Application\ method$		
	$Dipping$	$Spraying$	$Total$
1	4.0 4.5 4.3	5.4 4.9 5.6	28.7
2	5.6 4.9 5.4	5.8 6.1 6.3	34.1
3	3.8 3.7 4.0	5.5 5.0 5.0	27.0
$Total$	40.2	49.6	89.8

08

		OR	
11	a	Explain the general failure rate curve. Why is it called the bath tub curve? What is the significance of this curve?	08
	b	Studies of a single machine tool system showed that the time the machine operates before breaking down is exponentially distributed with a mean of 10 hours. Determine the failure rate and the reliability. Find the probability that machine operates for at least 12 hours before breaking down.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

STATISTICAL QUALITY CONTROL

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

14. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
15. Answer FIVE full questions from Part B.
16. Use of statistical and control chart tables permitted.

PART-A

1	1.1	The quality _____ are the means of translating the term 'fitness for use' into the language of technologists.	01
	1.2	The costs incurred to keep failure and appraisal costs to a minimum are called _____.	01
	1.3	When only in Huma of _____ are operating in the system, the process is said to be under statistical control.	01
	1.4	The 3σ limits in control chart for defects are based on _____ probability distribution.	01
	1.5	When a series of consecutive points fall above or below the central line of a '$\bar{X}$' chart, the control chart pattern is called _____..	01
	1.6	Concluding that the universe has not changed, when it has really changed is called _____ error.	01
	1.7	Wider the tolerance limits _____ is the probability of making tye-1 error.	01
	1.8	The natural tolerance of a process includes _____ percentage of measurements and it is called _____.	02
	1.9	When $(USL - LSL) = 6\sigma^1$, frequent use of control chart becomes _____.	01
	1.10	When destructive testing of a costly item is required in QA, the best acceptance sampling method to be used is _____.	01
	1.11	Fixed sample size, irrespective of lot size tends towards _____ quality protection.	01
	1.12	The quality level having a probability of acceptance of 50% for a given sampling plan represents _____ quality level.	01
	1.13	The region below 'Rejection line' and above 'Acceptance line' in a sequential sampling plan represents _____ region.	01
	1.14	_____ represents the maximum possible value of average percentage defectives in the out going products in an 'Acceptance-Rectification' scheme.	01
	1.15	In the experimental designs, the process in which every experimental unit has the same chance of being allocated to treatments is known as _____.	01

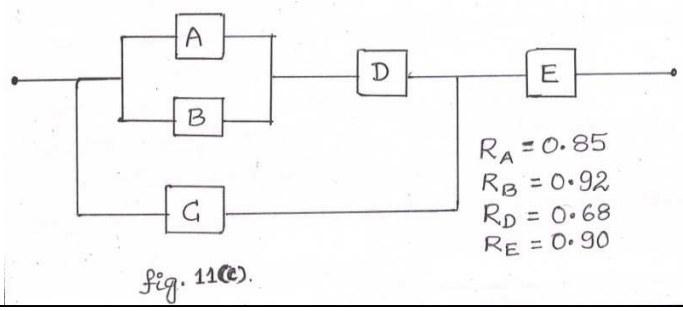
	1.16	If 'θ' is $MTBF$, then failure rate λ is equal to _____ and reliability of the unit over the time 't' is equal to _____.	02
	1.17	If there are 'm' levels for a factor 'A' and 'n levels for a factor 'B' in a factorial design of experiments, then there shall be _____ combinations.	01
	1.18	Stand-by redundancy produces _____ reliability compared to parallel redundancy.	01

PART-B

2	a	Define 'Quality'. Giving two examples each, explain the five quality characteristics.	08
	b	How are quality costs classified? Explain the 'Internal failure costs' giving its various components.	08
		OR	
3	a	Distinguish clearly between: i) Chance causes and assignable causes of variation ii) Type-I and Type-II errors.	08
	b	With neat sketches explain the following control chart patterns: i) Trends ii) Erratic fluctuations.	08
4		Control charts for '$\bar{X}$' and 'R' are to be maintained for a part which has a specification of $2.05^{\pm 0.02}$mm. The sample size is 4. The values of '$\bar{X}$' and 'R' totaled after 20 subgroups show the values at $41.283mm$ and $0.28mm$ respectively. Compute the trial control limits for both the charts. If the process is in control, what can you tell about the capacity of the process to meet the given specifications? What are your suggestions to improve the current situation?	16
		OR	
5	a	List at least six objectives of control charts for variable.	06
	b	Control charts for '$\bar{X}$' , 'R' and 'σ' are to be maintained on drawings from a bowl of chips. The distribution is approximately normal in this case. If the subgroup size is 5 with universal average and standard deviations at 60 and 8 respectively. Compute the control limits for the 3 charts. Estimate the 'process capability assuming that the process is in statistical control.	10
6	a	The number of defects recorded for 10 lots of production of a component are given below. Construct an appropriate control chart, using average lot size and offer your suggestions for future period.	10
	b	Compare and contrast control chart for variables with control chart for attributes on at least 6 counts.	06
		OR	

Lot Number	1	2	3	4	5	6	7	8	9	10
Units	46	43	49	45	45	47	44	46	45	42
Defects	118	96	125	87	99	123	104	123	112	98

7	a	An item is made in large lots and a sample of 200 items are drawn and given 100% inspection. The record sheet for first 25 samples showed a total of 75 defectives. i) Determine trial control limits for 'np' chart. ii) Assuming that all the points fall within control limits, what is the process average fraction defective? iii) If the 'p' remains constant, what is the probability that the 26th sample contains less than 4 defectives but more than 1 defective?	10
	b	What is 'process capability'? Explain the procedure of computing process capability using average range method?	06
8		The following data refers to a 'Double sampling plan; (with the usual notations): $N = 50{,}000;\ n_1 = 150;\ c_1 = 0;\ n_2 = 100;\ c_2 = 1$. If 1% defective lots are submitted, compute: i) P_a ii) AOQ iii) ASN iv) ATI v) AFI.	16
		OR	
9		A single sampling plan has the following specifications (with usual notations): $n = 300;\ c = 5$ i) Construct the OC curve of this plan and obtain P_a at 0.2% and 2.0% p values. i) Determine the $AOQL$ of this plan.	16
10	a	A single factor experiment is to be designed for estimating the tensile strength of a composite material. One of the constituents of the composite material varies in the combination at 5%, 10% and 20%. Six specimen are prepared for each combination. Explain the process of data tabulation for a completely randomized design.	08
	b	Explain the application of a "linear statistical model" and $ANOVA$ for the above case.	08
		OR	
11	a	Define 'Reliability'. Derive an expression for reliability of a component for time 't' $R(t)$.	06
	b	Define the terms: i) $MTTF$ ii) $MTBF$ iii) $MTTR$ iv) $MTBM$.	04

c	Find the reliability of the component '*c*' in the system shown in fig.11c such that the system reliability is 88%.  fig. 11(c).	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

STATISTICAL QUALITY CONTROL

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

7. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
8. Answer FIVE full questions from Part B.
9. Use of statistical and control chart tables permitted.

PART-A

1	Give the one string of words connotation for the following descriptive sentences, based on the understanding of the subject. Credit uses be given for the precise identification of the technical term and the usage of the word in that subject domain (Abbreviated terms needs the expansion form).	
1.1	The set of operational, managerial and engineering activities that a company uses to ensure that the quality characteristics of product are at the nominal or required levels.	01
1.2	Process monitoring tool and technique through which unusual sources of variability are detected as regards their presence in the system being studied.	01
1.3	The phase in the six sigma quality improvement program which leads quantification of the quality problem in terms of the defects per million opportunities.	01
1.4	The principle which specifies that samples should be selected so that if assignable causes are present, the chances for difference between subgroups will be maximized while the chance for differences due to these assignable cause with in a subgroup will be minimized.	01
1.5	A sequence of observations which fall on one side of the central line in a control chart – for example 19 to 25 points plot below the central line, while only 6 of them plot above. This being a nonrandom pattern in the control chart.	01
1.6	The tendency for the points to cluster artificially around the central line at the stage of interpretation of the $\bar{X} - R$ control chart. This pattern identified as out-of-control condition in the R-chart.	01
1.7	The control chart suited for the situation wherein multiple measurements are taken on the same unit of product such as measuring oxide thickness at different locations on a wafer in semiconductor manufacturing.	01
1.8	The expression of process capacity as the ratio of the specification width $(USL - LSL)$ to the spread of the process estimated as $6\hat{\sigma}\,.\hat{\sigma}$ being _____.	` 01

	1.9	The characteristic curves which are used to study the shifts and detect the same by estimating the probability of detecting the magnitude of shift (denoted as the β risk for various sample sizes).	01
	1.10	The expected number of samples taken before the shift is derived and is denoted as _____.	01
	1.11	The ratio of the number of nonconforming items in a population to the total number of items in the population.	01
	1.12	The chart in which the limits are established based on number nonconforming rather than the fraction nonconforming. The treatment being attribute type.	01
	1.13	The chart most suited for monitoring the number of workmanship related nonconformities observed in the final inspection of external hard disk assemblies, with data being available day wise for 10days. The total number of assemblies inspected and the total number of non conformities being recorded for each day.	01
	1.14	The proper type of control chart suited for the situation in which the process is in trouble and the control chart tool can be useful for diagnostic purposes (mobile shooting).	01
	1.15	The measure of the natural or inherent variability at a specified time that is 'Instantaneous' variability and the variability overtime. Measure of the uniformity of the process.	01
	1.16	The operating characteristic curve that discriminates perfectly between good and bad lots and that runs horizontally at a probability of acceptance $Pa = 1.00$ until a level of lot quality that is considered 'bad' is reached at which point the curve drops vertically to a probability of acceptance $Pa = 0.00$, and then the curve drops vertically to a probability of acceptance $Pa = 0.00$, and then the curve horizontally again for all lots fraction defectives grater than the undesirable level.	01
	1.17	The poorest level of Quality for the vendor's process that the customer/consumer ward consider to be acceptable as a process average. The context being continuing supply of component or raw materials supplied by vendors and evaluated using Acceptance sampling schemes.	01
	1.18	A measure related to rectifying inspection scheme, which gives the total amount of inspection required by the given sampling program denoted as _____.	01
	1.19	The exact sampling distribution of the number of defective items in the sample. The basis for plotting the type – A occurrences.	
	1.20	The tests or series of tests in which purposeful changes are made to the input variables of a process so that we may observe and identify corresponding changes in the output response.	01

PART-B

2	a	Distinguish between chance and assignable causes of Quality variation. Give the economical implication of these systems for cause identification from the perspective of a typical manufacturing process for statistical process control.	06
	b	Write briefly on the Analysis of patterns on control charts. Why is it necessary for the analyst to distinguish between systematic pattern (non-random pattern) and random pattern, from SPC point of view?	04

	c	Suppose it is necessary in a computing environment to track the time taken to process the algorithm in the processing unit of the computing environments. The CPU processing times are made available to you. Which tool will be useful in understanding the variability in the processing times? How would you use this tool to setup a SPC procedure?	04
	d	Define Cyclic pattern. What causes it?	02
		OR	
3	a	When drawing samples or subgroups from a process, do you want assignable causes occurring with in the subgroup or between them? Completely explain your answer with illustration.	04
	b	Mention the tools that are used in the measure phase of the $DMAIC$ approach to problem solving. Detail any one of these tools with respect to inputs, purpose and the output obtained through the application of the tool.	06
	c	How do the costs of sampling, the costs of producing an excessive number of defective units and the costs of searching for assignable causes impact the choice of parameters of a control chart?	04
	d	Define Pareto Principle.	02
4	a	Samples of size $n = 5$ are drawn from a manufacturing process every hour. A quality characteristic is measured and $\bar{X}$ and R are computed for each sample. After 25 samples have been analyzed we have $\sum_{i=1}^{25} \bar{x}_i = 662.50$ & $\sum_{i=1}^{25} \bar{R}_i = 9.00$. the quality characteristic is normally distributed. i) Find the control limits for the $\bar{X}$ & R charts. ii) Assume that both charts exhibit control. If the specifications are 26.40 ± 0.50. Estimate the fraction nonconforming. iii) If the mean of the process were set at 26.40, what fraction nonconforming would result?	08
	b	An $\bar{X}$ chart is to be established based on the standard values $\mu = 600$ & $\sigma = 12$, with $n = 9$. The control limits are to be based on an α risk of 0.01. What are the appropriate control limits?	05
	c	What is the purpose of the Shewart control chart for individual measurements? Give any situations which require the application of the control charts for individual measurements.	03
		OR	
5	a	A process is to be maintained with standard values $\mu = 10$ and $\sigma = 2.5$. The sample size is three i) Find the center line and the control limits for the $\bar{X}$ chart. ii) Find the center line and the control limits for R-chart iii) Find the center line and the control limits for the S-chart	08

	b	An X-chart is used to control the mean of a normally distributed quality characteristic. It is known that $\sigma = 6.0$ & $n = 4$. The central line $= 200, UCL = 209$ & $LCL = 191$. If the process mean shits to 188. Find the probability that the shift is detected on the first subsequent sample.	05
	c	Define a trend in process levels. Illustrate the trend for a control chart. What are the causes likely to generate a trend?	03
6	a	Define the following clearly: i) Attribute data ii) Sample fraction nonconforming iii) Trail control limits.	04
	b	A control chart for fraction nonconforming indicates that the current process average is 0.03. The sample size is constant at 200 units: i) Find the three-sigma control limits for the control chart. ii) What is the probability that a shift in the process average to 0.08 will be detected in the first subsequent sample? What is the probability that this shift will be detected at least by the forth sample following the shift?	08
	c	Mention any three example situations which will favor the choice of the variable control charts over a attribute control charts. Justify.	04
		OR	
7	a	A production line assemblies is related to electric clocks. The average number of nonconformities is estimated to eb 0.75. The quality engineer wishes to establish a C-chart fro this operation using an inspection unit of six clocks. Find the three-sigma limits for this chart.	06
	b	A control chart for nonconformities per unit uses 0.95 and 0.05 probability limits. The central line is at $V = 1.4$. Determine the control limits if the sample size is $n = 10$.	06
	c	Consider the results of monitoring from two processes as shown here: $Process\ A$: $\overline{X_A} = 100$, $\overline{\delta_A} = 3$ $Process\ B$: $\overline{X_B} = 105$, $\overline{\delta_B} = 1$ Specification are at 100 ± 10. Calculate C_P, C_{PK} & C_{PM} and interpret these ratios.	04
8		A single sampling plan is as follows $N = 3000, n = 80, C = 1$ a) Plot the OC curve for the above plan b) What is the producer's risk if AQL is 1.5%? c) What is the customer's risk if $LIPD$ is 4.5%? d) Determine the ATI for the above plan at 1.25% defective level of incoming lot.	16
		OR	

9	a	Which of the two following sampling plans would you prefer to increase the chance of getting the incoming lots accepted? $Plan\ A$: $Single\ sampling\ plan$, $N = 5000$, $n = 100$, $C = 2$ $Plan\ B$: $Double\ sampling\ plan$, $N = 5000$, $n_1 = 100$, $n_2 = 50$, $C_1 = 2$, $C_2 = 3$ Assume that a lot with 1% defective is submitted for inspection. Also calculate the ATI for the sampling plans.	12
	b	Write briefly on published sampling plans and their applications.	04
10	a	Explain the process characterization and process optimization steps for Quality Improvement, taking the example of Machining components using a CNC machine or any other example that you are familiar with.	06
	b	An experiment was conducted to investigate the effect of glass type and phosphor type on the brightness of a television tube. The response measured is the current necessary (in micro amps) to obtain a specified brightness level. The data are shown here. Analyze the data and draw conclusions. $Glass\ type$ / $Phosphor\ type$: 1 — 280, 300; 290, 310; 285, 295 — 2 — 230, 260; 235, 240; 240, 235	10
		OR	
11	a	Define the phases of failures as identified in the general failure rate characteristic curve or the bath tub curve. Mention the causes of failures in each of the phases of the bath tub curve.	06
	b	What is the distinction between $MTBF$ and $MTTF$? Give the application for each of these two?	04
	c	The failure rates of three components are 0.065×10^{-3}, 0.18×10^{-3} & 0.96×10^{-3} failures per hour. Evaluate the failure rate, $MTTF$ or the system. Estimate the reliability of the system for $500 hrs$ (Assume components are connected in series).	06

Engineering Economy and Financial Accounting

V Semester SEE Question Papers

Course codes: 10IM55

- Fast Track Jul 2016 (10IM55)

10IM55

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

ENGINEERING ECONOMY AND FINANCIAL ACCOUNTING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Use of Engineering Economy tables is permitted.
4. Cash flow diagram and formulation should be shown wherever required.

PART-A

1	1.1	Factors that cannot be readily be expressed in terms of financial units are known as _____.	01
	1.2	A uniform series of deposits intended to accumulate to a given future amount of money is known as _____ fund.	01
	1.3	The only difference between an increasing and decreasing gradient series is the sign infront of the _____ amount.	01
	1.4	When the compounding period is given without the interest rate identified as either nominal or effective, it is assumed to be a(an) _____ rate.	01
	1.5	The _____ of a finite life alternative can be obtained by calculating its annual worth over one life cycles then multiplying by the P/A factor for its life cycle.	01
	1.6	The _____ of an asset for life cycle is the same as that calculated over two, three or any other number of life cycles.	01
	1.7	If the single payment factors are used when interest is compounded continuously, there are an _____ number of interest rate-time period combinations that would yield the correct answer.	01
	1.8	The equation that is used to calculate a present worth from a single future amount is _____.	01
	1.9	The length of time required for money to quadruple in value at an interest rate of 6% per year is _____ years.	02
	1.10	The number of years from now that an initial investment of $Rs.1\,million$ would be recovered from uniform receipts of $Rs.1.31\,Lakhs/year$ beginning three years from now at an interest rate of 10% per year is _____.	02
	1.11	Ram has assets of $Rs\,10{,}000$ and liabilities of $.2000$. His capital would be _____.	01
	1.12	The liabilities of a firm are $Rs.6000$ and the capital of the proprietor is $Rs.4000$. The total asset are $Rs.$_____.	01
	1.13	Give the equation to find the asset in terms of capital and liabilities.	01
	1.14	Ledger is a book of _____ entry.	01
	1.15	Ledger folio column in the journal is filled at the time of _____.	01
	1.16	Patent right is _____ account.	01
	1.17	Good will is _____ asset.	01
	1.18	Return inwards in trial balance are deducted from _____.	01

PART-B

2	a	List and explain the principles of engineering economy.	07
	b	With a neat block diagram, explain problem solving process.	09
		OR	
3	a	With an example, explain the two main rates of interest which are commonly practiced.	08
	b	What is cash flow diagram? List some of the conventions used while drawing CFD.	08
4	a	State and explain the conditions of present worth comparison.	06
	b	A company is considering two alternatives for manufacturing a certain part. Method R will have a first cost of $Rs.40{,}000$, an annual operating cost of $Rs.25000$ and a salvage value of Rs. 10,000. Method S will have an initial cost of $Rs.1Lakh$, an annual operating cost of $Rs.15{,}000$ and 12,000 salvage value. R has a life of 5 $years$, S has a life of 10 $years$. At an interest rate of 12% $per\ year$,Find the present worth values if the two alternatives which alternative is selected? Justify your answer.	10
		OR	
5	a	Explain the following: i) Present worth equivalence ii) Net present worth.	06
	b	A dam will have a first cost of $Rs.5\ million$, an annual maintenance cost of $Rs.25{,}000$ and minor reconstruction costs of $Rs.1\ lakh$ every five years. At an interest rate of $8\%/year$, find the capitalized cost.	10
6	a	Explain the three situations for equivalent annual worth comparisons.	06
	b	Two stamping machines are under consideration for purchase by a metal recycling company. The manual model will cost $Rs.25{,}000$ to buy with an eight year life and $Rs.5000$ is its salvage value. Its annual operating costs will be $Rs.16{,}000$. A computer controlled model will cost $Rs.95{,}000$ to buy and it will have a twelve year life if upgraded at the end of year six for $Rs.15{,}000$. Its terminal salvage value will be $Rs.23{,}000$, with annual operating cost of $Rs.7500$ for labor and $Rs.2500$ for maintenance. The company's minimum attractive rate of return 18%. Find annual worth of the computer controlled machine.	10
		OR	
7	a	Define the following: i) Service life. ii) Accounting life. iii) Economic life.	06
	b	An asset was purchased 5 years ago for $Rs.52{,}000$.It was expected to have an economic life of 8 $years$ at which time its salvage value would be $Rs.4000$. If the function that the asset was serving is no longer needed, for what price must it be sold now to recover the invested capital when $i = 12\%$?.	10

8 Journalize the following transactions in the books of Shri Ganesh.

		Rs.
1	*Shri ganesh started his business with cash*	50,000
2	*Purchased goods*	5,000
3	*Purchased goods for cash*	10,000
4	*Purchased goods from Mr. Sun for cash*	15,000
5	*Sold goods*	6,000
6	*Sold goods for cash*	12,000
7	*Sold goods to Mr. Sky for cash*	18,000
8	*Purchased goods from Mr. Moon*	10,000
9	*Purchased goods from Mr. Star on credit*	20,000
10	*Sold goods to Mr. Sea*	12,500
11	*Sold goods to Mr. ocean on credit*	25,000
12	*Mr. Sea returned goods.*	2,500

16

OR

9 Transactions of Ramesh for April 20*X*1 are given below. Journalise them.

20*X*1 *April*		*Rs.*
1	*Ramesh started business with cash*	10,000
2	*Paid into bank*	7,000
3	*Brought goods for cash*	500
4	*Draw cash from bank for office*	100
13	*Sold to Krishna goods on credit*	150
20	*Bought of Shyam goods on credit*	225
24	*Received from Krishna*	145
	allowed him discount	5
28	*Paid shyam cash*	215
	discount allowed.	10
30	*Cash sales for the month*	800
	paid rent	50
	Paid salary	100

16

10 The following trial balance has been extracted from the books of Sri Santhosh kumar on 31st March 20*X*2:

Particulars	*Dr*(*Rs.*)	*Cr*(*Rs.*)
Plant and machinery	1,00,000	-
Furnitures and fixtures	12,000	-
Capital account/ drawings account	2,000	1,76,000
loose tools	20,000	-
good will	10,000	-
opening stock (1.4.20*X*1)	20,000	-
Returns	8,000	4,000
discount	-	6,000
Purchases/sales	2,12,000	4,68,000
Wages and other expenses	1,00,000	-
Provision for doubtful debts	-	2,000
Carriage inwards	12,000	-
Salaries	41,600	-
General expenses and insurance	72,000	-
Rent and Taxes	14,400	-

Particulars	Dr(Rs.)	Cr(Rs.)
Postage and telegrams	4,000	-
Bank overdraft	-	20,000
Sundry debtors/ creditors	56,000	24,000
B Barva	2,000	-
Cash and Bank balance	14,000	-
	7,00,000	7,00,000

The following additional information is available

a) Stock on March 31*st* 20*X* 2 was 30,800.
b) Depreciation is to be charged on plant and machinery at 5% *pa* and furniture and fixtures at 6% *pa*. Loose tools are revalued at *Rs*. 16,000.
c) Provision for doubtful debts is to be maintained at 5% on Sunday debtors.
d) Remuneration of *Rs*. 2,000 paid to Mr. B Barva, a temporary employee stands debited to his personal account and it is to be corrected.
e) Unexpired insurance was *Rs*. 400. 16

OR

11 The following trial balance extracted from the books of a merchant Mr. Nageswara Rao on 31.03.20*X*2.

Particulars	Dr(Rs)	Cr(Rs.)
Furniture and fittings	640	-
Motor vehicles	6250	-
Buildings	7500	-
Capital	-	12500
Bad debts	125	-
Provision for doubtful debts	-	200
Sundry debtors and creditors	3800	2500
Stock as on 1.4.20*X*1	3460	-
Purchases and sales	5475	15450
Bank overdraft	-	2850
Sales and purchases returns	200	125
Advertising	450	-
Interest on Bank overdraft	118	-
Commission	-	375
Cash	650	-
Taxes and insurance premium	782	-
General expenses	1250	-
Salaries	3300	-
	34,000	34,000

Adjustments

a) Stock on hand 31.03.20*X*2 *Rs*. 3250.
b) Depreciate buildings at 5% *pa* furniture at 10% *pa* ; Motor vehicles at 20% *pa*.
c) *Rs*. 85 is due for interest on bank overdraft.
d) Salaries *Rs*. 300 and taxes *Rs*. 200 are outstanding
e) Insurance premium amounting *Rs*. 100 prepaid
f) One third of the commission received is in respect of work to be done next year.
g) Write off a further sum of *Rs*. 100 as bad debts from debtors and create provision for doubtful debts. at 5% on debtors.

Prepare a trading and profit and loss account and the balance sheet. 16